

An HIV-specific patient-reported measure for functioning and social participation in Belgium

Development of an HIV-specific patient-reported
measurement module for people living with HIV
in Belgium. The SPADIS project.

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EXECUTIVE SUMMARY

A novel multidimensional patient-reported measure for functioning and social participation of people living with HIV in Belgium

This report presents the development of an HIV-specific module for the SPADIS Patient-Reported Measure on Functioning and Social Participation. The instrument captures the social participation and functioning, and experiences with environmental factors such as the health care sector, for people living with HIV in Belgium in Belgium. The work described in this report is part of the SPADIS project¹.

The SPADIS project

The SPADIS project (Measuring Social Participation of People Living with Chronic Diseases) aims to comprehensively measure social participation among people with chronic conditions in Belgium. It combines multiple approaches, including the development of patient-reported measures that capture participation, functioning, and related determinants (Work Package 3).

A standardized methodological approach

The field of patient-reported outcome and experience measures (PROMs and PREMs) is characterized by a large heterogeneity of available instruments as well as methodological approaches for their development.. This report provides a detailed overview of a novel, standardized approach to develop disease-specific patient-reported measures. While the methodology has been published elsewhere, this report details all phases and outcomes and provides the templates that were used to reach these outcomes.

An HIV-specific patient-reported measurement tool of the social participation of PLWH in Belgium

The steps outlined in this report resulted in the development of an HIV-specific patient-reported measurement module based on the perspectives of PLWH, building upon existing, high quality HIV-specific PROMs, and incorporating the perspectives of Belgian health professionals. This instrument is now ready for dissemination. This will help perform the first comprehensive measurement of social participation and its determinants, and allow for the fine-tuning of the measure based on psychometric testing.

¹ Measuring Social Participation Among People Living With Chronic Diseases | sciensano.be

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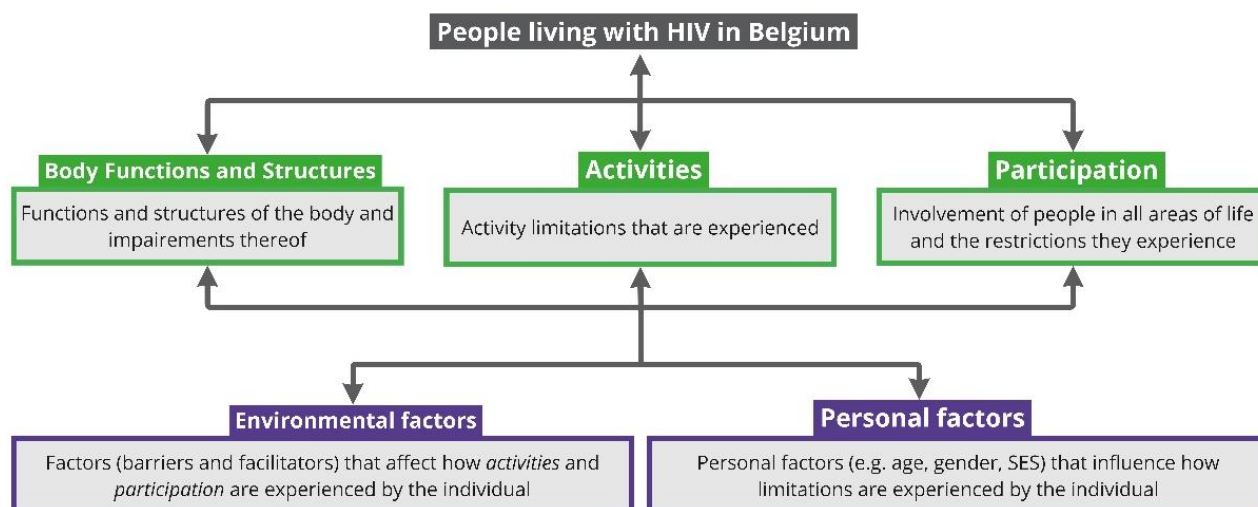
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I. Background, objectives and methodology

The Sciensano SPADIS project aims to measure and understand the social participation of people living with, among others, HIV in Belgium. **Social participation** refers to the extent to which individuals (can) participate in activities that allow for interactions with others in the society². Additionally, SPADIS aims to investigate potential determinants and inequalities in social participation.

The **International Classification of Functioning and Disability (ICF) model** was chosen as the guiding framework to define social participation (by the ICF component of 'participation') as well as potentially associated factors (ICF components of 'Body functions and structures', 'Activities', 'Personal Factors' and 'Environmental factors'). Figure 1 describes the ICF model and its components, which are conceptualized to dynamically interact with one another. SPADIS will investigate and quantify the associations between the ICF component of participation ("social participation") and the other ICF components. Although the ICF framework emphasizes functioning and disability, the model is suitable to measure health and well-being as a biopsychosocial concept while also taking into account contextual factors, at both the individual and societal or environmental level.

Figure 1 • The ICF model and its components



This report is framed within Work Package 3 of the SPADIS project in which **an HIV-specific patient reported measure (PRM)** is developed to quantify impairments, limitations and restrictions for the ICF components of Body Functions and Structures, Activities and participation, respectively. The SPADIS PRM will collect patient reported outcomes (PROs) as well as patient reported experiences (PREs). The latter will be used to measure the impact of environmental factors. The **content of the PRM is based on the cross-matching of three different perspectives**:

1. Results from **literature review** on existing PRMs that (1) measure health or health-related quality of life (HRQoL) in a comprehensive and multidimensional way, (2) are widely used and (3) are of high methodological quality (esp. high content validity) according to [COSMIN standards](#).
2. **Patient perspectives**, collected in two focus groups organized with people living with HIV in Belgium.
3. **Experts perspectives**, provided by the members of the HIV Reference Centre (HRC) QoL Working Group and by their own developed QoL PROM, as well as through an expert validation exercise with healthcare professionals involved in the care and support of PLWH in Belgium.

Importantly, this HIV-specific PRM will be accompanied by a generic PRM module aimed at capturing relevant Personal factors (Figure 1) as well as generic outcomes of HRQoL, social participation, and more specifically work participation in Belgium. A protocol paper outlining the methodological approach can be found [here](#).

² Levasseur /, Lussie-Therrien M, Biron ML, et al. Scoping study of definitions of social participation. Update and co-construction of an interdisciplinary consensual definition. *Age Ageing*. 2022;52(2):afab215. Doi:10.1093/ageing/afab215

1. Methodology

1.1. Thematic analysis: Finding a common language to compare perspectives of PLWH and results from the literature review

To facilitate the content development of the PRM, and more specifically the cross-matching of results from the literature review (content analysis of existing PRMs), the focus groups and the content of the HRC QoL PROM, the [ICF classification system](#) was applied. Across all ICF components, the ICF classification system provides approximately 1400 different ICF categories and codes to classify specific body functions, structures, types of activities at both the individual and societal or community level, and environmental factors which can affect an individual's functioning or disability experience.

Cieza's refined ICF linking rules³ were applied to link each item from the eligible PROMs identified in the literature review, each quote given by participants of the focus groups, and each item of the HRC QoL PROM (Table 1). Each item or quote may refer to multiple ICF codes. For example, the MOS-HIV item :*"During the past 4 weeks, how much did pain interfere with your normal work (or your normal activities, including work outside the home and housework)?"* can be linked to ICF codes within the ICF component of body functions and structures (b280 – Sensation of pain) as well as the ICF component of activities and participation (d850 – Remunerative employment, d230 – Carrying out a daily routine).

The level of detail provided by the ICF framework allows SPADIS researchers to comprehensively identify which aspects of functioning and disability are addressed by the PROMs from the literature, by the QoL PROM from the HRCs, and by the PLWH in the focus groups. The common ICF language is thus used as means of **thematic deductive analysis**.

In parallel, the collected perspectives are classified by performing a **thematic inductive analysis**. This more exploratory approach, in which the topics and subtopics emerge from the data itself, enables the classification of items and quotes that go beyond the ICF's more narrow focus on ability and medical functioning. Similarly to the domains and subdomains assessed by the HRC QoL PROM, the topics and subtopics from this inductive method reflect the (sub)domains of more common (HR)QoL models or conceptualizations such as 'emotional well-being', 'physical functioning', 'cognitions', etc. The topics and subtopics that were identified in the thematic inductive analysis, are presented in **Table 1**.

Acceptance	Meaningfulness	Social security
Accessibility of information	Being useful	Insurance
Activities of daily living	Pain	Symptoms
Household chores	Interference of pain	Digestive problems
Self-care	Physical environment	Dizziness
Appetite	Physical functioning	Fever, chills or sweat
Cognitions	Bending or lifting	General physical symptoms
Concentration	Mobility	Headaches
Education	Moderate physical activity	Muscle cramps
Emotional well-being	Stability	Numbness
Negative feelings	Vigorous physical activity	Shortness of breath
Positive feelings	Physical safety	Transport
Energy	Quality of life	Treatment-related
Fears, worries, stigma	Self-esteem	Stress
Fear of death	Body image	Treatment burden
Fear of future	Confidence	Weight maintenance
Stigma	Self-rated health	Work capacity
Stress	Sense of freedom	
Financial resources	Sexuality	
Financial system	Skin problems	
Health behavior	Sleep	
Alcohol	Sleep quality	
Drug use	Social activities	

³ Cieza A, Fayed N, Bickenbach J, Prodinger B. Refinements of the ICF Linking Rules to strengthen their potential for establishing comparability of health information. Disabil Rehabil. 2019 Mar;41(5):574-583. doi: 10.3109/09638288.2016.1145258. Epub 2016 Mar 17. PMID: 26984720.

Smoking	Leisure activities
Health care service	Social relationships
Accessibility	Acceptance
Knowledge of health professionals	Asking for help
Multidisciplinarity	Being open about disease
Relationship with health professionals	Friends
Health changes	Family
Home environment	Intimate relationships
	Loneliness
	Personal relationships
	Receiving support
	Supporting others

Table 1 • Overview of topics and subtopics from the thematic inductive analysis of eligible PROMs from literature, focus groups with PLWH, and the HRC QoL PROM

II. Results from the focus groups and literature review

This section of the report provides an overview of the results of the focus groups with PLWH as well as the literature review on existing HIV-specific PRMs, using the terminology as defined by the thematic inductive analysis (Table 1).

1. Results from the focus groups with PLWH

In total, 14 PLWH participated in the focus group discussions, which were guided by a semi-structured interview guide. They were **presented with the definition of social participation** (figure 1) and were asked **two main questions**:

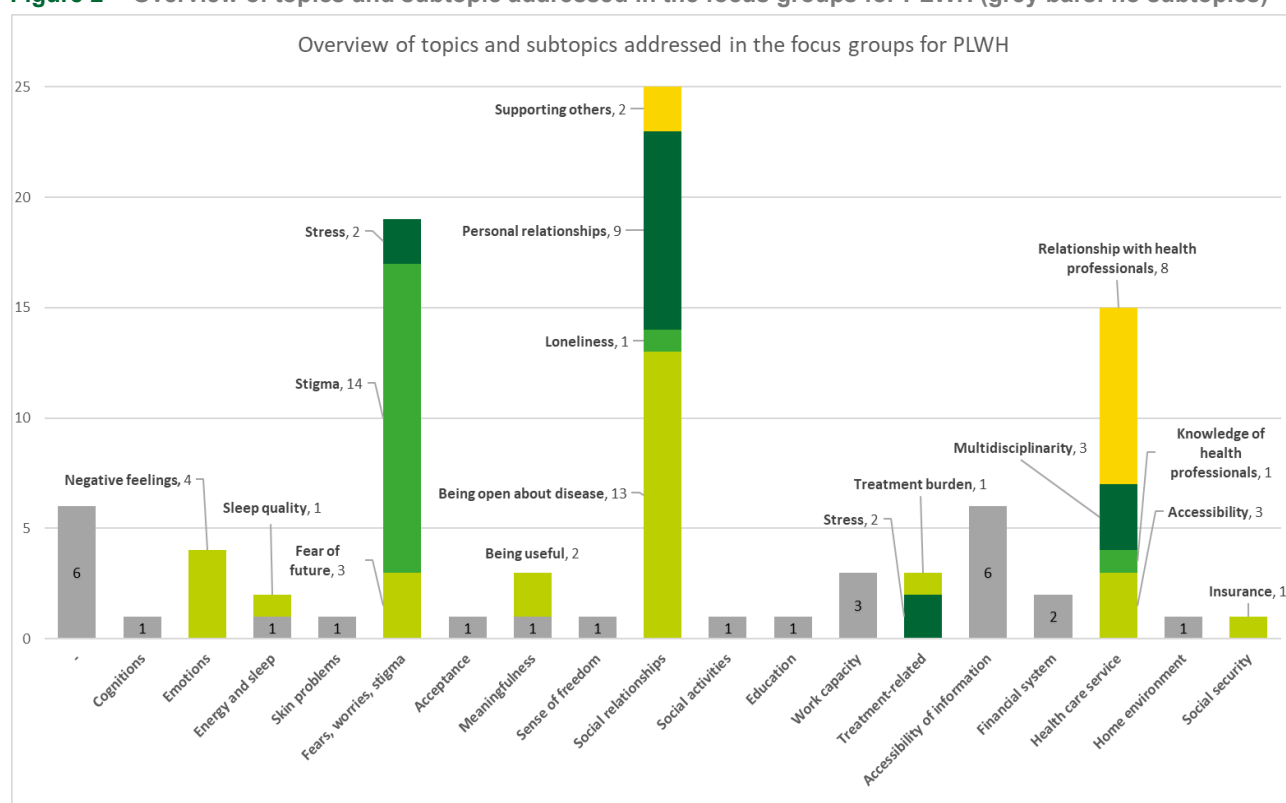
1. *“Which **aspects of social participation** are important for your (social) well-being and/or which aspects are impacted by your condition?”*
2. *“On a **physical and psychological level**, do you feel an impact of your condition? Do you function as well as you would like or need to on a physical and psychological level?”*

Related to each of these main questions, participants were asked a **subquestion**: *“If you think about your **environment and living conditions**, what did you experience as helpful or supporting, or as a barrier?”*

The goal of the focus group discussion was not to reach saturation. Rather, the goal was to identify which topics are the most important to PLWH in Belgium and, as a consequence, which topics the SPADIS PRM module for PLWH should emphasize.

Figure 2 provides an overview of the topics (x-axis) and the subtopics (stacked bar sections) that were quoted by the participants, as well as the amount of times they were quoted (y-axis). A clear emphasis was put on **fears, worries and stigma**, on **social relationships** and behaviours within these relationships and on aspects of **health care** that acted as either barriers or facilitators for their physical, psychological and social well-being.

Figure 2 • Overview of topics and subtopic addressed in the focus groups for PLWH (grey bars: no subtopics)



2. Results from the literature review on existing HIV-specific PROMs

Two SPADIS researchers searched the literature independently, using the same search strategy, to identify appropriate HIV-specific PROMs that assess functioning and disability or (HR)QoL in a multidimensional and comprehensive way. This included searching for systematic reviews of HIV-specific PROMs that could be used as a basis. Three such systematic reviews were identified, the most recent of which covered studies published up to February 2021. The PROMs addressed in these systematic review were discussed by the researchers to reach consensus on which PROMs to select (considering the eligibility criteria described in the study protocol). To capture newer instruments, SPADIS researchers additionally searched for studies published after February 2021. This resulted in the identification of two additional PROMs (see [annex 1](#) for list of all instruments identified in the literature review phase). After discussing the instruments among the two SPADIS researchers, the following PROMs were selected for further examination: the Short-form HIV Disability Questionnaire (**SF-HDQ**), the Medical Outcomes Study HIV Health Survey (**MOS-HIV**), the WHO Quality of Life HIV BREF (**WHO-QoL HIV BREF**), the Multidimensional quality of life Questionnaire for persons with HIV/AIDS (**MQOL-HIV**), the Clinic Screening tool HIV (**CST-HIV**), the Patient-Reported Outcomes Quality Of Life HIV instrument (**PROQOL-HIV**) and the **PoZQoL HIV**.

An **adapted COSMIN Study Design Checklist⁴ (SDG) that emphasizes content and face validity** (See [Annex 2](#) for adapted COSMIN checklist) of the PROMs was applied by two independent researchers to rate the seven instruments in terms of measurement properties and methodological quality in general. The results of which were discussed to build consensus. Two of the identified PROMs did not meet the standards set forth, mainly in terms of content validity: the **MQOL-HIV** and the **PoZQoL HIV**. The other PROMs were ranked in terms of methodological quality by summing the results of the COSMIN SDG 4-point rating scale (weighted to emphasize content validity). The five PROMs that met all the methodological quality requirements, ranked from highest to lowest quality are the following: WHO-QoL HIV BREF, MOS-HIV, CST-HIV, PROQOL-HIV, SF-HDQ. The complete and detailed results of the COSMIN rating exercise are presented in [Annex 3](#). Next, the refined **ICF** linking rules (thematic deductive analysis) and the thematic inductive analysis were applied to all items of the selected PROMs.

Items of the eligible PROMs were **linked to the domains of the HRC QoL PROM** to enhance the comparability of the present results with the exercise of the HRC Working group. An **additional domain of 'environmental factors'** was added.

Figure 3 presents the results of the inductive approach and shows which (sub)topics are most often addressed by the eligible PROMs for the HRC QoL PROM domain of psychosocial health and support. One should note that 10 out of the 11 treatment-related items come from the PROQOL HIV. Figure 4 describes the number of items across all eligible PROMs that were linked to the domain of environmental factors, which was added to the HRC QoL PROM domains. **Figure 5** presents similar results for the HRC QoL PROM domain of health and quality of life.

The items from eligible PROMs identified in the **literature mainly focus on fears, worries and stigma, on emotional well-being** (e.g. anxiety, feeling blue, despair, etc.) and on **social relationships**. This is in line with the perspectives of PLWH that participated in the focus groups. Unlike the results of the focus groups, the eligible PROMs also emphasize **symptom burden, cognitions, physical functioning**. Compared to the results of the focus groups, the eligible PROMs offer **few items that can be associated with environmental factors**, especially items related to the health care service such as satisfaction or experience with health care professionals and multidisciplinary of care and support. All PROMs include items for self-rated health and/or quality of life in general.

⁴ COSMIN Study Design checklist for Patient-reported outcome measurement instruments. ([URL](#))

Figure 3 • Overview of topics and subtopics addressed in the eligible PROMs, for the domain of psychosocial health and support (grey bars: no subtopics)

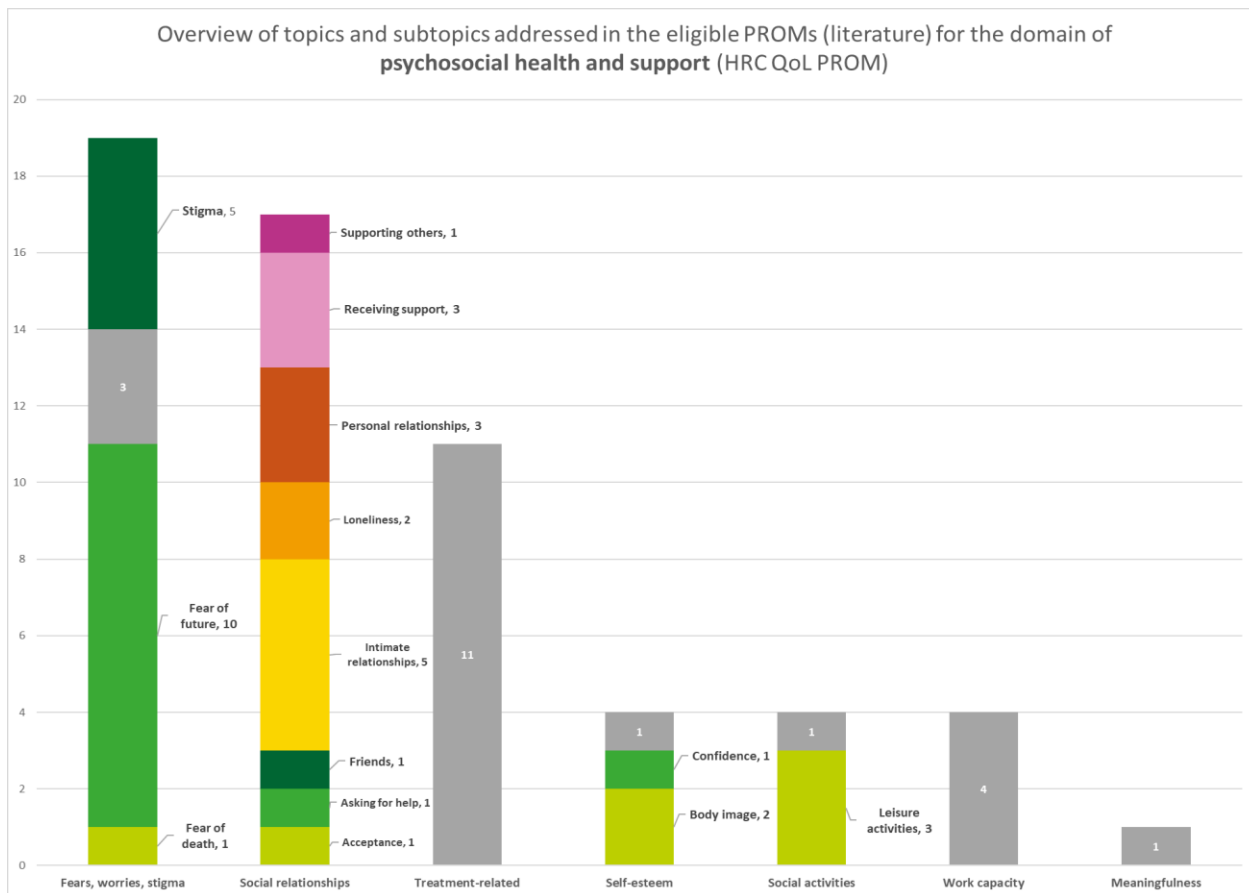


Figure 4 • Overview of topics and subtopic addressed in the eligible PROMs for the domain of Environmental factors (grey bars: no subtopic)

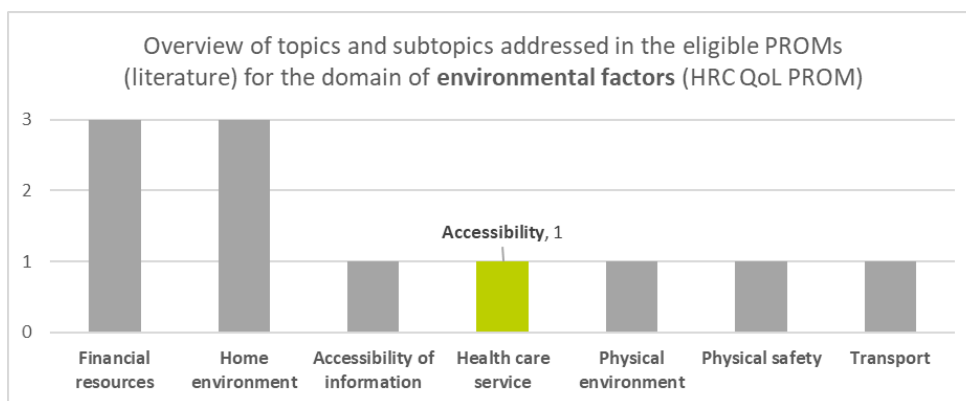
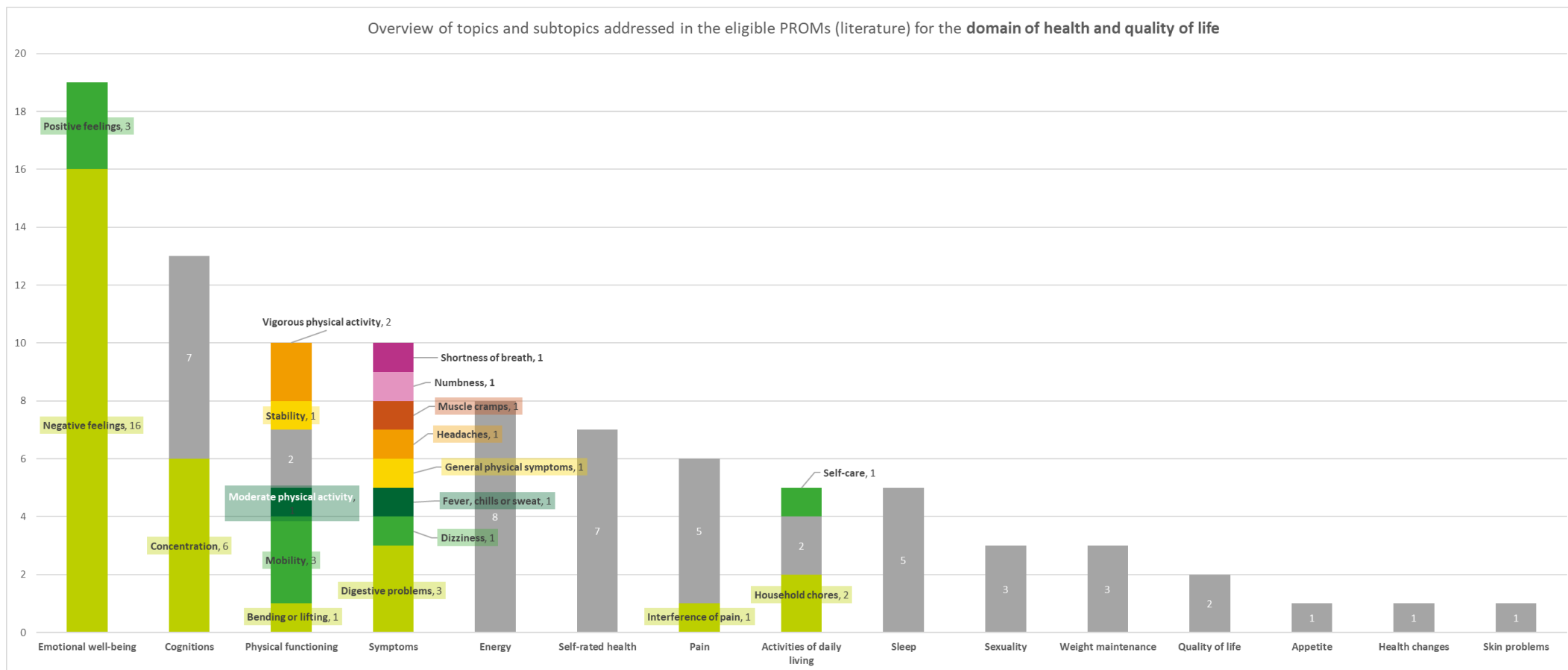


Figure 5 • Overview of topics and subtopic addressed in the eligible PROMs for the HRC QoL PROM domain of health and quality of life (grey bars: no subtopic)



3. Results from the cross-matching exercise

This section describes the (sub)topics that were identified for each of the HRC QoL PROM domains and the additional domain of environmental factors. In order to deduct which topics should be included in a first item set, we looked at whether or not the topics, subtopics, and ICF codes were addressed by PLWH in the FGs, i.e. **cross-matching**. This suggests that their inclusion in the SPADIS PRM item set is relevant (content validity). If (also) addressed by the majority of eligible PROMs (literature review) as well as in the HRC QoL PROM, the inclusion of the (sub)topics in the SPADIS PRM is justified. An overview of the topics from the thematic inductive analysis addressed in the FGs, the HRC QoL PROM and the eligible PROMs from literature is presented in Table 2.

Topic	FG	HRC QoL tool	#PROMs that cover topic	Eligible PROMs from the literature review, ranked from highest methodological quality to lowest.				
				WHO-HIV	MOS-HIV	PROQ oL-HIV	CST-HIV	SF-HDQ
Health and quality of life	Activities of daily living		4	x	x	x		x
	Appetite		1			x		
	Cognitions	x	5	x	x	x	x	x
	Emotional well-being	x	5	x	x	x	x	x
	Energy	x	5	x	x	x	x	x
	Health changes		1		x			
	Pain		5	x	x	x	x	x
	Physical functioning		4	x	x	x		x
	Quality of life		2	x	x			
	Self-rated health		3	x	x	x		
	Sexuality		3	x		x	x	
	Skin problems	x	1			x		
	Sleep		4	x		x	x	x
	Symptoms		4	x		x	x	x
	Weight maintenance		2			x	x	
Psychosocial health and support	Self-acceptance	x	0					
	Education	x	0					
	Fears, worries, stigma	x	5	x	x	x	x	x
	Meaningfulness	x	1	x				
	Self-esteem		3	x		x		x
	Sense of freedom	x	0					
	Social activities	x	3	x	x			x
	Social relationships	x	4	x		x	x	x
	Treatment-related	x	2	x		x		
	Work-capacity	x	3	x	x			x
Health behaviors		x	0					
Environmental factors*	Accessibility of information	x	1	x				
	Financial resources		2	x			x	
	Financial system	x	0					
	Health care service	x	1	x				
	Home environment	x	3	x			x	x
	Physical environment		1	x				
	Physical safety		1	x				
	Social security	x	0					
	Transport		1	x				

Table 2 • Overview of topics addressed in the focus groups for PLWH, the eligible PROMs found in literature, and the HRC QoL PROM, by domains of the HRC QoL PROM.

*the domain of environmental factors was added by SPADIS researchers

In what follows, for each domain, the topics and related subtopics that emerged from the thematic inductive analyses are described for the focus groups, as well as for the items of the eligible PROMs from the literature review and the HRC QoL PROM.

For each (sub)topic, the ICF codes which were linked to the related items are provided in a table format. The ICF codes allow us to identify which ICF component is being addressed and offer the necessary level of detail to distinguish between different perspectives adopted by items linked to one (sub)topic.

For (sub)topics and ICF codes that were addressed in the FGs, and/or were covered by the majority of eligible PROMs in literature, the corresponding item of the PROM that was ranked highest in terms of methodological quality is selected for the initial SPADIS PRM item set. Items from the eligible PROMs that match an item from the HRC QoL PROM are of higher preference. Where necessary, this approach for selecting the first item set can be altered to fit specific objectives of SPADIS and/or the HRC Working Group. This is explained in the text.

3.1. Health and quality of life

This domain of the HRC QoL PROM describes aspects related to general health and well-being or quality of life, general physical symptoms or impairments related to HIV, cognitive and emotional impairments, sleep quality and energy level, and sexual satisfaction. The topics from the thematic inductive analysis of the focus groups and literature review PROMs that align most with this domain of the HRC QoL PROM are described in Table 2. **Aspects within this domain mostly align with ICF codes related to the ICF component of body functions and structures and the ICF component of activities.**

3.1.1. Appetite, health changes, skin problems, and weight maintenance

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Appetite			1			1		
b1302 - Appetite			1			x		
Health changes			1		x			
<i>No ICF code available</i>			1		x			
Skin problems	x		1			x		
b840 - Sensation related to the skin	x		1			x		
Weight maintenance			2			x	x	
b530 - Weight maintenance functions			2			x	x	

The topics of **appetite**, **health changes** and **weight maintenance** were not addressed by the participants of the **FGs**. Only one participant referred to changes of the skin and did so in a limited way. The **HRC QoL PROM** also does not address these topics.

For each of the above described topics from the inductive thematic analysis, the **LR** showed that, at most, 2 out of 5 PROMs covered the topics described here. This is the case for weight maintenance related issues, which remain unaddressed by the most methodologically sound [high (content) validity] PROMs (WHO-QoL HIV BREF and MOS-HIV). Our exercise and methodological approach do not support the inclusion of these items in the SPADIS PRM for PLWH.

The SPADIS PRM will **not include items** related to **appetite**, **health changes**, **weight maintenance** and **skin-related aspects** (*sensation of the skin*).

3.1.2. Activities of daily living

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Activities of daily living			4	x	x	x		x
d230 - Carrying out daily routine			2	x		x		
d620 - Acquisition of goods and services			1					x
d629 - Acquisition of necessities, other specified and unspecified			1					x
Household chores			2	x				x
d630 - Preparing meals			1					x
d640 - Doing housework			2	x				x
Self-care			2		x	x		
d510 - Washing oneself			1		x			
d530 - Toileting			1		x			
d540 - Dressing			1		x			
d550 - Eating			1		x			

When asked to discuss functional impairments and limitations in activities, the participants from the **FGs** did not describe any limitations in their ability to perform *activities of daily living*. In the **HRC QoL PROM**, no item can be found that assesses limitations in activities of daily living.

However, the fact that **4 out of 5 eligible PROMs from the LR** assess limitations for activities of daily living warrants the inclusion of such an item in the SPADIS PRM for PLWH. The following items are included in the PROMs:

MOS-HIV

How much does your health now limit you in eating, dressing, bathing or using the toilet?

Yes, limited a lot - Yes, limited a little - No, not limited

PROQoL-HIV

During the last two weeks, because I am HIV positive, I have had difficulty with daily activities

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble doing household chores such as cleaning, doing dishes, laundry, and cooking".

Not at all - A little trouble - A moderate amount of trouble - A lot of trouble - I can't do this

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble getting out to do errands such as grocery shopping, banking, or doctor's appointments".

Not at all - A little trouble - A moderate amount of trouble - A lot of trouble - I can't do this

WHO-QoL HIV BREF

How satisfied are you with your ability to perform your daily living activities?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Items from the MOS-HIV and SF-HDQ offer more detail in terms of the specific daily living activities that are assessed. The items from the PROQoL-HIV and the WHO-QoL HIV BREF remain more general and do not specify which daily living activities are being assessed.

Item selection: adopt singular **WHO-QoL HIV BREF** item (highest methodological quality and content validity) – and **add examples** to illustrate aspects related to ICF codes for household chores and self-care:

"How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)."

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

3.1.3. Cognitions

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Cognitions	x	x	5	x	x	x	x	x
b144 - Memory functions		x	3		x	x		x
b160 - Thought functions	x	x	3		x		x	x
b1646 - Problem Solving		x	1		x			
d155 - Acquiring new skills			1				x	
d163 - Thinking		x	1		x			
d175 - Solving problems		x	1		x			
Concentration		x	5	x	x	x	x	x
b140 - Attention functions		x	5	x	x	x	x	x
d160 - Focusing attention		x	1		x			
d163 - Thinking	x		1		x			

During the **FGs**, the number of remarks that pertained to cognitive functioning remained limited. The few remarks related more specifically to thought functions (b160 – Thought functions) and the fact that reasoning and decision-making became more difficult (d163 – Thinking) with time (since diagnosis).

In contrast, 3 out of the 20 items from the **HRC QoL PROM** relate to cognitive functioning. They are the three screening questions for cognitive impairment of the European Aids Clinical Society (EACS) guidelines:

- Do you regularly experience memory loss (e.g. forget special events, recent events, appointments, et.)?
Not at all – A little – Moderately – Very much - Extremely
- Do you feel that you are slower at reasoning, planning activities or solving problems?
Not at all – A little – Moderately – Very much - Extremely
- Do you have difficulty maintaining your attention on something (e.g. a conversation, a book, a movie)?
Not at all – A little – Moderately – Very much – Extremely

These questions were developed by health professionals without the involvement of PLWH. Although they are frequently used in clinical practice and are very similar to questions in the LR PROMs, they have not been psychometrically validated and **their content validity can therefore not be assumed**.

The **LR** and analysis of eligible PROMs highlights the importance of assessing cognitive functioning (covered by all 5 PROMs) in PLWH. The results of the ICF linking exercise shows that the ability to focus **attention** (b140 – Attention function), **memory** (b144 – Memory functions), **thought functioning** and the activity of **thinking** (b160 – Thought functions, d163 – Thinking) are most often assessed with PRO items. The following items on cognitions are included in the eligible PROMs from the LR:

CST-HIV

During the past month, do you think it has been harder for you to learn new things?

None - Slightly - Somewhat - Quite - Extremely

During the past month, have you had any difficulty paying attention?

None - Slightly - Somewhat - Quite - Extremely

During the past month, have you had any difficulty thinking clearly?

None - Slightly - Somewhat - Quite - Extremely

MOS-HIV

How much of the time, during the past 4 weeks, did you forget things that happened recently, for example, where you put things and when you had appointments?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, did you have difficulty doing activities involving concentration and thinking?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, did you have trouble keeping your attention on any activity for long?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

PROQoL-HIV

During the last two weeks, because I am HIV positive, I have had difficulty concentrating or paying attention

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, I have had problems with my memory

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble concentrating".

None at all - A little trouble - A moderate amount of trouble - Very much trouble - Extreme trouble

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble remembering things like appointments and when to take my medication".

None at all - A little trouble - A moderate amount of trouble - Very much trouble - Extreme trouble

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble thinking clearly".

None at all - A little trouble - A moderate amount of trouble - Very much trouble - Extreme trouble

WHO-QoL HIV BREF

How well are you able to concentrate?

Not at all - A little - A moderate amount - Very much - Extremely

Although the WHO-QoL HIV BREF is the most eligible in terms of methodological quality and (content) validity, **a singular item to assess cognitive functioning that focuses solely on attention or concentration does not reflect the importance of (and heterogeneity in) assessing cognitive functioning**. The second highest rated PROM, the MOS-HIV, offers a more elaborate set of items to measure cognitive functioning.

Item selection: SPADIS researchers discussed the adoption of the EACS screening questions for cognitive impairment versus the adoption of the four items of the **MOS-HIV** (second highest rating for methodological quality). The lack of established validity for the EACS questions resulted in the choice for the MOS-HIV items on cognitive functioning. Two items of the MOS-HIV assess the same ICF code of b140 – Attention function, which led to the decision to **only include one of these items**.

MOS-HIV

- How much of the time, during the past 4 weeks, did you **forget things that happened recently**, for example, where you put things and when you had appointments?
- How much of the time, during the past 4 weeks, did you have **difficulty** doing **activities** involving **concentration** and **thinking**?
- How much of the time, during the past 4 weeks, did you have **difficulty reasoning and solving problems**, for example, making plans, making decisions, learning new things?
- ~~How much of the time, during the past 4 weeks, did you have **trouble keeping your attention on any activity for long**?~~

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

3.1.4. Emotional well-being

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Emotional well-being	x	x	5	x	x	x	x	x
Negative feelings	x	x	5	x	x	x	x	x
b152 - Emotional functions	x	x	5	x	x	x	x	x
d240 - Handling stress and other psychological demands	x		0					
Positive feelings			1		x			
b152 - Emotional functions			2	x	x			

FG remarks and items from the LR PROMs that relate to this topic refer to **anxiety, sadness/feeling blue, being happy, feeling peace and calm, despair, irritability, and depression**. These specific aspects of psychological well-being **cannot be accurately classified within the ICF coding system**. The ICF code by which they are classified in this exercise is b152 – Emotional functions. With the **inductive thematic analysis**, items under the topic of emotional well-being are

classified as either adopting a positive perspective, or rather assessing positive states of emotional well-being (positive feelings), or negative states of emotional well-being (negative feelings).

During the **FGs**, several participants indicated that negative feelings (loneliness, sadness, feeling blue or anxious, suicidal thoughts, feeling despair) weigh heavily on PLWH.

The **HRC QoL PROM** assesses these topics under the subdomain of 'depression' with two items, which were both linked to b152 – Emotional functions. One of these items was adopted from the WHO-QoL HIV BREF, the other item's origin is unknown and focuses more on symptoms of depression specifically:

"In the past month, have you lost interest or joy in activities that normally interested you or brought you joy?" *Never – Seldom – Quite often – Very often – Always*

The **LR** shows that all eligible PROMs assess states of emotional well-being. Only the MOS-HIV includes items that assess positive states of emotional well-being (being happy and feeling calm and peaceful). The items identified in the PROMs from the LR that assess emotional well-being are the following:

CST-HIV

During the past month, how often have you felt anxiety?

Never - Rarely - Sometimes - Frequently - Always

During the past month, how often have you felt sadness or discouraged?

Never - Rarely - Sometimes - Frequently - Always

MOS-HIV

How much of the time, during the past 4 weeks, have you been a happy person?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, have you been a very nervous person?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, have you felt calm and peaceful?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, have you felt downhearted and blue?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How much of the time, during the past 4 weeks, have you felt so down the dumps that nothing could cheer you up?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How often during the past four weeks did you feel despair over your health problems

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How often during the past four weeks did you feel weighed down by your health problems?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How often during the past four weeks were you discouraged by your health problems

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

PROQoL-HIV

During the last two weeks, because I am HIV positive, (unknown) - Anxiety

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Depression

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Irritability

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Sadness

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Stress

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel anxious".

Not at all - Slightly - Moderately - Very - Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel sad, down, or depressed".

Not at all - Slightly - Moderately - Very - Extremely

WHO-QoL HIV BREF

How often do you have negative feelings such as blue mood, despair, anxiety, depression?

Never - Seldom - Quite often - Very often - Always

Item selection: adopt the singular item from the **WHO-QoL HIV BREF** as it covers ‘blue mood’, ‘despair’, ‘anxiety’ and ‘depression’ in one item, and because it was the highest rated PROM in terms of methodological quality.

WHO-QoL HIV BREF:

- How often do you have negative feelings such as blue mood, despair, anxiety, depression?
Never - Seldom - Quite often - Very often – Always

Remark:

This item assesses multiple aspects of negative states of emotional well-being at once and may therefore be difficult to interpret. After internal discussions, it was decided that this item will be split up to assess these states separately within a question matrix:

[question matrix] How often do you have the following negative feelings: (a) blue mood, (b) despair, (c) anxiety, (d) depression?
Never - Seldom - Quite often - Very often – Always

3.1.5. Energy and sleep

Topics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Energy	x		5	x	x	x	x	x
b130 – Energy and drive functions	x		5	x	x	x	x	x
Sleep	x	x	4	x		x	x	x
b134 – Sleep function	x							
b1343 – Quality of sleep			2	x			x	

During the **FGs**, PLWH made few remarks related to energy and sleep. The **quality of sleep** (b134 – Sleep function, b1343 – Quality of sleep) was discussed in the context of having vivid and unsettling dreams, for which the association with HIV or HIV-related treatment was assumed by PLWH. **Physical fatigue** (b130 – Energy and drive functions) was highlighted as an issue for PLWH during the FGs.

The **HRC QoL PROM** adopted a singular WHO-QoL HIV BREF HIV item on satisfaction with sleep, but did not include the WHO-QoL HIV BREF item on energy and drive functions.

In contrast, the results from the **LR** and the ICF linking exercise illustrate the importance of assessing **sleep functions and/or quality** (covered by 4 out of 5 eligible PROMs) **as well as energy** (covered by all PROMs). The following items from the LR PROMs relate to sleep and energy:

CST-HIV

During the past month, have you had any sleep problems?

None - Slightly - Somewhat - Quite - Extremely

During the past month, how satisfied have you felt with the quality of your sleep?

None - Slightly - Somewhat - Quite - Extremely

During the past month, how tired have you felt?

None - Slightly - Somewhat - Quite - Extremely

MOS-HIV

How often during the past four weeks did you feel full of pep?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How often during the past four weeks did you feel tired?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How often during the past four weeks did you feel worn out?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

How often during the past four weeks did you have enough energy to do the things you wanted to do?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

PROQoL-HIV

During the last two weeks, because I am HIV positive, I have felt tired

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, I have had difficulty sleeping

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel fatigued (or tired) when doing my usual activities".

Not at all - Slightly - Moderately - Very - Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble sleeping".

None at all - A little trouble - A moderate amount of trouble - Very much trouble - Extreme trouble

WHO-QoL HIV BREF

Do you have enough energy for everyday life?

Not at all - A little - Moderately - Mostly - Completely

How satisfied are you with your sleep?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection: In line with the LR results, the SPADIS PRM will adopt the **two items** that assess **sleep quality** as well as **energy level** from the **WHO-QoL-HIV BREF**:

- Do you have enough energy for everyday life?
Not at all - A little - Moderately - Mostly – Completely
- How satisfied are you with your sleep?
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

3.1.6. Pain

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Pain			5	x	x	x	x	x
b280 – Sensation of pain			5	x	x	x	x	x
Interference of pain					x			
b280 – Sensation of pain			1		x			
d850 – remunerative employment			1		x			
d855 – Non-remunerative employment			1		x			

During the **FGs**, pain was **not addressed** by the participants. The **HRC QoL PROM** does not assess pain either.

In contrast, **all eligible PROMs** from the **LR** assess pain. Additionally, the MOS-HIV assesses the way in which pain interferes with 'work', which includes normal or daily living activities. The items from the PROMs of the LR that relate to pain are the following:

CST-HIV

During the past month, have you felt pain somewhere in your body (for example, headache, joint pain, muscle cramps)?

None - Slightly - Somewhat - Quite - Extremely

MOS-HIV

During the past 4 weeks, how much did pain interfere with your normal work (or your normal activities, including work outside the home and housework)?

Not at all - A little bit - Moderately - Quite a bit - Extremely

How much bodily pain have you generally had during the past 4 weeks?

None - Very mild - Mild - Moderate - Severe - Very severe

PROQoL-HIV

During the last two weeks, because I am HIV positive, I have been bothered by pain

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have aches or pains".

Not at all - A little - A moderate amount - Very much - An extreme amount

WHO-QoL HIV BREF

To what extent do you feel that physical pain prevents you from doing what you need to do?

Not at all - A little - A moderate amount - Very much - An extreme amount

Item selection: adopt the **two items** from the **MOS-HIV** to distinguish between the sensation of pain and the extent to which it interferes with day-to-day life. As SPADIS emphasizes the assessment of functional limitations, the interference of pain with *normal activities* is important to include in the SPADIS PRM. To not only focus on 'normal work activities', the item was slightly adapted to assess interference with 'normal activities', while specifying the work perspective by including **examples**.

MOS-HIV:

- How much bodily pain have you generally had during the past 4 weeks?
None - Very mild - Mild - Moderate - Severe - Very severe
- During the past 4 weeks, how much did pain interfere with your normal **work activities** (including work outside the home and housework)?
Not at all - A little - Moderately - Quite a bit - Extremely

3.1.7. Physical functioning

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Physical functioning	x	x	4	x	x	x		x
Bending or lifting			1		x			
d4105 - Bending			1		x			
d430 - Lifting and carrying objects			1		x			x
Mobility			4	x	x	x		x
d450 - Walking			2		x	x		x
d4501 - Walking long distances			2		x	x		
d451 - Going up and down stairs			3		x	x		x
d455 - Moving around			2	x		x		
d4551 - Climbing			3		x	x		x
Vigorous physical activity			3		x	x		x
b280 - Sensation of pain			1					x
d4552 - Running			2		x	x		
Stability			1					x
b240 - Sensations associated with hearing and vestibular functions			1					x
b715 - Stability of joint functions			1					x
d415 - Maintaining body position			1					x

Physical functioning refers to the extent to which PLWH are mobile, can perform physical features such as bending or lifting objects, and can perform moderate or vigorous physical activities (e.g. running long distances).

During the **FGs**, **none** of the participants indicated **limitations in physical functioning** (ability to perform physical activities such as running, walking, sports, etc.). On the contrary, several participants described their HIV diagnosis as a 'wake-up call' to start living healthier. As a result, their abilities in terms of physical activity and functioning increased. As these remarks did not constitute impairments of functions or limitations for activities, they were not analysed as such. The **HRC QoL PROM** does not assess physical functioning.

Out of the five PROMs from the **LR**, **four PROMs** included **items related to physical functioning** or activities. They are, however, quite heterogeneous in terms of which type of activities are assessed: they include items related to mobility (e.g. getting around, walking one block) and items related to more strenuous physical activity (carrying heavy objects, walking long distances, climbing several flights of stairs, etc.). The items that were identified are the following:

MOS-HIV

How much does your health now limit you in bending, lifting or stooping?

Yes, limited a lot - Yes, limited a little - No, not limited

How much does your health now limit you in the kinds or amounts of moderate activities you can do, like moving a table, carrying groceries or bowling

Yes, limited a lot - Yes, limited a little - No, not limited

How much does your health now limit you in the kinds or amounts of vigorous activities you can do, like lifting heavy objects, running or participating in strenuous sports

Yes, limited a lot - Yes, limited a little - No, not limited

How much does your health now limit you in walking one block?

Yes, limited a lot - Yes, limited a little - No, not limited

How much does your health now limit you in walking uphill or climbing (a few flights of stairs)?

Yes, limited a lot - Yes, limited a little - No, not limited

PROQoL-HIV

During the last two weeks, because I am HIV positive I have had difficulty with strenuous physical activities such as carrying heavy objects, running, or walking a long distance, climbing several flights of stairs

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I am unsteady on my feet".

Not at all - Slightly - Moderately - Very - Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble walking or climbing stairs".

Not at all - A little trouble - A moderate amount of trouble - A lot of trouble - I cannot walk or climb stairs

WHO-QoL HIV BREF

How well are you able to get around?

Very poor - Poor - Neither poor nor good - Good - Very good

Item selection: SPADIS researchers decided to adopt the **singular item** from the **WHO-QoL HIV BREF** that focuses on (low-level) **mobility** and not necessarily on moderate or vigorous physical activity (e.g. running short or long distances, lifting heavy objects, etc.). After internal discussions, SPADIS researchers decided to clarify the item by adding descriptions of ICF codes that represent low-level aspects of mobility as examples.

- How well are you able to get around (**walking short distances, climbing a few flights of stairs, bending or lifting**)?
Very poor – Poor – Neither poor nor good – Good – Very good

3.1.8. Quality of life and self-rated (or subjective) health

Topics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Quality of life <i>No ICF code</i>		x	2	x	x			
Self-rated health <i>No ICF code</i>		x	3	x	x	x		

More generic topics such as QoL and self-rated health (SRH) were not addressed during the **FGs** as the questions of the interview guide were designed for the elicitation of more specific functioning- and health-related issues.

The **HRC QoL PROM** includes one question on subjective health and one on quality of life:

- How would you rate your own health over the past month?
Very poor – Poor – Neither poor nor good – Good – Very good
- How would you rate your quality of life?
Very poor – Poor – Neither poor not good – Good – Very good

It is **similar** to the question asked in the WHO-QoL HIV BREF, which is the source of the QoL item of the HRC QoL PROM. The **LR** revealed that 2 out of 5 PROMs specifically assess quality of life and 3 out of 5 have items to assess SRH. They are the following:

For quality of life:

MOS-HIV

How has the quality of your life been during the past 4 weeks? That is, how have things been going for you?

Very well; could hardly be better - Pretty good - Good and bad parts about equal - Pretty bad - Very bad; could hardly be worse

WHO-QoL HIV BREF

How would you rate your quality of life?

Very poor - Poor - Neither poor nor good - Good - Very good

For self-rated health (or subjective health):

MOS-HIV

In general, would you say your health is...

Excellent - Very Good - Good - Fair - Poor

Is the following statement true or false for you: I am as healthy as anybody I know

Definitely true - Mostly true - Not sure - Mostly false - Definitely false

Is the following statement true or false for you: I am somewhat ill

Definitely true - Mostly true - Not sure - Mostly false - Definitely false

Is the following statement true or false for you: I have been feeling bad lately

Definitely true - Mostly true - Not sure - Mostly false - Definitely false

Is the following statement true or false for you: my health is excellent

Definitely true - Mostly true - Not sure - Mostly false - Definitely false

PROQoL-HIV

During the last two weeks, my overall health (both HIV- and non-HIV related) has been

Very good - Good - Fair - Poor - Very poor

WHO-QoL HIV BREF

How satisfied are you with your health?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection: adopt the **singular items** from the **WHO-QoL HIV BREF** on quality of life and self-rated health or 'subjective health'. They align well with the items of the HRC QoL PROM.

- How satisfied are you with your health?
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied
- How would you rate your quality of life?
Very poor - Poor - Neither poor nor good - Good - Very good

3.1.9. Sexuality

For the topic of sexuality, we **distinguish between sexual functioning** (b640- Sexual functions) which includes the capacity of sexual arousal (b6400 – Functions of sexual arousal phase) **and having sexual or intimate relationships** (d770 – Intimate relationships, d7702 – Sexual relationships).

Topics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Sexuality		x	4	x		x	x	x
b640 - Sexual functions			1				x	
b6400 - Functions of sexual arousal phase			2			x	x	
d770 - Intimate relationships			4	x		x	x	x
d7702 - Sexual relationships			4	x		x	x	x

During the **FGs**, impairments in terms of sexual functioning and resulting limitations in having intimate or sexual relationships were not discussed as such. However, the inability to be 'open about HIV' to your (sexual) partner was addressed by several participants as having an impact on the ability to initiate and maintain intimate relationships. These FG results are further discussed under the topic of social relationships and the subtopics of personal relationships and 'being open about disease'.

The **HRC QoL PROM** includes only one item related to sexuality (HRC QoL PROM subdomain of 'sexual satisfaction') that originates from the WHO-QoL HIV BREF.

Four out of five PROMs from the **LR** include items related to sexuality with most items focusing on limitations for having sexual or intimate relationships rather than impairments of sexual functions. The items related to sexuality are the following:

CST-HIV

During the past month, has HIV negatively affected your sex life?

None - Slightly - Somewhat - Quite - Extremely

During the past month, has your sex drive or interest in sex decreased?

None - Slightly - Somewhat - Quite - Extremely

During the past month, how satisfied have you felt with your sex life?

None - Slightly - Somewhat - Quite - Extremely

PROQoL-HIV

During the last two weeks, because I am HIV positive, (unknown) - Love life

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Sexual

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Sexual desire

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I find it hard to start new, intimate, sexual relationships living with HIV".

Not at all or Not applicable - Moderately hard - I can't do it

WHO-QoL HIV BREF

How satisfied are you with your sex life?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection: adopt **singular item** from the **WHO-QoL HIV BREF** (highest methodological quality) on satisfaction with sex life (in alignment with HRC QoL PROM) and **add an item from the CST-HIV** to also assess sexual functions – functions of sexual arousal phase:

WHO-QoL HIV BREF:

- How satisfied are you with your sex life?
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

CST-HIV:

In order to align the format of the CST-HIV item to that of the WHO-QoL HIV BREF, satisfaction with sex drive and interest will be assessed and **the CST-HIV item is adapted** accordingly. In terms of content validity, the subject under assessment remains valid while the phrasing of the item will be further validated within the SPADIS subsequent methodological steps.

- During the past month, has How satisfied are you with your sex drive or interest in sex decreased?**
Very dissatisfied – Dissatisfied – Neither satisfied nor dissatisfied – Satisfied – Very satisfied

3.1.10. Symptoms

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Symptoms		x	4	x		x	x	x
General physical symptoms		x	1	x				
<i>No ICF code</i>								
Digestive problems			3			x	x	x
b515 – Digestive functions			3			x	x	x
b525 – Defecation functions			2			x	x	
b5350 – Sensation of nausea			2			x		x
Dizziness			1					x
b240 - Sensations associated with hearing and vestibular functions			1					x
Fevers, chills or sweat			1					x
<i>No ICF code</i>								x
Headaches								
b280 - Sensation of pain			1					x
Muscle cramps			1					x
b735 - Muscle tone functions			1					x
Numbness			1					x
b265 - Touch function			1					x
Shortness of breath			1					x
b440 - Respiration functions			1					x
b460 - Sensations associated with cardiovascular and respiratory functions			1					x

During the **FGs**, none of the symptoms identified in the thematic inductive analysis of the PROM items were put forth by the participants (important to note is that skin problems, fatigue or energy, cognitive impairments, etc. are not discussed under this topic of 'symptoms' but were previously discussed separately).

The **HRC QoL PROM** adopted the singular WHO-QoL HIV BREF item on the extent to which an individual is bothered by 'physical problems related to HIV'.

The **LR** shows that while four out of five eligible PROMs assess symptoms, **only digestive symptoms** are included in **more than one PROM**. The items on symptoms that were identified in the PROMs from the LR are the following:

WHO-QoL HIV BREF

How much are you bothered by any physical problems related to your HIV infection?

Not at all - A little - A moderate amount - Very much - An extreme amount

PROQoL-HIV

During the last two weeks, because I am HIV positive I have been bothered by digestive problems (stomach ache, bloating, diarrhea, nausea or vomiting)

Never - Rarely - Sometimes - Often - Always

CST-HIV

During the past month have you suffered digestive problems? (stomach pain, flatulence, diarrhea, nausea, or vomiting)

Item selection: adopt **singular items** from the **WHO-QoL HIV BREF** on 'physical problems related to HIV' and **singular item** of **PROQoL-HIV** on digestive problems. Compared to the CST-HIV item, that also describes several digestive symptoms as examples, the PROQoL-HIV received a higher methodological rating.

WHO-QoL HIV BREF:

- How much are you bothered by any physical problems related to your HIV infection?
Not at all - A little - A moderate amount - Very much - An extreme amount

PROQoL-HIV:

- During the last two weeks, because I am HIV positive I have been bothered by digestive problems (stomach ache, bloating, diarrhoea, nausea or vomiting) – *These specific digestive symptoms require further expert validation*
Never – Rarely – Sometimes – Often - Always

None - Slightly - Somewhat - Quite - Extremely

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel dizzy".

Not at all - A little - Moderately - Very much - Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel nauseous".

Not at all - Slightly - Moderately - Very or Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have fever, chills or sweats".

Not at all - A little - A moderate amount - Very much - An extreme amount

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have headaches".

Not at all - A little - A moderate amount - Very much - An extreme amount

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have muscle cramps".

Not at all - A little - A moderate amount - Very much - An extreme amount

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have numbness or tingling in my hands or feet".

Not at all - A little - A moderate amount - Very much - An extreme amount

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have shortness of breath".

Not at all - A little - A moderate amount - Very much - An extreme amount

3.2. PSYCHOSOCIAL HEALTH AND SUPPORT

This domain of the **HRC QoL PROM** describes aspects related to social well-being and relationships (including support from friends and family), to the extent to which an individual with HIV is able to accept his/her situation, to stigma and discrimination, and to other types of support (such as financial support). Impersonal types of support (e.g. healthcare, financial, social security) are not discussed here but under 1.4.3. Environmental factors.

The topics from the thematic inductive analysis of the focus groups and literature review results that align most with this domain of the HRC QoL PROM are described in Table 2 of this report. Aspects within this domain mostly align with ICF codes related to the **ICF components of activities and environmental factors**, although several aspects relate to the ICF component of Body functions and Structures.

3.2.1. Self-acceptance, self-esteem, meaningfulness, and sense of freedom

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Self-acceptance	x	x	0					
<i>No ICF code</i>	x	x	0					
Self-esteem			3	x		x		x
b152 – emotional functions			1	x				
Body image			2	x		x		
b1801 – Body image			2	x		x		
Confidence			1					x
b126 – Temperament and personality functions			1					x
Meaningfulness	x		1	x				
b152 - Emotional functions			1	x				
b1800 - Experience of self			1	x				
d930 - Religion and spirituality	x		1	x				
Being useful	x		0				x	
d660 – Assisting others	x		0				x	
Sense of freedom	x		0					
<i>No ICF code</i>	x		0					

During the **FGs**, the difficulty of **accepting** the diagnosis was emphasized, as well as the related trauma were highlighted as having a large impact on sexual activity and satisfaction, social relationships, and health or well-being in general. The participants also stressed the need to have a '**meaningful** life', albeit by supporting others through, for example, volunteering, (d660 – Assisting others), or by *finding meaning* in religion, mass and spirituality (d930 – Religion and spirituality). In addition to the topics of meaningfulness and self-acceptance, a '**sense of freedom** to be able to do what

you want' and feel no restrictions because of an HIV diagnosis was put forth as an important aspect of psychosocial well-being.

The **HRC QoL PROM** includes one item related to the ability to accept the HIV diagnosis. No items related to meaningfulness, self-esteem or a sense of freedom are included.

The **LR** shows that none of the eligible PROMs include items on the topics of self-acceptance or sense of freedom. Three out of five PROMs however, include an item related to self-esteem, of which **two relate to body image** and **one relates to confidence** in general. Meaningfulness is assessed by only one PROM; the WHO-QoL HIV BREF. **One item** was identified in the LR (SF-HDQ) that relates to the ability to **support others** (being useful > d660 – Assisting others).

PROQoL-HIV

During the last two weeks, because I am HIV positive, (unknown) - Physical appearance

Never - Rarely - Sometimes - Often – Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I find it hard to meet the needs of those I care for (such as children, parents, grandparents, partners, or pets)".

Not at all or Not applicable - A little hard - Moderately hard - Really hard - I can't do it

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I lack confidence around others".

Not at all - Slightly - Moderately - Very – Extremely

WHO-QoL HIV BREF

Are you able to accept your bodily appearance?

Not at all - A little - Moderately - Mostly – Completely

How satisfied are you with yourself?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

To what extent do you feel your life to be meaningful?

Not at all - A little - A moderate amount - Very much - An extreme amount

Item selection: Because of the emphasis on requiring life to be or feel meaningful during the FG for PLWH, we will adopt the one item of the WHO-QoL HIV BREF on meaningfulness. To further assess if PLWH can still support others, we adopt the item from the SF-HDQ on the ability to meet the needs of those they care for.

WHO-QoL HIV BREF:

- To what extent do you feel your life to be meaningful? (**Meaningfulness**)
Not at all - A little - A moderate amount - Very much - An extreme amount

SF-HDQ:

To align the format of the SF-HDQ item to the other selected items of the SPADIS PRM, the **format is slightly adapted**. After internal discussions, SPADIS researchers felt this item is **more closely associated with the topic of 'Activities of daily living'**. In this report's item overviews that follow, it will be presented under this topic.

- Check the box that describes how you are feeling today:** "Do you find it hard to meet the needs of those you care for (such as children, parents, grandparents, partners, or pets)" (**Meaningfulness > Being useful**).
Not at all or Not applicable - A little hard - Moderately hard - Really hard - I can't do it

3.2.2. Fears, worries, stigma and discrimination

The ICF codes for the subtopic of stigma in the table overview below mainly **distinguish between the different types of individuals or systems in relation to whom stigma is experienced** (e.g. family, friends, acquaintances, peers, colleagues, strangers, health care professionals or system, social security, society in general). Yet, it the ICF classification importantly offers no detailed codes that capture nuances on how stigma impacts individuals and how it is experienced. This marks an important limitations of the ICF framework, particularly in the context of HIV.

During the **FGs**, one of the **most discussed** topics was that of **stigma** and **discrimination**, which was seen as having a very significant impact on PLWH, their health and well-being, their ability to participate fully in our society, and more specifically their ability to maintain or begin (in)formal relationships with close ones (friends and family), acquaintances, colleagues, strangers, etc.

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Fears, worries, stigma	x	x	5	x	x	x	x	x
b152 - Emotional functions			2		x	x		
d240 - Handling stress and other psychological demands			1		x			
Fear of death			1	x				
b152 - Emotional functions			1	x				
d240 - Handling stress and other psychological demands			1	x				
Fear of future	x		4	x		x	x	x
b152 - Emotional functions	x		4	x		x	x	x
d240 - Handling stress and other psychological demands			1	x				
Stigma	x	x	3	x		x	x	
b152 - Emotional functions			3	x		x	x	
d760 - Family relationships	x		0					
e310 - Immediate family	x		0					
e325 - Acquaintances, peers, colleagues	x		0					
e355 - Health professionals	x		0					
e410 - Individual attitudes of immediate family members	x	x	3	x		x	x	
e415 - Individual attitudes of extended family members	x	x	3	x		x	x	
e420 - Individual attitudes of friends	x	x	3	x		x	x	
e425 - Individual attitudes of acquaintances, peers, colleagues, neighbours and community members	x	x	3	x		x	x	
e445 - Individual attitudes of strangers			2			x	x	
e450 - Individual attitudes of health professionals	x	x	0					
e460 - Societal attitudes	x	x	2			x	x	
e585 - Education and training services, systems and policies	x		0					
e850 - Health services, systems and policies	x	x	0					
Stress	x		0					
b152 - Emotional functions	x		0					
d240 - Handling stress and other psychological demands								

Stigma and discrimination is experienced by the participants of the FGs at multiple levels in our society: health professionals and health care services (e.g. refusal of treatment by dentist), financial system and services (e.g. insurance), by strangers and co-workers who 'keep their distance' and 'mock', or by (close) family members and friends who refuse to maintain contact after learning about the diagnosis. According to the participants, the main reason for these types of discrimination is related to uninformed and negative attitudes (societal, family, friends, strangers, health professionals), misinformation, and a general lack of information. Living in a society where these types of stigma and discriminatory practices remain, increases the burden of disease for PLWH in Belgium, especially on a psychosocial level (stress, b152 – Emotional functions). Experiences of stigma and discriminations also affect PLWH in the sense that they are hesitant to 'open up about their diagnosis' to their environment. Participants deemed this essential in order to have good relationships

with close ones. Although this is not necessarily true for colleagues and acquaintances in the work environment. The ability to be open about HIV is further discussed under 1.4.2.5. social relationships.

During the **FGs**, other fears and worries with which PLWH are confronted relate to a **fear of death** and a **fear of the future** (due to uncertainty about HIV and its progression). Living with such fears, worries and uncertainties further impacts the health and (psychosocial) well-being of PLWH in Belgium (b152 – Emotional functions).

The **HRC QoL PROM** includes one item on 'stigma', one on 'discrimination', and one on 'self-stigma', although their exact origin is unclear:

- Have you ever experienced HIV **stigma** (e.g. stereotypes, negative attitudes)?
Never – Seldom – Quite often – Very often – Always
- Were you ever **treated differently** because of your HIV status (e.g. in health-care settings, work, friends or family, ...)?
Never – Seldom – Quite often – Very often – Always
- Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status (starting an intimate relationship, applying for a job, social activities, ...)? (**self-stigma**)
Never – Seldom – Quite often – Very often – Always

The PROMs that were identified in the **LR** all include items that can be associated with the topic of 'fears, worries and stigma'. Only the WHO-QoL HIV BREF includes a specific item related to 'fear of death'. Four out of five PROMs (excl. MOS-HIV) describe items related to 'fear of future'. Stigma (or 'self-stigma' or 'anticipated stigma') is assessed by three out of five PROMs from the LR. The items identified in the LR for the topic of 'fears, worries and stigma' are the following:

CST-HIV

During the past month, how often have you felt fearful for the future?

Never - Rarely - Sometimes - Frequently - Always

During the past month, to what extent have you been worried about people judging you if they learn you have HIV?

None - Slightly - Somewhat - Quite - Extremely

During the past month, to what extent have you been worried about people rejecting you for having HIV?

None - Slightly - Somewhat - Quite - Extremely

During the past month, to what extent have you been worried about telling someone you have HIV?

None - Slightly - Somewhat - Quite - Extremely

MOS-HIV

How often during the past four weeks were you afraid because of your health

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

PROQoL-HIV

During the last two weeks, because I am HIV positive, (unknown) - Catching infections

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Disease worsening

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - HIV disclosure

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Infecting others

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Result-related stress

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I am afraid for my future".

Not at all - Slightly - Moderately - Very - Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I worry about my future health living with HIV".

Not at all - A little - Somewhat - Very much - A lot

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I worry about my income or financial security living with HIV".

Not at all - A little - Somewhat - Very much - A lot

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I worry about my lab test results such as my CD4 count and viral load".

Not at all - A little - Somewhat - Very much - A lot

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I worry about the side effects of HIV treatments".

Not at all - A little - Somewhat - Very much - A lot

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I worry what might happen to my family and friends if I get sick".

Not at all - A little - Somewhat - Very much - A lot

WHO-QoL HIV BREF

How much do you fear the future?

Not at all - A little - A moderate amount - Very much - An extreme amount

How much do you worry about death?

Not at all - A little - A moderate amount - Very much - An extreme amount

To what extent are you bothered by people blaming you for your HIV status

Not at all - A little - A moderate amount - Very much - An extreme amount

Item selection: adopt the **three items** from the **WHO-QoL HIV BREF**, which include two questions on fears (death and future) and one on the extent to which PLWH are bothered by being blamed for their HIV status. We added a response options for those who do not generally disclose their HIV status. In addition, inspired by the items designed by the HRC QoL PROM, we designed one matrix question to assess matters from which PLWH might have withheld themselves for fear of reactions of others (self-stigma), and one matrix question that asks PLWH which individuals have shown negative attitudes or have treated PLWH differently because to their diagnosis (incl. response option to indicate that PLWH do not disclose their HIV status). The matters and individuals addressed in these two matrix questions are inspired by the comments given during the FGs.

WHO-QoL HIV BREF:

- How much do you fear the future? (*fear of future*)
Not at all - A little - A moderate amount - Very much - An extreme amount
- How much do you worry about death? (*fear of death*)
Not at all - A little - A moderate amount - Very much - An extreme amount
- To what extent are you bothered by people blaming you for your HIV status? (*stigma*)
Not at all - A little - A moderate amount - Very much - An extreme amount – I generally do not disclose my HIV status

HRC QoL PROM (Novel question matrices inspired by content and phrasing of HRC QoL PROM items):

- **[Matrix question]** Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following situations: (a) starting an intimate relationship, (b) applying for a job, (c) participating in social activities? (*self-stigma*)
Never – Seldom – Quite often – Very often – Always
- **[Matrix question]** Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following individuals? (a) friends, (b) partner, (c) colleagues or employers, (d) health professional in the HIV reference centre, (e) health professionals in a hospital that is not a HIV reference centre, (f) health professionals outside the hospital? (*experienced stigma*)
Never – Seldom – Quite often – Very often – Always – I do not disclose my HIV status to this/these individuals

In the item overviews that follow, we will distinguish between fears and worries on the one hand, and stigma (experienced and anticipated) on the other hand.

3.2.3. Social activities

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Social activities	x		3	x	x			x
d910 - Community life	x		1		x			
d920 - Recreation and leisure			1		x			
d9205 - Socializing	x		1		x			
Leisure activities			2	x				x
No ICF code			1	x				
d920 - Recreation and leisure			2	x				x
e165 - Assets			1					x

During the **FGs**, social activities were mainly discussed within the context of engagement in the community or society in general and the interactions with others that take place in the community or society (d910 – Community life, d9205 - Socializing). Taking part in support groups was mentioned as a specific social activity with a positive effect on the well-being of PLWH. This is closely linked to the ‘ability to be open about the HIV diagnosis’ to other PLWH and others in society.

The **HRC QoL PROM** contains **no item** specific to social activities, neither in general, nor for specific activities.

The **LR** on eligible PROMs shows that **3 out of 5 PROMs** include items related to social activities. The item from the MOS-HIV however, focuses on friends and family which are further discussed [under 1.4.2.5. social relationships](#). The first item of the SF-HDQ focuses on the financial situation of the individual and how this impacts their ability to participate in leisure activities.

MOS-HIV

How much of the time, during the past 4 weeks, has your health limited your social activities (like visiting with friends or close relatives)?

All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "Because of my financial situation, I find it hard to take part in leisure or recreational activities such as going to the movies, out to dinner, or on vacation".

Not hard at all - A little hard - Moderately hard - Really hard - I can't do it

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I have trouble taking part in leisure or recreational activities, such as exercise or dancing".

Not at all - A little trouble - A moderate amount of trouble - A lot of trouble - I can't do this

WHO-QoL HIV BREF

To what extent do you have the opportunity for leisure activities?

Not at all - A little - Moderately - Mostly - Completely

Item selection: the **singular item** from the **WHO-QoL HIV BREF** on the ‘opportunity for leisure activities’ is not sufficiently specific and unclear as to what constitutes ‘opportunities’. Instead, we choose the MOS-HIV item on social limitations due to health, and we specify leisure opportunities in the included examples.

MOS-HIV:

- How much of the time, during the past 4 weeks, has your health limited your social activities (like visiting with friends **or participating in leisure activities**)?
All of the time – Most of the time – A good bit of the time – Some of the time – A little of the time – None of the time

Within the theoretical framework of SPADIS (ICF model), social activities are closely related to participation of work activities [both are aspects of (social) participation]. Items on work are discussed below (3.2.4). **More importantly, the majority of question on social activities are included in the generic module of the SPADIS PRM and will be assessed in all SPADIS populations.**

3.2.4. Education and work

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Work (capacity)	x		3	x	x			x
d640 - Doing housework			1		x			
d845 - Acquiring, keeping and terminating a job			1		x			
d850 - Remunerative employment	x	x	3	x	x			x
d855 - Non-remunerative employment	x		3	x	x			x
e425 - Individual attitudes of acquaintances, peers, colleagues, neighbours and community members	x	x	0					
Education	x		1		x			
d820 - School education	x		1		x			
d830 - Higher education	x		1		x			

When discussing work and education, some **FG** participants explained how **educational activities** were put on hold for some time when they received their diagnosis. Noticeably, several of the 'older' FG participants of the first FG were not actively employed at the time of the FG. When discussing **work and the capacity to work**, the main barrier or issue that was put forth related to the attitudes of colleagues and others in their work environment. The participants stressed the need to 'feel tolerated' and be 'treated normally'. The sense of 'being useful' by doing volunteer work (d855 - Non-remunerative employment) was also reiterated when discussing work.

The **HRC QoL PROM** contains two items in which **work** is given as an example of a context where **stigma and discrimination** may be experienced, they are discussed above. No items that focus solely on the work environment are included, although it should be noted that attitudes and tolerance of colleagues is also an important factor addressed by the participants of the **FGs**. In that sense, the inclusion of items on stigma and discrimination at work is valid for SPADIS purposes as well. Such items are described above. The **HRC QoL PROM** has **no items** related to **education**.

Three out of five PROMs identified in the LR, include items related to **work**. Only **one PROM** (MOS-HIV) assesses **education** (in a limited way). The items from the LR on work and education are the following:

MOS-HIV

Does your health keep you from working at a job, doing work around the house or going to school?

Yes - No

Have you been unable to do certain kinds or amounts of work, housework, or schoolwork because of your health?

Yes - No

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel that my work or volunteer performance is limited because of my illness".

Not at all or Not applicable - Slightly - Moderately - Very much - Extremely

WHO-QoL HIV BREF

How satisfied are you with your capacity for work?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection: Importantly, items on work participation and educational participation are broadly assessed in the SPADIS PRM generic module. The items from the MOS-HIV and the SF-HDQ directly assess whether their health has limited their capacity to perform work (incl. work at home or at school). As the MOS-HIV is of higher methodological quality, we **adopt the first MOS-HIV item**. The second MOS-HIV item is discarded due to high thematic overlap with the first one. **Its response options were also adapted** from a binary response option to a 5 point Likert scale to align the item with the other items of the SPADIS PRM.

MOS-HIV:

- Does your health keep you from working at a job, doing work around the house or going to school?
Not at all – A little – Moderately – Mostly – Completely

For the SPADIS PRM, more descriptive questions on labour market positions (before and after diagnosis), work participation, job or work environment characteristics, etc. will be assessed in the generic module of the SPADIS PRM.

3.2.5. Social relationships

The following is a restricted table that only describes the subtopics and not the ICF codes that were linked. The full table can be found in [Annex 4](#). **Importantly, the ICF codes that were linked to the subtopics of the table below mostly illustrate to which individuals the subtopics apply: friends, family, colleagues, acquaintances, strangers, etc.**

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Social relationships	x	x	4	x		x	x	x
Acceptance by others	x							
Asking for help			1					x
Being open about disease	x		1				x	
Intimate relationships	x	x	4	x		x	x	x
Loneliness	x		1					x
Personal relationships (family, friends)	x	x	2	x		x		
Support	x	x	1				x	x
Supporting others	x							

The topic of **social relationships** was by far the **most discussed topic** of the **FGs**. This should be reflected as such in the SPADIS PRM for PLWH. Participants talked about informal relationships with friends, family members, (sexual) partners and their importance for psychosocial well-being and in avoiding loneliness. They also discussed how stigma and discrimination is experienced in these relationships, often due to misinformation and unjustified negative attitudes. They also addressed the need to be able to disclose the HIV situation without judgement in these relationships and how this is not easy. **Three aspects of social relationships seem important: (1) having informal social relationships, (2) being able to disclose their HIV diagnosis and feel accepted within these relationships (attitudes), and (3) receiving support within these relationships.**

The **HRC QoL PROM** assesses the last two of these three aspects by assessing stigma and discrimination and providing relationships with friends and family as examples (see above), and by assessing the satisfaction with support from their family and friends.

Four out of five PROMs from the **LR** assess aspects which can be related to social relationships. More specifically, four out of five PROMs assess **intimate relationships**. These are questions which are already described above under sexuality. The items related to social relationships (excluding intimate relationships due to overlap) are the following:

CST-HIV

During the past month, how often have people made you feel loved?

Never - Rarely - Sometimes - Frequently - Always

During the past month, how often have you had people around you whom you can lean on in case of need?

Never - Rarely - Sometimes - Frequently - Always

During the past month, how often have you had someone you trust to speak to about your problems?

Never - Rarely - Sometimes - Frequently - Always

PROQoL-HIV

During the last two weeks, because I am HIV positive I have felt restricted in my relationships with my family or friends

Never - Rarely - Sometimes - Often - Always

During the last two weeks, because I am HIV positive, (unknown) - Social life

Never - Rarely - Sometimes - Often - Always

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I feel lonely".

Not at all - Slightly - Moderately - Very - Extremely

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I find it hard to ask others for help when I go through an episode of illness".

Not hard at all - A little hard - Moderately hard - Really hard - I can't do it

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I find it hard to meet the needs of those I care for (such as children, parents, grandparents, partners, or pets)".

Not at all or Not applicable - A little hard - Moderately hard - Really hard - I can't do it

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I tend to isolate myself from others because I am HIV positive".

Not at all - A little - Somewhat - Very much - Extremely

WHO-QoL HIV BREF

How satisfied are you with the support you get from your friends?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

How satisfied are you with your personal relationships?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection: adopt the **two items** from the **WHO-QoL HIV BREF** and add two **CST-HIV items**. Items were **slightly altered (examples were added)** to cover the ICF codes for the types of relationships which were most often assessed in the PROMs identified in the LR (friends and family). One of the CST-HIV items was originally coded under the topic of fears, worries, stigma, but has been reused here due to its assessment of worrying to disclose an HIV status.

WHO-QoL HIV BREF:

- How satisfied are you with the support you get from your friends **and family**? (cfr. HRC-QoL PROM)
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied
- How satisfied are you with your personal relationships (**friends and family**)?
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

CST-HIV (item added that was classified under stigma):

- During the past month, how often have you had someone you trust to speak to about your problems?
Never - Rarely - Sometimes - Frequently - Always
- During the past month, to what extent have you been worried about telling someone you have HIV? (**added to assess 'being open about disease'**).
None - Slightly - Somewhat - Quite - Extremely

3.2.6. Treatment-burden

Treatment-burden cannot be coded within the ICF classification system, or at least in a way that captures patient experiences and narratives. There is the ICF environmental factors of 'drugs' (e1101 – Drugs) but this does not offer the required level of detail.

During the **FGs**, participants addressed the burden of managing their disease, and the stress it causes. Also, going from oral medication to injections felt like a relief for several participants: daily intakes serves as daily reminders of having HIV. The stress of having to take your medication each day and sometimes doubting whether or not you took it further increases the treatment burden felt by the participants. Side effects of treatment were also discussed during the FGs, but the extent to which these are really linked to the medication is unclear. Given the psychological nature of the treatment-burden, this topic is described within this domain of psychological health and support.

The **HRC QoL PROM** includes one item on treatment-**adherence** that asks how difficult the individual finds it to take their medication properly.

The **LR** shows that only 2 PROMs provide questions on treatment adherence and burden. A singular item from the WHO-QoL HIV BREF pertains to how much medication is needed to function in daily life. Yet, after internal discussions, the SPADIS researchers felt that this might be difficult for PLWH to objectively quantify and would be too dependent on assumptions. The **PRO-QoL HIV provides 10 individual items related to treatment burden**: daily intake, impact of treatment, satisfaction, side effects, etc. However, these remain quite general in their phrasing and seem too numerous.

Item selection: adopt the **singular item** by the **HRC QoL PROM** that assess treatment adherence:

- How often do you find it difficult to take your HIV medication properly?
Never – Seldom – quite often – Very often - Always

3.3. ENVIRONMENTAL FACTORS

This domain is not included in the HRC QoL PROM but is essential for SPADIS purposes. Environmental factors include, among others, the attitudes of (and relationships with) **family, friends, acquaintances, colleagues, strangers, etc.** These aspects are, however, **discussed under** Social relationships and Fears, worries, stigma and discrimination.

Environmental factors that will be discussed here, relate to **accessibility of information, financial resources and system, health care professionals and system, the home environment, social security system, and transport.** Surprisingly, only three out of five PROMs have items that address one or more environmental factors. As described under education and work, environmental factors related to the work environment (excl. HIV-specific attitudes of colleagues) will be assessed in the generic SPADIS PRM module.

Most of the aspects discussed within the domain of environmental factors can be linked to the ICF component of Environmental factors.

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	CST-HIV	SF-HDQ
Accessibility of information	x		1	x		
<i>No ICF code</i>	x		1	x		
Financial resources		x	2	x	x	
d920 - Recreation and leisure			1		x	
e165 - Assets		x	2	x	x	
e1650 - Financial assets			1		x	
Financial system	x		0			
e565 - Economic services, systems and policies	x		0			
Health care service	x	x	1	x		
Accessibility	x		1	x		
e355 - Health professionals	x		0			
e850 - Health services, systems and policies	x		1	x		
Knowledge of health professionals	x		0			
e355 - Health professionals	x		0			
Multidisciplinarity	x		0			
e850 - Health services, systems and policies	x		0			
Relationship with health professionals	x	x	0			
e355 - Health professionals	x	x	0			
e450 - Individual attitudes of health professionals	x		0			
e850 - Health services, systems and policies	x	x	0			
Home environment	x		3	x	x	x
d610 - Acquiring a place to live			1			x
d650 - Caring for household objects			2	x		x
e525 - Housing services, systems and policies	x		1		x	
Physical environment			1	x		
<i>No ICF code</i>			1	x		
Physical safety			1	x		
<i>No ICF code</i>			1	x		
Social security (health insurance)			0			
e570 - Social security services, systems and policies	x		0			
Transport			1	x		
d470 - Using transportation			1	x		
e540 - Transportation services, systems and policies			1	x		

During the **FGs**, the participants frequently addressed the **lack** of easily **accessible information** on HIV and its treatment: a lot of confusion still exists among PLWH, and the uncertainty this causes has an impact on their psychosocial well-being. Also at a societal level, misconceptions and misinformation surrounding HIV are very apparent and lead to further stigmatization and discrimination.

For the **financial system**, a concrete example was given in relation to the difficulties that PLWH are confronted with when trying to get a **mortgage credit** (e565 - Economic services, systems and policies). Access to **absenteeism insurance**

was obstructed by similar difficulties (e570 - Social security services, systems and policies). Although not necessarily linked to HIV, participants also mentioned that an unstable **home or living arrangement** through **social programs** leads to further psychosocial distress and a negative impact on well-being (e525 - Housing services, systems and policies).

Most comments of the FGs related to the **health care system, services** (e850 - Health services, systems and policies) and **professionals** (e355 - Health professionals, e450 - Individual attitudes of health professionals). Participants reported difficult relationships with health professionals in primary care such as dentists and GPs and associated this with a lack of knowledge on HIV and its treatment. PLWH describe both good and bad examples of how health professionals provide care and support, distinguishing between those who give plenty of **clear information** and those who do not provide clear explanations on current and future treatments or health changes. The **multidisciplinarity of care** and support that they receive was quite heterogenous for the participants of the FGs, with some reporting having access to all types of professionals. Others described how nobody explained the availability of, for example, psychologists in hospitals that offer support to PLWH. The experience and satisfaction of their **relationship with their health professionals was seen as having an important impact on functioning and health and functioning**.

The **few HRC QoL PROM items** related to these environmental factors focus on discrimination which includes the health care setting in its examples, and on financial resources ("*Have you enough money to meet your needs?*" originating from the **WHO-QoL HIV BREF**).

The three PROMs from the LR that assess environmental factors other than those described under [1.4.2.2](#) and [1.4.2.5](#) include the following items:

CST-HIV

During the past month have you had enough money to meet your needs?

None - Slightly - Somewhat - Quite - Extremely

During the past month, have you had money for leisure activities?

None - Slightly - Somewhat - Quite - Extremely

During the past month, how satisfied have you been with the quality of the place where you live?

None - Slightly - Somewhat - Quite - Extremely

SF-HDQ

"Check the box that describes how you are feeling today" (Response 1) and "has this challenge fluctuated over the past week?" (Response 2) for "I struggle to maintain safe and stable housing living with HIV".

Not at all - Moderately - Extremely

WHO-QoL HIV BREF

Have you enough money to meet your needs?

Not at all - A little - Moderately - Mostly - Completely

How available to you is the information that you need in your day-to-day life?

Not at all - A little - Moderately - Mostly - Completely

How healthy is your physical environment?

Not at all - A little - A moderate amount - Very much - Extremely

How safe do you feel in your daily life?

Not at all - A little - A moderate amount - Very much - Extremely

How satisfied are you with the conditions of your living place?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

How satisfied are you with your access to health services?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

How satisfied are you with your transport?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection (part 1): adopt the **items from the WHO-QoL HIV BREF** related to financial resources, accessibility of information, home environment or living place conditions and accessibility of health care services. The item on accessibility of information originally assesses 'availability'. We rephrased this item to assess more directly assess 'accessibility'. For this last item, we restructured the format to assess satisfaction in regard to the accessibility of specific health care services or professionals by designing a **matrix question**. We also added a response option so PLWH can specify that they do not require access.

- Have you enough money to meet your needs? (**financial resources**)
Not at all - A little - Moderately - Mostly - Completely
- How available to you is the information that you need in your day-to-day life? (**accessibility of information**)
Not at all - A little - Moderately - Mostly - Completely
- How satisfied are you with the conditions of your living place? (**home environment**)
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied
- **[Matrix question]** How satisfied are you with your access to the following healthcare services or professionals? (a) Infectiologist, (b) Psychologist, (c) social support / worker, (d) Sexologist, (e) General practitioner, (f) Dentist, (g) (only for women) Gynecologist, (h) HIV nurse (**accessibility of health care services**)
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied - I do not require access to this service or professional

Due to its significance for PLWH that participated in the FGs, **experiences with healthcare professionals** and the **accessibility of health care services**, and more specifically **multidisciplinary care**, are to be included in the SPADIS PRM. The same reasoning applies for **accessibility of information**. The fact that three out of five PROMs from the LR assess **home environment** warrants the inclusion of such an item in the SPADIS PRM as well. **Financial resources** were not addressed during the FGs but is covered by two out of five PROMs and the HRC QoL PROM. Limited financial resources can also be seen as an important proxy of socioeconomic status for PLWH. To investigate its association with other aspects of functioning or HRQoL, the SPADIS PRM will include an item on financial resources. In regard to **accessibility of health care services**, the available **item of the WHO-QoL HIV BREF** is quite general and **does not specify types of services of health professionals**.

The PROMs from the LR offer **no items** on **satisfaction with multidisciplinary care** and **relationships with health care professionals**. One reason for this might be that such items are considered patient-reported experience measures (**PREMs**) rather than outcome measures (PROMs). The current scientific literature offers no suitable HIV-specific PREMs that can be used in this regard. To address this gap, the SPADIS researchers consulted **generic PREMs** for potential items on satisfaction of multidisciplinary care (incl. flexibility of care and involvement in decision-making), and relationships with health professionals (incl. caring attitude and communication style).

As these aspects relate to well-coordinated and patient-centred care for chronically-ill individuals, the P3C experience measure (P3CEQ) can be used. **The P3CEQ is an 11 item measure designed to assess patient experiences of person centred coordinated care**. It has been well-validated in people living with chronic conditions, although not specifically for PLWH. The tool is available in English, German, Norwegian, Dutch, Estonian, Spanish, and Catalan. Several items of this tool refer to the use of personalized care plans and are not applicable in the current context although it can be argued that PLWH receive such a plan even though it is not specified as such. Furthermore, an adapted version of this questionnaire is included in the OECD's PaRIS Patient Questionnaire (**PaRIS-PQ**), which has been widely disseminated (incl. in Belgium) and would offer meaningful comparisons for the results of the SPADIS cross-sectional analyses. The P3CEQ (modified) items from the PaRIS questionnaire are the following:

P3CEQ1	Do you discuss with the health care professionals involved in your care what is most important for you in managing your own health and wellbeing? <i>Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure</i>
P3CEQ2	Are you involved as much as you want to be in decisions about your care? <i>Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure</i>
P3CEQ3	Are you considered as a 'whole person' rather than just a disease/condition in relation to your care? <i>Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure</i>
P3CEQ4	Were there times when you had to repeat information that should have been in your care records? <i>Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure</i>
P3CEQ5	Is your health care organised in a way that works for you? <i>Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure</i>
P3CEQ6	Do you have a single professional who takes responsibility for coordinating your care across the services that you use?

	<i>Yes – No – I do not receive care from more than one service</i>
P3CEQ7a	Do you have a care plan that takes into account all your health and wellbeing needs?
	<i>Yes – No – Not sure</i>
P3CEQ7b (if 7a = Y)	Is this care plan available to you?
	<i>Yes – No – Not sure</i>
P3CEQ7c (if 7a = Y)	To what extent have you found your care plan useful for you to manage your health and wellbeing?
	<i>Not at all – To some extent – To a large extent – Completely – Not sure</i>
P3CEQ7d (if 7a = Y)	To what extent do all the professionals involved in your care appear to be following the same care plan?
	<i>Not at all – To some extent – To a large extent – Completely – Not sure</i>
P3CEQ8	Do you have enough support from your health care professionals to help you manage your own health and wellbeing?
	<i>I do not need support – I have had no support – I sometimes have enough support – I often have enough support – I always have enough support – Not relevant – Not sure</i>
P3CEQ9	To what extent do you receive useful information at the time you need it to help you manage your own health and wellbeing?
	<i>I do not receive any information – I sometimes receive enough information – I often receive enough information – I receive too much information – Not relevant – Not sure</i>
P3CEQ10	How confident are you that you can manage your own health and well-being?
	<i>Very confident – Confident – Somewhat confident – Not confident at all</i>
P3CEQ11b	Do your health care professionals involve your family, friends and/or carers as much as you want them to be in decisions about your care?
	<i>No, definitely not – No, not really – Yes, to some extent – Yes, definitely – I do not want my family, friends or carers to be involved – Not applicable</i>

Several items of the P3CEQ measure could be used to assess important patient experiences and satisfaction in regard to the care that PLWH receive. The SPADIS PRM measure could adopt several items from the P3CEQ that measure patient experiences and satisfaction in regard to shared-decision making or involvement in care decisions, and multidisciplinary of care. However, to limit the total number of items, SPADIS researchers - after internal discussion - decided to adopt one item on shared decision making (**P3CEQ2**) and one item on multidisciplinary care (**P3CEQ8**). For this last item, SPADIS researcher **reformatted** the item into a **matrix question** to provide more detail on specific health professionals.

The **P3CEQ** measure does not include questions that enable the assessment of the PLWH experienced relationships with health professionals, which is deemed important by the participants of the FGs. Moreover, no such item was identified during the literature review phase of the SPADIS methodology. Therefore, we constructed a novel item to assess satisfaction of PLWH for the relationships with health professionals involved in their care:

How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?

Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied

Item selection (2): several items from the **P3CEQ** measure can be adopted and/or slightly altered to fit the needs of the SPADIS PRM in terms of environmental factors. One novel item was created by the SPADIS research team to assess PLWH satisfaction with the relationships with health care professionals.

P3CEQ:

- Are you involved as much as you want to be in decisions about your care?
Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure
- [Matrix question]** Do you have enough support from the following health professionals to help you manage your own health and wellbeing? (a) Infectiologist, (b) Psychologist, (c) social support / worker, (d) Sexologist, (e) General practioner, (f) Dentist, (g) (only for women) Gynecologist, (h) HIV nurse
I do not need support – I have had no support – I sometimes have enough support – I often have enough support – I always have enough support – Not relevant – Not sure

Novel item:

- How satisfied are you with your relationships with the following health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?
Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied
+ **Open text answer option to explain response**

III. Developing the initial HIV-specific PRM module

1. Structuring the HIV-specific PRM module

The **cross-matching exercise described in the sections above resulted in the selection and/or development of 61 items** for the HIV-specific module of the SPADIS PRM that assess the following subtopics: **activities of daily living** (2 items), **cognitions** (3 items), **emotional well-being** (matrix question with 4 items), **energy and sleep** (2 items), **pain** (2 items), **physical functioning** (1 item), **quality of life and self-rated health** (2 items), **sexuality** (2 items), **symptoms** (2 items), **meaningfulness** (1 item), **fears and worries** (2 items), **stigma and discrimination** (1 item and two matrix questions with respectively 3 and 6 items), **social activities** (1 item), **education and work** (1 item), **social relationships** (4 items), **treatment adherence** (1 item), **financial resources** (1 item), **accessibility of information** (1 item), **home environment** (1 item), **accessibility of health care service** (matrix question with 8 items), **multidisciplinarity of care** (matrix question with 8 items), **relationships with health professionals** (1 item), **involvement in decision about care** (1 item).

To facilitate the filling out of the electronic version of the SPADIS PRM and to achieve higher response rates, **items will be reordered by time frame and by type of response options**. After internal discussions, SPADIS researchers decided to **adopt the 4 week timeframe** used by the MOS-HIV, with several notable exceptions mostly related to environmental factors. These refer to, for example, aspects of care with which PLWH do not come into contact every 4 weeks, or items referring to environmental factors which are more constant in time and less amenable to change in the short-term. Examples are the items on home environment, accessibility of health care services, multidisciplinarity of care, relationships with (specialist) health professionals and involvement in care. The timeframes for each of the items are described below.

Across the selected items, we distinguish between three main types of response options:

- **Items that assess frequency** assess 'how often' or 'how much of the time' PLWH experience something. The associated response options that were identified in the cross-matching exercise are: '*seldom – quite often – very often – always*' (WHO QoL HIV BREF), '*all of the time – most of the time – a good bit of the time – some of the time – a little of the time – none of the time*' (MOS-HIV), '*never – rarely – sometimes – often always*' (PROQoL-HIV).
- **Items that assess intensity** assess how much or how completely PLWH have experienced things or were able to do certain things. The associated response options that were identified in the cross-matching exercise are: '*not at all – a little – moderately – mostly – completely*' (WHO QoL HIV BREF), '*not at all – a little – a moderate amount – very much – an extreme amount*' (WHO QoL HIV BREF), '*very poor – poor – neither poor nor good – very good*' (WHO QoL HIV BREF), '*none – very mild – mild – moderate – severe – very severe*' (MOS-HIV), '*I do not need support – I have had no support – I sometimes have enough support – I often have enough support – I always have enough support – Not relevant – Not sure*' (P3CEQ).
- **Items that assess appraisal** assess the extent to which individuals are **satisfied** with certain body functions, abilities to perform tasks, experiences or environmental factors such as contact with health professionals. The associated response options that were identified in the cross-matching exercise are: '*very dissatisfied – dissatisfied – neither satisfied nor dissatisfied – satisfied – very satisfied*' (WHO-QoL HIV BREF).

In order to facilitate responding to the items or to decrease the response burden, **items that fit within the same type of response option** (frequency vs. intensity vs. appraisal) need - to the highest possible extent - to have the same or similar response options. As our COSMIN rating exercise indicated that the WHO QoL HIV BREF tool was of the highest methodological quality (in terms of content validity), we will adopt the response options from this tool when possible. There are a few notable exceptions. For example, items from the MOS-HIV on frequency have 6-point Likert scales, compared to the 5-point Likert scales of the WHO QoL HIV BREF and will be included in its original form as the original items have not validated with, or calibrated on, 5-point Likert scales. Taking into account the context of the items, some Likert scales need to be accompanied by a response option that enables PLWH to specify that the item is '*not applicable*' to their situation or that they are unsure about their response. For example: when asking if PLWH experienced stigma from colleagues because of their HIV diagnosis, PLWH should be able to specify whether or not they have disclosed their HIV status to their colleagues. Similarly, for the item on different types of health care support, we require PLWH to indicate whether or not they require the respective types of support.

2. Reformatting and -ordering the HIV-specific PRM module

Within the final PRM, items that assess **frequency over the past four weeks** will be introduced by a phrase that is similar to the one used by the WHO-QoL HIV BREF but that specifies the four week time frame: “*The following questions refer to how often you have felt or experienced certain things in the last four weeks*”. Items on **appraisal over the past four weeks** are introduced by the following phrase: “*The following questions ask you how good or satisfied you have felt about various aspects of your life over the last four weeks*”. Items that assess **intensity over the past four weeks** are introduced by the following phrases: “*The following questions ask about how completely you experience or were able to do certain things in the last 4 weeks*” and “*The following questions ask about how much you have experienced certain things in the last 4 weeks*”. For the **items which do not adopt the four week time frame**, the introductory phrase is the following: “*The following questions ask how much you have experienced certain things, or how good or satisfied you have felt about various aspects of your life, not restricted to the last 4 weeks*”.

The following tables provide an overview of the changes that were made to the original items in the thematic order that was used throughout the text above (Health and quality of life - [Table 3](#), Psychosocial health and support – [Table 4](#), Environmental factors – [Table 5](#)), a thematically ordered list of the final items ([Table 6](#)), and the HIV-specific module with the final items, in its final structure and order (by type of response option and timeframe – [Table 7](#)) – ready for being subjected to the expert validation exercise.

2.1. Overview of original items, changes made to the items, and final items pre-expert validation

2.1.1. Health and quality of life - Overview of original items, changes made to the items and final items of the HIV-specific PRM module

Subtopic	Original Item			(adapted) SPADIS HIV PRM items				
	Source	Original item and response options	Time-frame	#	Justification of adaptation/deletion	Time-frame	Novel item and response options	ICF codes assessed by items
Activities of daily living	WHO-QoL HIV BREF	How satisfied are you with your ability to perform your daily living activities? <i>Very dissatisfied – Dissatisfied – Neither satisfied nor dissatisfied – Satisfied – Very satisfied</i>	“think about your life in the last 2 weeks”	1	Examples were added to assess specific ICF codes that are addressed in FGs and other eligible PROMs from the LR. Timeframe adapted.	“think about your life in the last 4 weeks”	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)? <i>Very dissatisfied – Dissatisfied – Neither satisfied nor dissatisfied – Satisfied – Very satisfied</i>	d230 – Carrying out a daily routine, d510 – Washing oneself, d540 – Dressing, d620 – Acquisition of goods and services, d629 – Acquisition of necessities, d640 – Doing housework, d540 – Dressing
	SF-HDQ	I find it hard to meet the needs of those I care for (such as children, parents, grandparents, partners, or pets): <i>Not at all or Not applicable - A little hard - Moderately hard - Really hard - I can't do it</i>	“Check the box that describes how you are feeling today”	2	Timeframe adapted. Item phrased as question rather than statement. Response categories adapted to align with WHO response categories, and ‘not applicable’ option added.	“think about your life in the last 4 weeks”	Do you find it hard to meet the needs of those you care for (such as children, parents, grandparents, partners, or pets) <i>Not at all or not applicable – A little – A moderate amount – Very much – Extremely</i>	d660 – Assisting others
Cognitions	MOS-HIV	How much of the time, during the past 4 weeks, did you forget things that happened recently, for example, where you put things and when you had appointments? <i>All of the time – Most of the time – A good bit of the time – Some of the time – A little of the time – None of the time</i>	“..during the past 4 weeks...”	3	/	“..during the past 4 weeks...”	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments? <i>All of the time – Most of the time – A good bit of the time – Some of the time – A little of the time – None of the time</i>	b144 – Memory functions
	MOS-HIV	How much of the time, during the past 4 weeks, did you have difficulty doing activities involving concentration and thinking?	“..during the past 4 weeks...”	4	/	“..during the past 4 weeks...”	How much of the time did you have difficulty doing activities involving concentration and thinking?	b140 – Attention functions, b160 – Thought functions, d160 – Focusing attention

		<i>All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time</i>					<i>All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time</i>	
	MOS-HIV	How much of the time, during the past 4 weeks, did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?	“..during the past 4 weeks...”	5	/	“..during the past 4 weeks...”	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?	b160 – Thought functions, b1646 – Problem solving, d155 – Acquiring new skills, d163 – Thinking, d175 – d175 – Solving problems
		<i>All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time</i>					<i>All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time</i>	
Emotional well-being	WHO-QoL HIV BREF	How often do you have negative feelings such as blue mood, despair, anxiety, depression?	“think about your life in the last 2 weeks”	6, 7, 8, 9	This item assess multiple important aspects of emotional well-being at once. We need to split them up and create a matrix. Timeframe adapted.	“think about your life in the last 4 weeks”	How often do you have the following negative feelings: (Matrix options:) (5) blue mood, (6) despair, (7) anxiety, (8) depression?	b152 – Emotional functions, d240 – Handling stress and other psychological demands
		<i>Never - Seldom - Quite often - Very often – Always</i>					<i>Never - Seldom - Quite often - Very often – Always</i>	
Energy and sleep	WHO-QoL HIV BREF	Do you have enough energy for everyday life?	“think about your life in the last 2 weeks”	10	Timeframe adapted.	“think about your life in the last 4 weeks”	Do you have enough energy for everyday life?	b130 – Energy and drive functions
		<i>Not at all – A little – Moderately – Mostly - Completely</i>					<i>Not at all - A little - Moderately - Mostly – Completely</i>	
		How satisfied are you with your sleep?	“think about your life in the last 2 weeks”	11	Timeframe adapted.	“think about your life in the last 4 weeks”	How satisfied are you with your sleep?	b134 – Sleep function, b1343 – Quality of sleep
		<i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied</i>					<i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied</i>	
Pain	MOS-HIV	How much bodily pain have you generally had during the past 4 weeks?	“..during the past 4 weeks...”	12	/	“..during the past 4 weeks...”	How much bodily pain have you generally had during the past 4 weeks?	b280 – Sensation of pain
		<i>None - Very mild - Mild - Moderate - Severe - Very severe</i>					<i>None - Very mild - Mild - Moderate - Severe - Very severe</i>	

	MOS-HIV	During the past 4 weeks, how much did pain interfere with your normal work (or your normal activities, including work outside the home and housework)?	“..during the past 4 weeks...”	13	To not only focus on ‘normal work’, the item was slightly adapted to assess interference with ‘normal activities’, while specifying the work perspective by including the original examples. WHO response options were adopted.	“..during the past 4 weeks...”	How much did pain interfere with your normal activities (including work outside the home and housework)?	b280 – Sensation of pain, d640 – Doing housework, d850 – Remunerative employment, d855 – Non-remunerative employment,
		Not at all – A little bit – Moderately – Quite a bit – extremely					Not at all – A little – Moderately – Mostly – Completely	
Physical functioning	WHO-QoL HIV BREF	How well are you able to get around?	“think about your life in the last 2 weeks”	14	Examples were added to assess specific ICF codes that are addressed in other eligible PROMs from the LR. Timeframe adapted.	“think about your life in the last 4 weeks”	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	d4105 – Bending, d430 – Lifting and carrying objects, d455 – Moving around, d4500 – Walking short distances, d4551 – Climbing
		Very poor – Poor – Neither poor nor good – Good – Very good					Very poor – Poor – Neither poor nor good – Good – Very good	
Quality of life and self-rated health	WHO-QoL HIV BREF	How satisfied are you with your health?	“think about your life in the last 2 weeks”	15	Timeframe adapted.	“think about your life in the last 4 weeks”	How satisfied are you with your health?	No ICF codes
		Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied					Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	
	WHO-QoL HIV BREF	How would you rate your quality of life?	“think about your life in the last 2 weeks”	16	Timeframe adapted.	“think about your life in the last 4 weeks”	How would you rate your quality of life?	No ICF codes
		Very poor - Poor - Neither poor nor good - Good - Very good					Very poor - Poor - Neither poor nor good - Good - Very good	
Sexuality	WHO-QoL HIV BREF	How satisfied are you with your sex life?	“think about your life in the last 2 weeks”	17	Timeframe adapted.	“think about your life in the last 4 weeks”	How satisfied are you with your sex life?	d770 – Intimate relationships, d7702 – Sexual relationships
		Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied					Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	
	CST-HIV	During the past month, has your sex drive or interest in sex decreased?	“During the past month...”	18	Align phrasing & response categories to WHO	“think about your life	How satisfied are you with your sex drive or interest in sex?	b640 – Sexual functions, b6400 – Functions of sexual arousal phase

		None – Slightly – Somewhat – Quite - Extremely			QoL HIV BREF format on sexuality: assess PLWH's satisfaction with sex drive/interest. Do not assess changes in sex drive.	in the last 4 weeks"	Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	
Symptoms	WHO-QoL HIV BREF	How much are you bothered by any physical problems related to your HIV infection?	"think about your life in the last 2 weeks"	19	Timeframe adapted.	"think about your life in the last 4 weeks"	How much are you bothered by any physical problems related to your HIV infection?	No ICF code (physical problems is too general for the ICF classification system)
		Not at all - A little - A moderate amount - Very much - An extreme amount					Not at all - A little - A moderate amount - Very much - An extreme amount	
	PROQoL-HIV	During the last two weeks, because I am HIV positive I have been bothered by digestive problems (stomach ache, bloating, diarrhoea, nausea or vomiting)	'During the last two weeks, ...'	20	Timeframe adapted. Phrasing and response categories adapted to align with WHO QoL HIV BREF items that assess frequency. Link with HIV discarded in line with other questions.	"think about your life in the last 4 weeks"	How often are you bothered by digestive problems (stomach ache, bloating, diarrhoea, nausea or vomiting)?	d515 – Digestive functions, d525 – Defecation functions, b5350 – Sensation of nausea
		Never – Rarely – Sometimes – Often - Always					Never - Seldom - Quite often - Very often – Always	

Table 3 • Health and quality of life - Overview of original items, changes made to the items and final items of the HIV-specific module of the SPADIS PRM

2.1.2. Psychosocial health and support - Overview of original items, changes made to the items and final items of the HIV-specific module of the SPADIS PRM

Subdomain	Original Item			(adapted) SPADIS HIV PRM items				
	Source	Original item and response options	Time-frame	#	Justification of adaptation	Time-frame	Novel item and response options	ICF codes assessed by item
Meaningfulness	WHO-QoL HIV BREF	To what extent do you feel your life to be meaningful?	"think about your life in the last 2 weeks"	21	Timeframe adapted.	"think about your life in the last 4 weeks"	To what extent do you feel your life to be meaningful?	d930 – Religion and spirituality,
		Not at all - A little - A moderate amount - Very much - An extreme amount					Not at all - A little - A moderate amount - Very much - An extreme amount	

Fears and worries (split from stigma and discrimination)	WHO-QoL HIV BREF	How much do you fear the future? <i>Not at all - A little - A moderate amount - Very much - An extreme amount</i>	“think about your life in the last 2 weeks”	22	Timeframe adapted.	“think about your life in the last 4 weeks”	How much do you fear the future? <i>Not at all - A little - A moderate amount - Very much - An extreme amount</i>	b152 – Emotional functions, d240 – Handling stress and other psychological demands
	WHO-QoL HIV BREF	How much do you worry about death? <i>Not at all - A little - A moderate amount - Very much - An extreme amount</i>	“think about your life in the last 2 weeks”	23	Timeframe adapted.	“think about your life in the last 4 weeks”	How much do you worry about death? <i>Not at all - A little - A moderate amount - Very much - An extreme amount</i>	b152 – Emotional functions, d240 – Handling stress and other psychological demands
	WHO-QoL HIV BREF	To what extent are you bothered by people blaming you for your HIV status? <i>Not at all - A little - A moderate amount - Very much - An extreme amount</i>	“think about your life in the last 2 weeks”	24	Timeframe adapted. Added response category for PLWH who do not disclose their status.	“think about your life in the last 4 weeks”	To what extent are you bothered by people blaming you for your HIV status? <i>Not at all - A little - A moderate amount - Very much - An extreme amount – I generally do not disclose my HIV status</i>	None or ICF codes for 'people': e410 - Individual attitudes of immediate family members, e415 - Individual attitudes of extended family members e420 - Individual attitudes of friends, e425 - Individual attitudes of acquaintances, peers, colleagues, neighbours and community members
	HRC QoL PROM	Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status (starting an intimate relationship, applying for a job, social activities, ...)? <i>Never – Seldom – Quite often – Very often – Always</i>	“Have you ever ..”	25, 26, 27	Create matrix to assess different 'important matters'.	“Have you ever...”	Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following contexts or situations: (matrix options:) (25) starting an intimate relationship, (26) applying for a job, (27) participating in social activities <i>Never – Seldom – Quite often – Very often – Always</i>	d9205 – Socializing, d770 – Intimate relationships, d850 – remunerative employment
Stigma (experienced and anticipated) and discrimination	HRC QoL PROM	Were you ever treated differently because of your HIV status (e.g.; in health-care settings, work, friends or family, ...)?	“Were you ever...”	28, 29, 30, 32, 33, 34	Create matrix to assess discrimination separately for different types of health professionals. Rephrased to assess both 'different treatment' and	“Have you ever...”	Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following individuals? (matrix options): (28) Friends, (29) Family, (30) partner, (31) colleagues or employers (32) health professionals in the HIV reference centre, (33) health professionals in a hospital that is not a HIV reference centre, (34) health professionals outside the hospital	d770 - Intimate relationships, d740 - Formal relationships, d750 - Informal social relationships, e320 – Friends, e310 - Immediate family, e315 - Extended family, e325 – Acquaintances, peers, colleagues, neighbours and community members, e355 –Health professionals, e410 - Individual attitudes of immediate family members, e415 - Individual attitudes of extended family members, e420 - Individual attitudes of friends, e425 -

		Never - Seldom - Quite often - Very often - Always			'negative attitudes'. Add response category for 'I do not disclose my HIV status to this/these individuals'.		Never – Seldom – Quite often – Very often – Always – <i>I do not disclose my HIV status to this/these individuals.</i>	Individual attitudes of acquaintances, peers, colleagues, neighbours and community members, e445 – Individual attitudes of health professionals.
Social activities	MOS-HIV	How much of the time, during the past 4 weeks, has your health limited your social activities (like visiting with friends or close relatives)? All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time	"...during the past 4 weeks, ..."	35	Example added to question to also include participation in leisure activities (in alignment with WHO QoL HIV BREF item on social activities).	"...during the past 4 weeks, ..."	How much of the time, during the past 4 weeks, has your health limited your social activities (like visiting with friends or participating in leisure activities)? All of the time - Most of the time - A good bit of the time - Some of the time - A little of the time - None of the time	d750 – Informal social relationships, d910 – Community life, d920 – Recreation and leisure, d9205 – Socializing,
Education and work	MOS-HIV	Does your health keep you from working at a job, doing work around the house or going to school? Yes - No	Not specified	36	Adapted to avoid overlap with daily activities items. Response categories adapted to align with WHO-QoL HIV BREF.	"think about your life in the last 4 weeks"	Does your health keep you from working at a job or going to school? Not at all - A little - Moderately - Mostly – Completely	d850 – Remunerative employment (, d855 – Non-remunerative employment), d820 – School education, (d825 – Vocational education, d830 – Higher education, d835 – Education life)
Social relationships	WHO-QoL HIV BREF	How satisfied are you with your personal relationships? Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	"think about your life in the last 2 weeks"	37	Added friends and family as examples of personal relationships. Timeframe adapted.	"think about your life in the last 4 weeks"	How satisfied are you with your personal relationships (friends and family)? Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	d750 – Informal social relationships
	WHO-QoL HIV BREF	How satisfied are you with the support you get from your friends? Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	"think about your life in the last 2 weeks"	38	Added family support besides support from friends. Timeframe adapted.	"think about your life in the last 4 weeks"	How satisfied are you with the support you get from your friends and family? Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied	e320 – Friends, e310 - Immediate family, e315 - Extended family
	CST-HIV	During the past month, how often have you had someone you trust to speak to about your problems? Never - Rarely - Sometimes - Frequently - Always	"During the past month, ..."	39	Phrasing of timeframe adapted to align with other items. Response categories of WHO-	"During the past 4 weeks, ..."	During the past 4 weeks, how often have you had someone you trust to speak to about your problems? Never - Seldom - Quite often - Very often – Always	None or e320 – Friends, e310 – Immediate family, e315 – Extended family

					QoL HIV BREF are added.			
	CST-HIV	During the past month to what extent have you been worried about telling someone you have HIV? <i>None - Slightly - Somewhat - Quite - Extremely</i>	“During the past month, ...”	40	Phrasing of timeframe adapted to align with other items. Response categories of WHO-QoL HIV BREF are added.	“During the past 4 weeks, ...”	During the past 4 weeks, to what extent have you been worried about telling someone you have HIV? <i>Not at all - A little - A moderate amount - Very much - Extremely</i>	No ICF codes
Treatment adherence	HRC QoL PROM	How often do you find it difficult to take your HIV medication properly? <i>Never - Seldom - Quite often - Very often - Always</i>	“think about your life in the last 2 weeks”	41	Timeframe adapted.	“think about your life in the last 4 weeks”	How often do you find it difficult to take your HIV medication properly? <i>Never - Seldom - Quite often - Very often - Always</i>	e1101 - Drugs

Table 4 • Psychosocial health and support - Overview of original items, changes made to the items and final items of the HIV-specific module of the SPADIS PRM

2.1.3. Environmental factors - Overview of original items, changes made to the items and final items of the HIV-specific module of the SPADIS PRM

Subdomain	Original Item				(adapted) SPADIS HIV PRM items			
	Source	Original item and response options	Time-frame	#	Justification of adaptation	Time-frame	Novel item and response options	ICF codes assessed by item
Financial resources	WHO-QoL HIV BREF	Have you enough money to meet your needs? <i>Not at all - A little - Moderately - Mostly - Completely</i>	“think about your life in the last 2 weeks”	42	Timeframe adapted.	“think about your life in the last 4 weeks”	Have you enough money to meet your needs? <i>Not at all - A little - Moderately - Mostly - Completely</i>	e165 – Assets, e1650 – Financial assets (d870 – Economic self-sufficiency: although this item relates to ability to handle resources available to people)
Accessibility of information	WHO-QoL HIV BREF	How available to you is the information that you need in your day-to-day life? <i>Not at all - A little - Moderately - Mostly - Completely</i>	“think about your life in the last 2 weeks”	43	Timeframe adapted. ‘Accessibility’ was used instead of ‘availability’ in line with comments of the FG.	“think about your life in the last 4 weeks”	How accessible to you is the information that you need in your day-to-day life? <i>Not at all - A little - Moderately - Mostly - Completely</i>	No ICF codes
Home environment	WHO-QoL HIV BREF	How satisfied are you with the conditions of your living place? <i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied</i>	“think about your life in the last 2 weeks”	44	Timeframe adapted.	No timeframe	How satisfied are you with the conditions of your living place? <i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied</i>	d650 – Caring for household objects, e525 – Housing services, systems and policies, e1651 – Tangible assets

Accessibility of health care services	WHO-QoL HIV BREF	How satisfied are you with your access to healthcare services? <i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied</i>	“think about your life in the last 2 weeks”	45, 46, 47, 48, 49, 50, 51, 52	Timeframe adapted. Create question matrix to assess access to different types of healthcare services. Added response category to indicate not needing the type of care/support .	No timeframe	How satisfied are you with your access to the following healthcare services or professionals? (matrix options): (45) Infectiologist, (46) Psychologist, (47) Social support/worker, (48) Sexologist, (49) General Practitioner, (50) Dentist (51) gynaecologist (<i>for women only</i>), (52) HIV nurse <i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied – I do not require this type of care or support</i>	e355 – Health professionals, e850 – Health services, systems and policies
Multidisciplinary of care	P3CEQ	Do you have enough support from your health care professionals to help you manage your own health and wellbeing? <i>I do not need support – I have had no support – I sometimes have enough support – I often have enough support – I always have enough support – Not relevant – Not sure</i>	No timeframe	53, 54, 55, 56, 57, 58, 59, 60	Create question matrix to adequacy of support for different types of health professionals.	No timeframe	Do you have enough support from the following health care professionals to help you manage your own health and wellbeing? (matrix options): (53) Infectiologist, (54) Psychologist, (55) Social support/worker, (56) Sexologist, (57) General Practitioner, (58) Dentist, (59) gynaecologist (<i>for women only</i>) (60) HIV nurse <i>I do not need support – I have had no support – I sometimes have enough support – I often have enough support – I always have enough support – Not relevant – Not sure</i>	e355 – Health professionals, e445 – Individual attitudes of health professionals
Relationship with (specialist) health professionals	Novel item.			60	This item will be accompanied by an open text question in which PLWH can explain their response	No timeframe	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)? <i>Very dissatisfied - Dissatisfied - Neither satisfied nor dissatisfied - Satisfied - Very satisfied</i>	e355 – Health professionals, e445 – Individual attitudes of health professionals
Involvement in care	P3CEQ	Are you involved as much as you want in decisions about your HIV care? <i>Yes, definitely – Yes, to some extent – No, not really – No, definitely not – Not sure</i>	No timeframe	61	Adapted response categories to align with WHO-QoL HIV BREF, while keeping the original ‘not sure’ response option.	No timeframe	Are you involved as much as you want in decisions about your HIV care? <i>Not at all - A little - Moderately - Mostly – Completely – Not sure</i>	e355 – Health professionals, e445 – Individual attitudes of health professionals

Table 5 • Environmental factors - Overview of original items, changes made to the items and final items of the HIV-specific module of the SPADIS PRM

2.1.4. Overview of items of the HIV-specific module of the SPADIS PRM, thematically ordered

(sub)topic	#	Timeframe	Item	Item response options
Activities of daily living	1	4 weeks	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>
	2	4 weeks	Do you find it hard to meet the needs of those you care for (such as children, parents, grandparents, partners, or pets)	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>Moderately</i> 4. <i>Mostly</i> 5. <i>Completely</i> 6. <i>Not applicable</i>
Cognitions	3	4 weeks	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments?	1. <i>All of the time</i> 2. <i>Most of the time</i> 3. <i>A good bit of the time</i> 4. <i>Some of the time</i> 5. <i>A little of the time</i> 6. <i>None of the time</i>
	4	4 weeks	How much of the time did you have difficulty doing activities involving concentration and thinking?	
	5	4 weeks	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?	
Emotional well-being	6-9	4 weeks	[Matrix question] How often do you have the following negative feelings: (6) blue mood, (7) despair, (8) anxiety, (9) depression	1. <i>Never</i> 2. <i>Seldom</i> 3. <i>Quite often</i> 4. <i>Very often</i> 5. <i>Always</i>
Energy and sleep	10	4 weeks	Do you have enough energy for everyday life?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>Moderately</i> 4. <i>Mostly</i> 5. <i>Completely</i>
	11	4 weeks	How satisfied are you with your sleep?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i>

				5. <i>Very satisfied</i>
Pain	12	4 weeks	How much bodily pain have you generally had during the past 4 weeks?	1. <i>None</i> 2. <i>Very mild</i> 3. <i>Mild</i> 4. <i>Moderate</i> 5. <i>Severe</i> 6. <i>Very severe</i>
	13	4 weeks	How much did pain interfere with your normal activities (including work outside the home and housework)?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>Moderately</i> 4. <i>Mostly</i> 5. <i>Completely</i>
Physical functioning	14	4 weeks	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	1. <i>Very poor</i> 2. <i>Poor</i> 3. <i>Neither poor nor good</i> 4. <i>Good</i> 5. <i>Very good</i>
Quality of life and self-rated health	15	4 weeks	How satisfied are you with your health?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>
	16	4 weeks	How would you rate your quality of life?	1. <i>Very poor</i> 2. <i>Poor</i> 3. <i>Neither poor nor good</i> 4. <i>Good</i> 5. <i>Very good</i>
Sexuality	17	4 weeks	How satisfied are you with your sex life?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>
	18	4 weeks	How satisfied are you with your sex drive or interest in sex?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>A moderate amount</i> 4. <i>Very much</i> 5. <i>An extreme amount</i>
Symptoms	19	4 weeks	How much are you bothered by any physical problems related to your HIV infection?	1. <i>Never</i> 2. <i>Seldom</i> 3. <i>Quite often</i> 4. <i>Very often</i>
	20	4 weeks	How often are you bothered by digestive problems (stomach ache, bloating, diarrhoea, nausea or vomiting)?	

				5. <i>Always</i>
Meaningfulness	21	4 weeks	To what extent do you feel your life to be meaningful?	1. <i>Not at all</i> 2. <i>A little</i>
Fears and worries	22	4 weeks	How much do you fear the future?	3. <i>A moderate amount</i> 4. <i>Very much</i>
	23	4 weeks	How much do you worry about death?	5. <i>An extreme amount</i>
Stigma (experienced and anticipated) and discrimination	24	4 weeks	To what extent are you bothered by people blaming you for your HIV status?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>A moderate amount</i> 4. <i>Very much</i> 5. <i>An extreme amount</i> 6. <i>I generally do not disclose my HIV status</i>
	25, 26, 27	-	[Matrix question] Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following contexts or situations: (25) starting an intimate relationship, (26) applying for a job, (27) participating in social activities	1. <i>Never</i> 2. <i>Seldom</i> 3. <i>Quite often</i> 4. <i>Very often</i> 5. <i>Always</i>
	28, 29, 30, 31, 32, 33, 34	-	[Matrix question] Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following individuals? (matrix options): (28) Friends, (29) Family, (30) partner, (31) colleagues or employers (32) health professionals in the HIV reference centre, (33) health professionals in a hospital that is not a HIV reference centre, (34) health professionals outside the hospital	1. <i>Never</i> 2. <i>Seldom</i> 3. <i>Quite often</i> 4. <i>Very often</i> 5. <i>Always</i> 6. <i>I do not disclose my HIV status to this/these individuals.</i>
Social activities	35	4 weeks	How much of the time, during the past 4 weeks, has your health limited your social activities (like visiting with friends or participating in leisure activities)?	1. <i>All of the time</i> 2. <i>Most of the time</i> 3. <i>A good bit of the time</i> 4. <i>Some of the time</i> 5. <i>A little of the time</i> 6. <i>None of the time</i>
Education and work	36	4 weeks	Does your health keep you from working at a job or going to school?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>Moderately</i> 4. <i>Mostly</i> 5. <i>Completely</i>
Social relationships	37	4 weeks	How satisfied are you with your personal relationships (friends and family)?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i>

	38	4 weeks	How satisfied are you with the support you get from your friends and family?	4. <i>Satisfied</i> 5. <i>Very satisfied</i>
	39	4 weeks	How often have you had someone you trust to speak to about your problems?	1. <i>Never</i> 2. <i>Seldom</i> 3. <i>Quite often</i> 4. <i>Very often</i> 5. <i>Always</i>
	40	4 weeks	To what extent have you been worried about telling someone you have HIV?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>A moderate amount</i> 4. <i>Very much</i> 5. <i>Extremely</i>
Treatment adherence	41	4 weeks	How often do you find it difficult to take your HIV medication properly?	1. <i>Never</i> 2. <i>Seldom</i> 3. <i>Quite often</i> 4. <i>Very often</i> 5. <i>Always</i>
Financial resources	42	4 weeks	Have you enough money to meet your needs?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>Moderately</i> 4. <i>Mostly</i> 5. <i>Completely</i>
Accessibility of information	43	4 weeks	How accessible to you is the information that you need in you day-to-day life?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>
Home environment	44	x	How satisfied are you with the conditions of your living place?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>
Accessibility of health care services	45, 46, 47, 48, 49, 50, 51, 52	x	[Matrix question] How satisfied are you with your access to the following healthcare services or professionals? (45) Infectiologist, (46) Psychologist, (47) Social support/worker, (48) Sexologist, (49) General Practitioner, (50) Dentist, (51) gynaecologist (for women only), (52) HIV nurse	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i> 6. <i>I do not require this type of care or support</i>
Multidisciplinarity of care	53, 54, 55, 56	x	[Matrix question] Do you have enough support from the following health care professionals to help you manage your own health and wellbeing? (53) Infectiologist, (54) Psychologist, (55) Social	1. <i>I do not need support</i> 2. <i>I have had no support</i> 3. <i>I sometimes have enough support</i> 4. <i>I often have enough support</i>

	57, 58, 59		support/worker, (56) Sexologist, (57) General Practitioner, (58) Dentist, gynaecologist (only for women), (59) HIV nurse	5. <i>I always have enough support</i> 6. <i>Not relevant</i> 7. <i>Not sure</i>
Relationships with (specialist) health professionals	60	x	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>
Involvement in care decisions	61	x	Are you involved as much as you want in decisions about your HIV care?	1. Not at all 2. A little 3. Moderately 4. Mostly 5. Completely 6. Not sure

Table 6 • HIV-specific module of the SPADIS PRM – Overview of items, thematically ordered

2.2. Structure and overview of the pre-exert validation HIV-specific PRM module – Ordered by time-frame and response options

The table below offers an overview of the final structure and item order for the HIV-specific module of the SPADIS PRM. The first column refers to the numbers of the items as they are reported in Table 5 and 6. The second column refers to the order in which the items will be presented to the PLWH.

Ref. #	#	Item	Response options
Introduction: “The following questions refer to how often you have felt or experienced certain things in the last four weeks.”			
Matrix Q	How often do you have the following negative feelings?		1. Never 2. Seldom 3. Quite often 4. Very often 5. Always
Matrix	6	1 Blue mood	
	7	2 Despair	
	8	3 Anxiety	
	9	4 Depression	
20	5	How often are you bothered by digestive problems (stomach ache, bloating, diarrhoea, nausea or vomiting)?	
39	6	How often have you had someone you trust to speak to about your problems?	
41	7	How often do you find it difficult to take your HIV medication properly?	
Matrix Q	Have you ever withheld yourself from important matters , because of potential reactions of others to your HIV-status , in one of the following contexts or situations:		

Matrix	25	8	Starting an intimate relationship	
	26	9	Applying for a job	
	27	10	Participating in social activities	
Matrix Q			Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by the following individuals?	1. Never 2. Seldom 3. Quite often 4. Very often 5. Always 6. I do not disclose my HIV status to this/these individuals
Matrix	28	11	Friends	
	29	12	Family	
	30	13	Partner	
	31	14	Colleagues or employer	
	32	15	Health professionals in the hospital	
	33	16	Health professionals outside of the hospital	
	34	17		6.
3	18	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments?		1. All of the time 2. Most of the time 3. A good bit of the time 4. Some of the time 5. A little of the time 6. None of the time
4	19	How much of the time did you have difficulty doing activities involving concentration and thinking?		
5	20	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?		
35	21	How much of the time has your health limited your social activities (like visiting with friends or participating in leisure activities)?		
Introduction: "The following questions ask you how good or satisfied you have felt about various aspects of your life over the <u>past four weeks</u>:"				
1	22	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?		1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied
11	23	How satisfied are you with your sleep?		
15	24	How satisfied are you with your health?		
17	25	How satisfied are you with your sex life?		
18	26	How satisfied are you with your sex drive or interest in sex?		
37	27	How satisfied are you with your personal relationships (friends and family)?		
38	28	How satisfied are you with the support you get from your friends and family?		
Introduction: "The following questions ask about how completely you experienced or were able to do certain things in the <u>last four weeks</u>:"				
10	29	Do you have enough energy for everyday life?		1. Not at all 2. A little 3. Moderately 4. Mostly
36	30	Does your health keep you from working at a job or going to school?		
42	31	Have you enough money to meet your needs?		
42	32	How accessible to you is the information that you need in you day-to-day life?		

13	33	How much did pain interfere with your normal activities (including work outside the home and housework)?	5. Completely
Introduction: “The following questions ask about how much you have experienced certain things in the <u>last four weeks</u>.”			
12	34	How much bodily pain have you generally had during the past 4 weeks?	1. None 2. Very mild 3. Mild 4. Moderate 5. Severe 6. Very severe
14	35	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	1. Very poor 2. Poor 3. Neither poor nor good 4. Good 5. Very good
16	36	How would you rate your quality of life?	
19	37	How much are you bothered by any physical problems related to your HIV infection?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. An extreme amount
21	38	To what extent do you feel your life to be meaningful?	
22	39	How much do you fear the future?	
23	40	How much do you worry about death?	
24	41	To what extent are you bothered by people blaming you for your HIV status?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. An extreme amount 6. I generally do not disclose my HIV status
40	42	To what extent have you been worried about telling someone you have HIV?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. Extremely
2	43	Do you find it hard to meet the needs of those you care for (such as children, parents, grandparents, partners, or pets)?	1. Not applicable 2. Not at all 3. A little 4. A moderate amount 5. Very much 6. Extremely
Introduction: “The following questions ask about how completely you experienced certain things, <u>not restricted to the last four weeks</u>.”			
44	44	How satisfied are you with the conditions of your living place?	1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied

60	45	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?		4. Satisfied 5. Very satisfied + open text box following item 44: "Please describe any experiences you want to share:"
Matrix Q		How satisfied are you with your access to the following healthcare services or professionals?		1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied
Matrix	45	46	Infectiologist	
	46	47	Psychologist	
	47	48	Social support / worker	
	48	49	Sexologist	
	49	50	General practioner	
	50	51	Dentist	
	51	52	Gynaecologist (<i>only for women</i>)	
52	53	HIV nurse		
Matrix Q		Do you have enough support from the following health care professionals to help you manage your own health and wellbeing?		1. I do not need support 2. I have had no support 3. I sometimes have enough support 4. I often have enough support 5. I always have enough support 6. Not relevant 7. Not sure
Matrix	53	54	Infectiologist	
	54	55	Psychologist	
	55	56	Social support / worker	
	56	57	Sexologist	
	57	58	General practioner	
	58	59	Dentist	
	59	60	Gynaecologist (<i>only for women</i>)	
60	61	HIV nurse		
57		Are you involved as much as you want in decisions about your HIV care?		1. Not at all 2. A little 3. Moderately 4. Mostly 5. Completely 6. Not sure

Table 7 • Structure and overview of the HIV-specific module of the SPADIS PRM – Ordered by time-frame and response options

3. Expert validation of the HIV-specific PRM module

An expert validation exercise was organized to further validate the items of the HIV specific PRM module and their relevance for measuring social participation and associated factors or determinants for people living with HIV in Belgium.

3.1. EXPERT VALIDATION: METHODOLOGY

Representatives (multidisciplinary) from all HIV reference centres in Belgium were invited to participate in this expert validation exercise. The exercise was informed by established international guidelines on expert validation and consisted of two parts:

1. A survey ([Annex 5](#)) was disseminated in which all items were described and accompanied by a scale that assesses the relevance of each item (essential vs. useful but not essential vs. not necessary) for the measurement of either social participation or its potential determinants. Based on these results we calculated the Content Validity Ratio (CVR) originally proposed by Lawshe. The CVR is a linear transformation of a proportional level of agreement on how many experts rate an item as essential. This is calculated in the following way: $CVR = (n_e - (\frac{N}{2})) / (N/2)$ where n_e is the number of panel members indicating the essential rating and N is the number of respondents or panel members. The calculated CVR's were evaluated based on the critical values defined by Baghestani et al.'s method⁵ based on Bayesian statistics. These evaluations informed the second part of the expert validation exercise. Also, open survey questions allowed the expert respondents to give feedback in an unrestricted manner. The results are discussed under Expert validation: results.
2. An online expert validation session was organized in which the development of the HIV-specific PRM module up until then was discussed as well as its specific items. This session was guided by the results (CVR's and comments) of the expert validation survey. The results are discussed under Expert validation: results.

3.2. EXPERT VALIDATION: RESULTS

Representatives from all Belgian HIV reference centres were invited to participate. In the invitation, the representatives, mostly infectiologists, were asked to fill out the survey as well as disseminate it among their colleagues to collect more multidisciplinary responses. They were also invited to participate in the online session by confirming their attendance in advance.

A total of 11 experts responded to the expert validation survey, with 10 experts providing complete responses to all survey questions. Among them were 4 infectiologists (medical doctors), 3 psychologists, 2 nurses, 1 study coordinator, and 1 researcher.

In the survey, we combined matrix items and presented them as singular items. This resulted in the **inclusion of 37 items**. Based on the critical values for the CVR³ and the number of respondents, the following decision were made:

- Items for which $\geq 9/11$ or $8/10$ respondents provided the 'essential' rating were included by default. This resulted in the inclusion of 12 of 37 items.
- Items for which $> 2/11$ or $2/10$ respondents provided the 'not necessary' rating were excluded by default. This resulted in the exclusion of 4 out of 37 items.
- Items for which $\geq 9/11$ or $8/10$ respondents provided the 'essential' OR 'useful but not essential' rating were marked as the **focus of the online expert session** discussions. There were 21 items that were discussed.

The results of the survey were discussed in two rounds of expert panel discussions with a total of 10 participating experts, mostly infectiologists. **Table 8** provides an overview of the results of the expert validation exercise, with the items ordered thematically (see Figure 6). In **orange**, changes or additions based on the expert validation exercise are described, while text in **red** marks deletions. Table 9 provides an overview of the final, thematically ordered items post expert validation.

⁵ A review of Lawshe's method for calculating content validity in the social sciences. [URL](#).

A limited amount of changes were made. The item on the difficulty to meet the needs of *those they care for* was first excluded based on the expert validation survey but was revisited during the expert validation session. Upon discussion with the experts, it was decided that this item is to be included with a slightly altered phrasing. Similarly, the item on worrying about death was included although the survey results suggested otherwise. The items under the matrix question on negative feelings were slightly altered for enhanced clarity. Social activities were added as an example for the item that assess the extent to which pain interferes with activities. Similarly, 'partner' was added to the examples of the items on satisfaction with personal relationship and the support they receive. Gynaecologist was added as an item for the matrix questions on satisfaction with accessibility of health care professionals and adequacy of support received.

In total, three items were deleted based on the results of the expert validation exercise. These items relate to satisfaction with sex drive or interest in sex, frequency of bothersome digestive symptoms and accessibility of information.

Expert rating (CVR) and feedback		
Number of experts who rated this item as 'Essential'	8	
Number of experts who rated this item as 'Useful, but not essential'	3	Useful because these limitations may affect SP, but the limitations are not expected.
Number of experts who rated this item as 'Not necessary'	0	
	5	May depend very much on individual situation, would discard given limited selection,
	3	"how much time is left for patient /herself?"
	3	
	8	With respect to the growing age of the overall patient population and cognitive functioning, this is very important. There may be direct and indirect links to social participation. + Important to question cognitive functions
	3	
	0	Important to question cognitive functions
	8	
	3	Important to question cognitive functions
	7	
	4	Difficult to assess the four in one question.
	0	
	9	Mental health plays a major role in social participation
	1	
	0	ADD LONELINESS?

Figure 6 • Visual representation of how to interpret the expert validation results in Table 8

(sub)topic	Time-frame	Item	Item response options	Expert rating (CVR) and feedback
Activities of daily living	4 weeks	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i>	8 3 0 Useful because these limitations may affect SP, but the limitations are not expected.
	4 weeks	Do you find it hard to care for those who depend on you (such as children, parents, grandparents, partners, or pets)?	7. <i>Not at all</i> 8. <i>A little</i> 9. <i>Moderately</i> 10. <i>Mostly</i> 11. <i>Completely</i> 12. <i>Not applicable</i>	5 3 3 May depend very much on individual situation, would discard given limited selection, "how much time is left for patient /herself?"
Cognitions	4 weeks	How much of the time did you forget things that happened recently , for example, where you put things and when you had appointments?	7. <i>All of the time</i> 8. <i>Most of the time</i> 9. <i>A good bit of the time</i> 10. <i>Some of the time</i> 11. <i>A little of the time</i> 12. <i>None of the time</i>	8 3 0 With respect to the growing age of the overall patient population and cognitive functioning, this is very important. There may be direct and indirect links to social participation. + Important to question cognitive functions
	4 weeks	How much of the time did you have difficulty doing activities involving concentration and thinking ?		8 3 0 Important to question cognitive functions
	4 weeks	How much of the time did you have difficulty reasoning and solving problems , for example, making plans, making decisions, learning new things?		7 4 0 Important to question cognitive functions
Emotional well-being	4 weeks	[Matrix question] How often do you have the following negative feelings: (a) blue mood , (b) despair , (c) feelings of anxiety , (d) depressive feelings , (e) loneliness ?	6. <i>Never</i> 7. <i>Seldom</i> 8. <i>Quite often</i> 9. <i>Very often</i> 10. <i>Always</i>	9 1 0 Difficult to assess the four in one question. Mental health plays a major role in social participation ADD LONELINESS?
Energy and sleep	4 weeks	Do you have enough energy for everyday life?	6. <i>Not at all</i> 7. <i>A little</i> 8. <i>Moderately</i> 9. <i>Mostly</i> 10. <i>Completely</i>	9 1 1 when daily activities are already difficult, social participation will not be easy either + I deem this very essential, as it is may not only be related to physical but also to mental health.
	4 weeks	How satisfied are you with your sleep ?	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i>	8 3 0 Sleeping problems may create a vicious cycle affecting physical and mental health
Pain	4 weeks	How much bodily pain have you generally had?	7. <i>None</i> 8. <i>Very mild</i> 9. <i>Mild</i>	5 6 Difficult question, maybe rephrase? it is important to know to what extent a patient can

			10. <i>Moderate</i> 11. <i>Severe</i> 12. <i>Very severe</i>	0	cope with daily life and pain before they can do any social activities at all
	4 weeks	how much did pain interfere with your normal activities (including work outside the home and housework and social activities)?	6. <i>Not at all</i> 7. <i>A little</i> 8. <i>Moderately</i> 9. <i>Mostly</i> 10. <i>Completely</i>	7 4 0	Same comments as above. Present item only to those who indicated having pain (previous question) OR only keep this item
Physical functioning	4 weeks	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	6. <i>Very poor</i> 7. <i>Poor</i> 8. <i>Neither poor nor good</i> 9. <i>Good</i> 10. <i>Very good</i>	8 3 0	
Quality of life and self-rated health	4 weeks	How satisfied are you with your health ?	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i>	9 0 1	Very much related to next question on QoL. Satisfaction might be very subjective and not necessarily be linked to being socially active.
	4 weeks	How would you rate your quality of life ?	6. <i>Very poor</i> 7. <i>Poor</i> 8. <i>Neither poor nor good</i> 9. <i>Good</i> 10. <i>Very good</i>	9 1 0	Quality of life very subjective and relation to social participation is unclear.
Sexuality	4 weeks	How satisfied are you with your sex life ?		7 2 1	Don't see link with social participation. There are patients without a partner.
	4 weeks	How satisfied are you with your sex drive or interest in sex?	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i>	1 7 2	Don't see the link between sex drive and social participation, unless between two people. There are different aspects to how people perceive their sex life, sex drive is only one (quality of sexual experience, achieving organisms, etc.), which may also be gender-specific, I would not pick out only one aspect. More often we hear that PLWH (and especially women) have problems in finding new partners and maintaining trustful intimate relationships , with sexuality being only one aspect of it...
Symptoms	4 weeks	How much are you bothered by any physical problems related to your HIV infection?	6. <i>Not at all</i> 7. <i>A little</i> 8. <i>A moderate amount</i> 9. <i>Very much</i> 10. <i>An extreme amount</i>	8 2 0	Physical problems may interfere with social activities, whether related to HIV or not
			6. <i>Never</i>	3	

	4 weeks	How often are you bothered by digestive problems (stomach-ache, bloating, diarrhoea, nausea or vomiting)?	7. <i>Seldom</i> 8. <i>Quite often</i> 9. <i>Very often</i> 10. <i>Always</i>	6 1	2 comments with doubts on whether or not singling out 1 organ is not too specific
Meaningfulness	4 weeks	To what extent do you feel your life to be meaningful ?		6 3 1	-
Fears and worries	4 weeks	How much do you fear the future ?	7. <i>Not at all</i> 8. <i>A little</i> 9. <i>A moderate amount</i> 10. <i>Very much</i> 11. <i>An extreme amount</i>	4 4 2	For this one, I would say it is useful, but not essential: PLWH are usually confronted with uncertainty of the future, but it will depend very much on where they are in their trajectory of living and coping with HIV , how much they actually will "fear" the future.
	4 weeks	How much do you worry about death ?		3 4 3	Overlap with the previous item (fear of future) was identified and discussed. In the second session, the decision was made to include this item for the first dissemination;
Stigma (experienced and anticipated) and discrimination	4 weeks	To what extent are you bothered by people blaming you for your HIV status ?	7. <i>Not at all</i> 8. <i>A little</i> 9. <i>A moderate amount</i> 10. <i>Very much</i> 11. <i>An extreme amount</i> 12. <i>I generally do not disclose my HIV status</i>	8 2 0	Comments recognize this as important. Yet one comment questions the use of ' bothered by people ', saying it feels 'strange'.
	-	[Matrix question] Have you ever withheld yourself from important matters , because of potential reactions of others to your HIV-status , in one of the following contexts or situations: (25) starting an intimate relationship, (26) applying for a job, (27) participating in social activities	6. <i>Never</i> 7. <i>Seldom</i> 8. <i>Quite often</i> 9. <i>Very often</i> 10. <i>Always</i>	10 0 0	Comments stress the importance of the question. Yet categories may be too broad . And the use of ' ever ' is questioned.
	-	[Matrix question] Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following individuals? (matrix options): (28) Friends, (29) Family, (30) partner, (31) colleagues or employers (32) health professionals in the HIV reference centre, (33) health professionals in a hospital that is not a HIV reference centre, (34) health professionals outside the hospital	7. <i>Never</i> 8. <i>Seldom</i> 9. <i>Quite often</i> 10. <i>Very often</i> 11. <i>Always</i> 12. <i>I do not disclose my HIV status to this/these individuals.</i>	8 1 1	Included – Comments stress the importance of the question. Yet categories may be too broad . And the use of ' ever ' is questioned.
Social activities	4 weeks	How much of the time has living with HIV limited your social activities (like visiting with friends or participating in leisure activities)?	7. <i>All of the time</i> 8. <i>Most of the time</i> 9. <i>A good bit of the time</i> 10. <i>Some of the time</i> 11. <i>A little of the time</i> 12. <i>None of the time</i>	7 3 0	TBD – Health is vague, maybe we should specify 'HIV positive status' or use 'feeling ill' instead of 'your health'.
			6. <i>Not at all</i>	8	Included

Education and work	4 weeks	Does your health keep you from working at a job or going to school?	7. <i>A little</i> 8. <i>Moderately</i> 9. <i>Mostly</i> 10. <i>Completely</i>	2 0	
Social relationships	4 weeks	How satisfied are you with your personal relationships (partner , friends and family)?	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i>	7 2 1 8 2 0	
	4 weeks	How satisfied are you with the support you get from your partner, friends and family?			"Which kind of support?"
	4 weeks	How often have you had someone you trust to speak to about your problems?	6. <i>Never</i> 7. <i>Seldom</i> 8. <i>Quite often</i> 9. <i>Very often</i> 10. <i>Always</i>	6 3 1	Comment on the wording which may be a bit too difficult → Can make to have a clear phrase when translating to FR and NL
	4 weeks	To what extent have you been worried about telling someone you have HIV?	6. <i>Not at all</i> 7. <i>A little</i> 8. <i>A moderate amount</i> 9. <i>Very much</i> 10. <i>Extremely</i>	5 5 0	2 comments on lack of specificity ("someone") with expectations that all PLWH will have had worries if it remains broad.
Treatment adherence	4 weeks	How often do you find it difficult to take your HIV medication properly?	6. <i>Never</i> 7. <i>Seldom</i> 8. <i>Quite often</i> 9. <i>Very often</i> 10. <i>Always</i>	9 1 0	Included
Financial resources	4 weeks	Have you enough money to meet your needs?	6. <i>Not at all</i> 7. <i>A little</i> 8. <i>Moderately</i> 9. <i>Mostly</i> 10. <i>Completely</i>	8 2 0	Included
Home environment	4 weeks	How satisfied are you with the conditions of your living place?	1. <i>Very dissatisfied</i> 2. <i>Dissatisfied</i> 3. <i>Neither satisfied nor dissatisfied</i> 4. <i>Satisfied</i> 5. <i>Very satisfied</i>	8 2 0	Relevance was confirmed during the expert sessions
Accessibility of health care services	x	[Matrix question] How satisfied are you with your access to the following healthcare services or professionals? (a) Infectiologist, (b) Psychologist, (c) Social support/worker, (d) Sexologist, (e) General Practitioner, (f) Dentist, (g) gynaecologist (women only) (h) HIV nurse	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i> 11. <i>I do not require this type of care or support</i>	4 4 2	Please add: gynaecologist for women, therapy counsellor, chemsex counselor (not a specific profession, I know, but asking about access to a professional who can provide support for chemsex related problems would be important) To be discussed: more generally: drug-related problems/drug counsellor, Health promotion more generally? (e.g. smoking, diet, physical exercise, adopting healthy lifestyles)?

					But the question makes more sense if there is also a question in relation to people's need in this regard. What is the link with social participation? Is dentist necessary?
Multidisciplinary of care	x	[Matrix question] Do you have enough support from the following health care professionals to help you manage your own health and wellbeing? (a) Infectiologist, (b) Psychologist, (c) Social support/worker, (d) Sexologist, (e) General Practitioner, (f) Dentist, (g) Gynaecologist (for women) , (h) HIV nurse	8. <i>I do not need support</i> 9. <i>I have had no support</i> 10. <i>I sometimes have enough support</i> 11. <i>I often have enough support</i> 12. <i>I always have enough support</i> 13. <i>Not relevant</i> 14. <i>Not sure</i>	7 3 0	See previous comment. So maybe this one is sufficient? Add gynaecologist for women, therapy counsellor, and specific support for chemsex (drug-) related problems. Difference between "not relevant" and "I do not need support" is unclear.
Relationships with (specialist) health professionals	x	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?	6. <i>Very dissatisfied</i> 7. <i>Dissatisfied</i> 8. <i>Neither satisfied nor dissatisfied</i> 9. <i>Satisfied</i> 10. <i>Very satisfied</i>	4 5 1	
Involvement in care decisions	x	Are you involved as much as you want in decisions about your HIV care?	7. <i>Not at all</i> 8. <i>A little</i> 9. <i>Moderately</i> 10. <i>Mostly</i> 11. <i>Completely</i> 12. <i>Not sure</i>	5 5 0	Could ask more directly about patient involvement and participation, without asking about "as much as you want"? Ideally, this should be two questions... Very subjective, requires a lot of knowledge
Accessibility of information	x	How accessible is the information that you need in day-to-day life?	1. <i>Not at all</i> 2. <i>A little</i> 3. <i>Moderately</i> 4. <i>Mostly</i> 5. <i>Completely</i>	6 1 4	What does information refer to? It is unclear what responses to this question may add in terms of value.

Table 8 • Overview of results from the expert validation exercise of the HIV-specific SPADIS PRM module

3.3. STRUCTURE AND OVERVIEW OF THE HIV-SPECIFIC PRM MODULE – POST-EXPERT VALIDATION

Ref. #	#	Item	Response options
Introduction: "The following questions refer to how often you have felt or experienced certain things in the last four weeks:"			
Matrix Q		How often do you have the following negative feelings?	
Ma	6	1 Blue mood	1. Never
	7	2 Despair	2. Seldom
			3. Quite often

	8	3	Feelings of anxiety	4. Very often 5. Always	
	9	4	Depressive feelings		
	*	5	Loneliness		
38	6	How often have you had someone you trust to speak to about your problems?			
40	7	How often do you find it difficult to take your HIV medication properly?			
Matrix Q			Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following contexts or situations:		
Matrix	25	8	Starting an intimate relationship	1. Never 2. Seldom 3. Quite often 4. Very often 5. Always 6. I do not disclose my HIV status to this/these individuals	
	26	9	Applying for a job		
	27	10	Participating in social activities		
Matrix Q			Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by the following individuals?		
Matrix	28	11	Friends		
	29	12	Family		
	30	13	Partner		
	31	14	Colleagues or employer		
	32	15	Health professionals in the hospital		
	33	16	Health professionals outside of the hospital		
3	17	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments?		1. All of the time 2. Most of the time 3. A good bit of the time 4. Some of the time 5. A little of the time 6. None of the time	
4	18	How much of the time did you have difficulty doing activities involving concentration and thinking?			
5	19	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?			
34	20	How much of the time has living with HIV limited your social activities (like visiting with friends or participating in leisure activities)?			
Introduction: "The following questions ask you how good or satisfied you have felt about various aspects of your life over the past four weeks:"					
1	21	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?		1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied	
11	22	How satisfied are you with your sleep?			
15	23	How satisfied are you with your health?			
17	24	How satisfied are you with your sex life?			
36	25	How satisfied are you with your personal relationships (partner, friends and family)?			
37	26	How satisfied are you with the support you get from your partner, friends and family?			
Introduction: "The following questions ask about how completely you experienced or were able to do certain things in the last four weeks:"					

10	27	Do you have enough energy for everyday life?	1. Not at all 2. A little 3. Moderately 4. Mostly 5. Completely
35	28	Does your health keep you from working at a job or going to school?	
41	29	Have you enough money to meet your needs?	
13	30	How much did pain interfere with your normal activities (including work outside the home, housework and social activities)?	
Introduction: “The following questions ask about how much you have experienced certain things in the <u>last four weeks</u> :”			
12	31	How much bodily pain have you generally had during the past 4 weeks?	1. None 2. Very mild 3. Mild 4. Moderate 5. Severe 6. Very severe
14	32	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	1. Very poor 2. Poor 3. Neither poor nor good 4. Good 5. Very good
16	33	How would you rate you quality of life?	
19	34	How much are you bothered by any physical problems related to your HIV infection?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. An extreme amount
21	35	To what extent do your feel your life to be meaningful?	
22	36	How much do you fear the future?	
23	37	How much do you worry about death?	
24	38	To what extent are you bothered by people blaming you for your HIV status?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. An extreme amount 6. I generally do not disclose my HIV status
39	39	To what extent have you been worried about telling someone you have HIV?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. Extremely
2	40	Do you find it hard to meet the needs of those you care for (such as children, parents, grandparents, partners, or pets)?	1. Not applicable 2. Not at all 3. A little 4. A moderate amount 5. Very much 6. Extremely
Introduction: “The following questions ask about how completely you experienced certain things, <u>not restricted to the last four weeks</u> :”			

43	41	How satisfied are you with the conditions of your living place?	1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied
56	42	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?	1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied
Matrix Q		How satisfied are you with your access to the following healthcare services or professionals?	+ open text box following item 42: "Please describe any experiences you want to share:"
Matrix	44	43 Infectiologist	1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied
	45	44 Psychologist	
	46	45 Social support / worker	
	47	46 Sexologist	
	48	47 General practioner	
	49	48 Dentist	
	*	49 Gynaecologist (<i>for women only</i>)	
Matrix Q		Do you have enough support from the following health care professionals to help you manage your own health and wellbeing?	
Matrix	50	50 Infectiologist	1. I do not need support 2. I have had no support 3. I sometimes have enough support 4. I often have enough support 5. I always have enough support 6. Not sure
	51	51 Psychologist	
	52	52 Social support / worker	
	53	53 Sexologist	
	54	54 General practioner	
	55	55 Dentist	
	*	56 Gynaecologist (<i>for women only</i>)	
57	57	Are you involved as much as you want in decisions about your HIV care?	7. Not at all 8. A little 9. Moderately 10. Mostly 11. Completely 12. Not sure

Table 9 • Post-Expert validation Structure and overview of the HIV-specific module of the SPADIS PRM – Ordered by time-frame and response options.

*Novel item added based on perspectives gathered during the expert validation exercise.

4. Forward-backward translation with reconciliation

The HIV specific module of the SPADIS PRM (Table 9) is to be translated into the two main Belgian languages: French and Dutch. For several of the items, validated version already exist in French and/or Dutch. For the remaining items, a forward-backward translation with reconciliation process⁶ is applied. Figure 7 provides an overview of the 7 steps that this process entails. This process, which is further detailed below, requires the following translators:

- Two translators (T1 and T2 in Figure 1) fluent in the source language (English) and native speakers of the target language (French or Dutch).
- A third translator (T3 in Figure 1) fluent in the target language (French or Dutch) and native speaker of the source language (English).

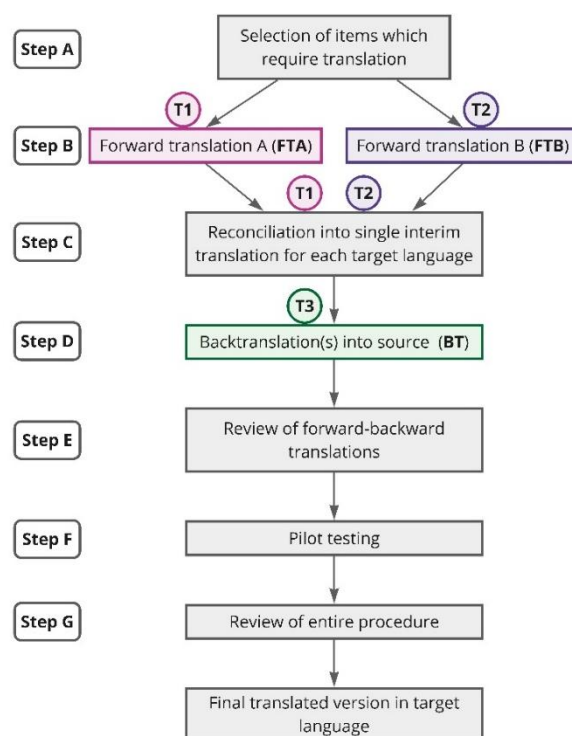


Figure 7 • Overview of steps for the forward-backward translation with reconciliation process

Step A: Selection of items which require translation

The source version of the eligible PROMs identified in the literature (2.2), from which most items were adopted, were developed in English. Several of these PROMs have previously been translated and validated in Dutch and/or French. The present translation process will only be applied to items which have not yet been translated to the target language (see Annex 6).

Step B: Forward translations

Two independent translators (T1 and T2) translate the items into the target language. These translators have expertise in health-related content and an understanding of PROMs. The translation aims for conceptual rather than literal equivalence, conveying the meaning and purpose of each item. Language should be simple and clear at the literary level of the target population. Furthermore, translations should ensure that questions remain neutrally phrased and avoid the unintentional suggestion of desirable answers. In Annex 6, the template used for the forward translations to Dutch and French can be found.

Step C: Reconciliation

For each translated item, the two independent translators (T1 and T2) discuss their respective translations based on several **decision criteria**:

1. Source and comprehensibility

1.1 Best reflects the conceptual definitions and meaning of the source text

⁶ The process of reconciliation: evaluation of guidelines for translating quality-of-life questionnaires. [URL](#).

- 1.2 Best reflects the stress of the source text (i.e., what the main point of the source text is)
- 1.3 Is understandable for a lay person without medical knowledge
- 1.4 Is understandable to a population of varied education levels
- 1.5 Is as close as possible to the source text
- 1.6 Reads more naturally in the target language

2. Cultural

- 2.1 Is culturally appropriate in the scope of sensitive topics
- 2.2 Is culturally appropriate in the scope of cultural differences of life

3. Grammatical

- 3.1 The syntax is correct
- 3.2 The verbal forms and tenses are correct
- 3.3 Gender and number are adapted and correct
- 3.4 Other elements are correct (especially articles and prepositions)

4. Terminology

- 4.1 Includes all the keywords
- 4.2 Is semantically precise
- 4.3 Vocabulary/terminology is consistent throughout the translation

Based on these decision criteria, the two independent translators have several **decision options** to reach consensus on how the translations for each item can be reconciled:

1. Forward translation A as it is
2. Forward translation B as it is
3. A with minor modifications
4. B with minor modifications
5. Merge A and B as they are, with A adapted to B
6. Merge A and B as they are, with B adapted to A
7. Merge A and B with modifications/additions, with A adapted to B
8. Merge A and B with modifications/additions, with B adapted to A
9. Prepare a completely new translation C

Each step until is to be **documented in its entirety**: forward translation A and B, reconciled translation and decision options and criteria. [Annex 6](#) provides the template that was used for documenting the reconciliation process.

Step D: Backward translation

A third independent translator, fluent in the target language and a native English speaker, translates the reconciled translated items back to English. The back-translated version is then compared with the items in the reconciled translated version (step C) to identify any discrepancies, ensuring that the translated PROM accurately reflects the original content.

Step E: Review of forward-backward translations

After having completed step A to D, the French or Dutch item sets are reviewed in their entirety to assess whether or not they make sense considering the (conceptual) objectives of the SPADIS PRMs.

Step F: Pilot testing

The French or Dutch item sets are presented to individuals of the target population during cognitive interviewing phase (see [Cognitive interviewing](#)) to identify any readily apparent issues (clarity of items and the concepts covered by the items). If required, adaptations are made by the SPADIS research team to address identified problems.

Step G: Review of entire procedure

The output of each step of the forward-backward translation (incl. reconciliation) process are reviewed by the SPADIS research team, leading to final translated item sets for the SPADIS PRMs.

[Annex 6](#) describes the resulting NL and FR versions of the reconciliation exercises and the backward translations. The final output (NL and FR version of the item set) can be found in the tables that documented the cognitive interviewing process (see below).

5. Cognitive interviewing

5.1. BACKGROUND

The steps described in this document are based on the guidelines put forth by Gordon B. Willis in his ‘*Cognitive interviewing – A “how to” guide*’, his Cognitive Interviewing Reporting Framework (CRIF) and J.S. Balza’s ‘*Effective questionnaire design: How to use cognitive interviews to refine questionnaire items*’. In this phase of the SPADIS PRM development, the cognitive interviews will serve **to identify issues that occur in the cognitive processes** defined by Tourangeau (1984) **when responding to the SPADIS disease-specific PRM modules**.

The cognitive processes under investigation are the following:

1. Comprehension of the question

- a. **Question intent**: What does the respondent believe the question to be asking?
- b. **Meaning of terms**: What do specific words and phrases in the question mean to the respondent?

2. Retrieval from memory of relevant information

- a. **Recallability of information**: What types of information does the respondent need to recall in order to answer the question?
- b. **Recall strategy**: What type of strategies are used to retrieve information? For example, does the respondent tend to count event by recalling each one individually, or does he/she use an estimation strategy?

3. Decision processes

- a. **Motivation**: Does the respondent devote sufficient mental effort to answer the question accurately and thoughtfully?
- b. **Sensitivity/social desirability**: Does the respondent want to tell the truth? Does he/she say something that makes him/her look “better”?

4. Response processes

- a. **Mapping the response**: Can the respondent match his or her internally generated answer to the response categories given by the survey question?

5.2. SELECTION OF COGNITIVE INTERVIEWING APPROACHES AND TECHNIQUES

Based on the supporting guidelines the **following choices in terms of cognitive interviewing approaches** were made:

- The use of **Verbal Probing** techniques instead of the think-aloud technique.

While the “think-aloud” approach instructs interviewees to “think aloud” as they answer the survey questions and the interviewers note the cognitive processes that drive the interviewee’s responses, the verbal probing technique involves the interviewer asking specific questions to the interviewee relevant to the question, or to the specific answers given. The verbal probing technique was chosen given that it provides more **control over the interview** allowing the interviewer to focus on particular areas of relevance, and it requires **little training of the subject or interviewee**. There are several categories of basic cognitive probes that can be used. They are the following:

Basic cognitive probe category	Example
Comprehension / Interpretation probe	<i>What does the term “outpatient” mean to you?</i>
Paraphrasing	<i>Can you repeat the question I just asked in your own words?</i>
Confidence judgment	<i>How sure are you that your health insurance covers drug and alcohol treatment?</i>
Recall probe	<i>How do you remember that you went to the doctor five times in the past 12 months?</i>
Specific probe?	<i>Why do you think cancer is the most serious health problem?</i>
General probes?	<i>How did you arrive at that answer? Was that easy or hard to answer? I noticed that you hesitated – tell me what were you thinking?</i>

- The use of **concurrent probing** instead of retrospective probing

While the retrospective probing technique involves asking the probe questions upon completion of the entire survey, the concurrent probing technique is characterized by (1) the interviewer asking the survey question, (2) the subject answering the question, (3) the interviewer asking a probe question, (4) the subject answering the probe question and (5) possibly further cycles of (1)-(4).

The concurrent probing technique was chosen given that information elicited by the probe questions is still **fresh in the interviewee's mind** and therefore less subjected to potential fabrications.

- The use of **scripted probes** with the possibility of spontaneous probes as the interview progresses

Scripted probes are developed before the interview commences. For example, when for certain terms one anticipates difficulty in terms of comprehension or readability, interviewers can be instructed to apply the probe: "What does [term] mean to you?". These probes are often typed directly into the questionnaire draft used by the interviewers (see tables below) The use of scripted probes is particularly useful when there is sufficient time to prepare for interviews, resources exist to plan and execute a fairly standardized testing approach, and when some interviewers are relatively inexperienced and would benefit from the guidance provided by a structured protocol.

Admittedly, the most interesting forms of probing frequently develop over the course of the interview and cannot be developed in advance of the interview. As such, the most effective interviews may consist of a combination of scripted and spontaneous probes. Therefore, in the present approach, **spontaneous follow-up probes that arise after the interviewer responds to the scripted probes may be asked by the interviewer.**

5.3. DEFINING SCRIPTED CONCURRENT PROBES FOR THE SPADIS DISEASE-SPECIFIC PRM MODULES

Throughout the development of the SPADIS disease-specific module progresses, potential issues were identified that require assessment during the present cognitive interviewing phase. This occurred in several phases:

- Upon completion of the cross-matching exercise in which the results of the focus groups and the literature review on eligible PRMs were compared, items were first selected and developed by SPADIS researchers in collaboration with internal HIV-experts. Several uncertainties in terms of item phrasing and terminology were identified in this phase. Scripted probing questions will focus on these uncertainties to ascertain whether or not individuals of the target populations encounter difficulties in the cognitive processes required to respond to the respective items.
- Similarly, the **expert validation** phase comprehensively collected expert perspectives regarding items and their phrasing/terminology that could potentially be unclear to individuals of the target population. Scripted probes will similarly focus on these potential issues.
- Throughout the **translation procedure** (forward-backward translation with reconciliation), the forward translators identified potential issues for Dutch and French item versions of the SPADIS PRMs. The scripted probes will serve to evaluate the cognitive processes used by individuals to respond to the items for which these potential issues were identified.
- In **preparation of the cognitive interviews**, scripted probes were freely developed by the SPADIS researchers.

5.4. COGNITIVE INTERVIEWING FOR THE DUTCH HIV-SPECIFIC SPADIS PRM MODULE - SCRIPTED PROBES AND TEMPLATE

The Dutch HIV-specific SPADIS PRM module is divided into 5 sections, each introduced by a distinct instructing phrase. The tables below report the items and response categories which will be read to the interviewees, and the scripted probes that will be used to evaluate the cognitive processes involved in answering the questions. The tables are used as templates for recording and reporting the results of the cognitive interviews.

The cognitive interviews are introduced with general information on the exercise and a description of the objectives:

- Explain that this constitutes one of the final phases of the development of a questionnaire that assesses the well-being and functioning of people living with HIV in Belgium. This questionnaire or measurement instrument can be used in-

hospital for monitoring purposes and to inform tailored health care and support. At a population level, these tools can also be used for monitoring purposes to inform policy-making in the field of health care and social security.

- Over the course of the interview, the questions will be read out loud and a print-out of the response options for the respective questions is presented to the interviewee.
- The interviewee is instructed to carefully think about the question and provide a response.
- After each question, a scripted probe (see tables below) or question is presented to the interviewee. These probes are designed to investigate the cognitive processes or thought processes that occur when the interviewee responds to each questions. By doing this, SPADIS researchers can evaluate whether or not the questions (and response options) are suitable for their respective purposes.

The following remarks are also given at the start of the interview:

- Interviewees can give any feedback they want about each question and their response options if something is not clear. Interviewee responses are not restricted to the scripted probes.
- Responses to the PRM questions are not recorded in any way, ensuring anonymity.

Nederlandse vertaling:

- *Dit interview hoort bij een van de laatste stappen in de ontwikkeling van een vragenlijst die het welzijn en functioneren van mensen met HIV in België in kaart brengt. Dit meetinstrument kan in ziekenhuizen worden ingezet voor monitoring en om zorg en ondersteuning beter af te stemmen op individuele behoeften. Op populatieniveau kunnen de verzamelde gegevens tevens worden gebruikt voor beleidsontwikkeling binnen de gezondheidszorg en sociale zekerheid.*
- *Tijdens het interview worden de vragen hardop voorgelezen, terwijl de geïnterviewde een geprinte versie van de antwoordopties ontvangt.*
- *De deelnemer wordt gevraagd zorgvuldig over elke vraag na te denken en vervolgens een antwoord te geven.*
- *Na elke vraag wordt een vooraf vastgestelde probe (zie tabellen hieronder) of een aanvullende vraag gesteld. Deze probes zijn bedoeld om inzicht te krijgen in de cognitieve processen die meespelen bij het beantwoorden van de vragen. Op deze manier kunnen SPADIS-onderzoekers beoordelen of zowel de vragen als de antwoordopties geschikt en doeltreffend zijn voor hun beoogde doeleinden.*

Aan het begin van het interview worden ook de volgende opmerkingen meegegeven:

- *De geïnterviewden kunnen openlijk feedback geven over elke vraag en de bijbehorende antwoordopties, vooral als iets onduidelijk is. Hun reacties zijn niet beperkt tot de vooraf vastgestelde probes.*
- *Antwoorden op de PRM-vragen worden op geen enkele manier geregistreerd, waardoor anonimiteit gegarandeerd blijft.*

5.4.1. Section 1

Introduction : « The following questions refer to how often you have felt or experienced certain things in the last four weeks »

Dutch translation of introduction : « De volgende vragen verwijzen naar hoe vaak u bepaalde dingen heeft gevoeld of ervaren in de afgelopen vier weken.»

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Frequency1] o Nooit o Zelden o Redelijk vaak o Zeer vaak o Altijd	[Intro - Matrix] Hoe vaak heb je de volgende negatieve gevoelens ? :	-	-	-	
	Een sombere stemming	1	"Was this difficult to answer?"	3a	Int1: Clear answer, not difficult to answer. Slight language barrier, for the term [<i>sombere</i> stemming]. Repeating the items made it clear. Int2: Clear answer, good interpretation – not limited to feelings linked to HIV.
	Wanhoop	2	"For you, what does [<i>wanhoop</i>] mean?"	1b	Int1: refers to limitations in terms of functioning which translate to not being able to envision doing everything the person wants. Int2: clear interpretation of the concept.
	Angstgevoelens	3	"When answering this question, do you think back on specific moments where you felt anxious, or do you look back at last four weeks in a more general manner?"	2a – b	Int1: Refers mostly to specific instances instead of general feeling. Int2: Rather a general feeling.
	Depressieve gevoelens	4	"When we talk about [<i>depressieve gevoelens</i>], what kind of feelings does this refer to according to you? "	1b	Int1: slight miscomprehension of terminology. Thinks about functional limitations that impact possibility to do things the interviewee wants. Int2: Clear interpretation of concept.
	Eenzaamheid	5	"When answering this question, do you think back on specific moments where you felt lonely, or do you approach this in a more general manner?"	2a – b	Int1: general feeling. Links it heavily to living with a chronic disease. Int2: Slight misapprehension, thinks about life in general.
	Hoe vaak was er iemand die je vertrouwt om te praten over je problemen?	6	"Can you explain in your own words what we are asking with this question?"	1a	Int1: Clear and straight answer. Clear comprehension of the intent of the question. Int2: Good comprehension of topic.
	Hoe vaak vind je het moeilijk om je hiv-medicatie op de juiste manier in te nemen?	7	[General probe] "Was this hard or easy to answer?"	[General probe]	Int1: For other types of medication easier to answer than for HIV. Although interviewee gives large explanation, when it comes to answering the question, the objective is clear. Int2: Slight misapprehension, difficulty refers to ingestion method (with vs. no food).

	[Intro - Matrix] Heb je jezelf ooit weerhouden van belangrijke zaken vanwege mogelijke reacties van anderen op uw hiv-status, in een van de volgende situaties	8	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS] “How did you arrive at those answers?” [possible follow-up probe if interviewee cannot respond: “Did you think about specific instances or do you answer based on a general feeling?”]	2b	Int1: Clear answers and straight-forward mapping of responses. Thinks on specific instances rather than a general feeling. Int2: Clear answers and straight-forward mapping of responses.
	Het beginnen van een intieme relatie				
	Solliciteren voor werk				
[Frequency2] o Nooit o Zelden o Redelijk vaak o Zeer vaak o Altijd o Ik deel mijn hiv-status niet met deze perso(o)n(en)	Deelnemen aan sociale activiteiten zoals vrienden bezoeken en vrijetijdsactiviteiten				
	[Intro - Matrix] Heb je ooit negatieve houdingen ervaren of bent je anders behandeld vanwege je Hiv-status door een van de volgende personen?	9	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS] “How did you arrive at those answers?” [possible follow-up probe if interviewee cannot respond: “Did you think about specific instances or do you answer based on a general feeling?”]	2b	Int2: In general, this question will elicit more interesting responses when not limited to the last four weeks.
	Vrienden	-	-		Int1: Clear and straight answer. Int2: Clear and straight answer.
	Familie	-	-		Int1: Clear and straight answer. Int2: Clear and straight answer.
	Partner	-	-		Int1: has no partner which blocked the interviewee's responses. Int2: Clear and straight answer.
	Collega's of werkgevers	-	-		Int1: Difficulty answering, didn't see the last option. Int2: Clear and straight answer.
	Zorgverleners in het hiv referentiecentrum	10	“When we say [hiv referentiecentrum], can you explain the setting you think we refer to?”	1b	Int1: Clear and correct comprehension of terminology.
	Zorgverleners in een ziekenhuis (niet in een hiv referentiecentrum)	-	-		Int1: Clear and straight-forward selection of response.
	Zorgverleners buiten het ziekenhuis	11	“Can you give us an example of the [zorgverleners] that you think of when hearing this question?”	1a	Int1: Clear and straight-forward answer. With correct interpretation of question's intent. Int: Clear and straight-forward answer. With correct interpretation of question's intent
[Frequency3]	Hoe vaak bent u dingen vergeten die recent geleden gebeurd zijn (bv. waar u dingen gelegd had en wanneer u afspraken had)?	12	[General probe] “Was everything about this question clear?”	[general probe]	Int1: Clear and straight answer. Int2: clear and straight answer

- o Altijd - o Meestal - o Vaak - o Soms - o Zelden - o Nooit → Switch order to align more with other response options	Hoe vaak hebt u moeite gehad bij activiteiten waarbij u zich moest concentreren en moest nadenken?	13	[General probe] "Was everything about this question clear?"	[general probe]	Int1: Clear and straight answer. Int2: Clear and straight answer.
	Hoe vaak hebt u moeite gehad om te redeneren en problemen op te lossen (bv. plannen maken, beslissingen nemen, nieuwe dingen leren)?	14	[General probe] "Was everything about this question clear?"	[general probe]	Int1: Clear and straight answer. Int2: Clear and straight answer.
	Hoe vaak heeft je gezondheid je gehinderd bij je sociale activiteiten (zoals vriendenbezoeken of vrijetijdsactiviteiten)?	15	"Could you explain in your own words how you interpret this question?"	1a	Int1: Straight-forward answer, with recall of specific instances rather than general feeling. Correct comprehension of question's intent. Int2: Straight forward and clear answer; good comprehension of the concept.

5.4.2. Section 2

Introduction : « The following questions ask you how good or satisfied you have felt about various aspects of your life over the past four weeks »

French translation of introduction : « De volgende vragen gaan over hoe goed of tevreden u zich heeft gevoeld over verschillende aspecten van uw leven in de afgelopen vier weken »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Satisfaction1] - o Erg ontevreden - o Ontevreden - o Tevreden noch ontevreden - o Tevreden - o Zeer tevreden	Ben je tevreden met je vermogen capaciteit om alledaagse activiteiten te verrichten (bijvoorbeeld uitvoeren van huishoudelijke taken, boodschappen doen, of zelfzorg zoals aankleden en wassen, enz.)?	16	"In your own life, what kinds of tasks do you think about when you hear this question"?	1a	Int1: Clear and quick answer. Correct interpretation and comprehension of question's intent. Int2: 'Vermogen' interpreted as "assets", "wealth"; Adjustment: vermogen changed to 'capaciteit'.
	Hoe tevreden bent u met uw slaap?	-	[General probe can be used]	[General probe]	Int1: Clear and quick mapping of response. Int2: Clear and straight response.
	Hoe tevreden bent u met uw gezondheid?	-	[General probe can be used]	[general probe]	Int1: Clear and quick mapping of response. Int2: Clear and quick mapping of response; with good comprehension of 'gezondheid'; not restricted to HIV.
	In welke mate bent u tevreden met uw seksuele leven?	-	[General probe can be used]	[general probe]	Int1: Clear and quickly mapped the response. Int2: Clear and quickly mapped responses.
	Hoe tevreden ben je met je persoonlijke relaties (partner, vrienden en familie)?	17	"Can you explain to us what you thought of before deciding to answer [answer given by interviewee]?"	4	Int1: Clear quick mapping of the response. Int2:
	Hoe tevreden ben je met de steun die je krijgt van je partner, vrienden en familie?	18	"Can you tell us how you arrived at your answer?"	4	Int1: confusing thought processes. Wants more support but is also linked to asking for support.

					Int2: clear comprehension of concept and not necessarily linked to HIV; thinks of rather specific instances.
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5.4.3. Section 3

Introduction : « The following questions ask about how completely you experienced or were able to do certain things in the last four weeks »

French translation of introduction : « De volgende vragen gaan over hoe volledig u bepaalde dingen heeft ervaren of heeft kunnen doen in de afgelopen vier weken »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Intensity1] o Helemaal niet o Bijna niet o Gemiddeld o Grotendeels o Helemaal	Hebt u genoeg energie voor het leven van alledag?	-	[General probe can be used]	[general probe]	Int1: Clear and quick response. Int2: Clear and quick response
	Verhindert uw gezondheid dat u kunt gaan werken of naar school gaan?	19	"What do you think of when you hear us ask about [uw gezondheid]"	1b	Int1: Clear and quick response, with correct interpretation / comprehension in question's intent. Education is seen mostly as secondary / tertiary education. Int2: Good comprehension of topic or concepts.
		20	"When you hear [naar school gaan], what types of education do you think about?"	1b	
	Hebt u genoeg geld om in uw behoeften te voorzien?	21	"Could you tell us in your own words what you think this questions refers to?"	1a	Int1: Clear comprehension / interpretation of the item's intent -
	In welke mate ben je door pijn gehinderd in je normale activiteiten (zowel werk buitenshuis als huishoudelijk werk en sociale activiteiten zoals vrienden bezoeken of vrijetijdsactiviteiten)?	22	[possible follow-up probe if interviewee cannot respond: "Could you tell us how you arrived at your answer? Did you recall specific moments or do you base your answer more on a general feeling?"	3a [2b]	Int1: Difficulty providing one confident answer for work and social activities together. Int2: Thinks on general feeling than about specific instances. Adjustment: split up into matrix item.

5.4.4. Section 4

Introduction : « The following questions ask about how much you have experienced certain things in the last four weeks »

Dutch translation of introduction : « De volgende vragen gaan over de mate waarin u bepaalde dingen heeft ervaren in de afgelopen vier weken »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Intensity2] o Geen o Heel licht o Licht o Nogal o Ernstig o Zeer ernstig	Hoeveel lichamelijke pijn heb je?	-	-	-	Int1: Clear and quick mapping of response. Int2: Clear answer and quick, although expressed doubts about response options: - Heel licht en licht is misschien weinig verschil. → Andere response optie toevoegen.

[Intensity3] o Erg slecht o Tamelijk slecht o Goed noch slecht o Tamelijk goed o Erg goed	Hoe goed kan je jezelf verplaatsen (korte afstanden wandelen, enkele trappen oplopen, bukken of tillen)?	23	"When hearing this question, is it clear what we mean with <i>[jezelf verplaatsen]</i> ?"	1a	Int1: correct interpretation / comprehension of the item's intent. Int2: correct interpretation of topic; focused on physical mobility.
	Hoe zou u uw kwaliteit van leven inschatten?	-	-	-	Int1: Clear and quick mapping of response. Int2: Clear and quick mapping of response.
[Intensity4] o Helemaal niet o Weinig o Middelmatig o Hevig o Een extreme hoeveelheid	Hoeveel last heb je van lichamelijke problemen die verband houden met je hiv-infectie?	24	"Can you repeat the question I just asked in your own words?"	1a	Int1: Clear and quick mapping of response with good comprehension / interpretation of the item's intent. Int2: Clear and quick mapping of response;
	In welke mate voelt u dat uw leven betekenisvol is?	25	[General probe] "Was it easy or hard to answer this question?"	[General probe]	Int1: Slightly tripped over the question at first, but repeating it made it instantly clear. Int2: Clear answer and good comprehension of concept.
	In welke mate heb je angst voor de toekomst?	-	-		Int1: Clear distinction with previous question on <i>[Angst voor de toekomst]</i> . Int2: Clear comprehension of context -
	In welke mate maak je je zorgen over de dood?	-			Int1: clear comprehension of topic. Int2: clear comprehension of topic.
	In welke mate maak je je zorgen om iemand te vertellen dat je hiv hebt?	26	"Can you explain to us how you arrived at your answer and who you think of when hearing <i>[iemand]</i> ?"	1a – 4	Int1: Clear and quick mapping of response with good comprehension / interpretation of the item's intent.
[Intensity5] o Helemaal niet o Weinig o Middelmatig o Hevig o Een extreme hoeveelheid o In het algemeen deel ik mijn hiv status niet	In welke mate heb je last van mensen die je de schuld geven van je hiv-status?	27	"Can you explain to us how you arrived at your answer and which persons you think of when trying to answer this question?"	1a – 4	Int1: Clear and quick mapping of response with good comprehension / interpretation of the item's intent. Int2: Clear and quick response; good comprehension of topics.
[Intensity6] o Niet van toepassing (use as last response option) o Helemaal niet o Bijna niet o Gemiddeld	Vind je het moeilijk om te zorgen voor degenen die van jou afhankelijk zijn (zoals kinderen, ouders, grootouders, partners of huisdieren)?	-	-	-	Int1: Clear question and quick mapping of responses. Int2: Clear question and quick mapping of responses.

o Grotendeels				
o Helemaal				

5.4.5. Section 5

Introduction : « The following questions ask about how completely you experienced certain things, not restricted to the last four weeks »

French translation of introduction : « De volgende vragen gaan over hoe volledig u bepaalde dingen heeft ervaren, niet beperkt tot de afgelopen vier weken »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Satisfaction1] o Erg ontevreden o Ontevreden o Tevreden noch ontevreden o Tevreden o Zeer tevreden o Ik heb geen contact gezocht met deze zorgverlener	Hoe tevreden ben je over de omstandigheden van de plaats waar je leeft?	28	"When you hear [<i>omstandigheden van de plaats waar je leeft</i>], what do you think about ? »	1a	Int1: Interviewee thinks about this in a broad sense, not restricted to the home environment or home address. → Clear and correct comprehension and interpretation of item's intent. Int2: Thinks about the city rather than the home itself.
	Hoe tevreden ben je over uw relaties met de zorgverleners die betrokken zijn bij je zorg (bijvoorbeeld zorgzame houding, communicatie met zorgverleners)?	-	[General probe] "Was it easy or hard to answer this question"?	[General probe]	Int2: Clear response; wanted us to reiterate that
	[Intro - Matrix] Hoe tevreden ben je over je toegang tot de volgende zorgverleners?	29	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS] "How did you arrive at those answers?" [possible follow-up probe if interviewee cannot respond: "Did you think about specific instances or do you answer based on a general feeling?"	2b	Int1: Indicated by interviewee that some health professionals were never involved. Int2: Adjustment: Add response option "Ik heb geen contact gezocht met deze zorgverlener".
	Infectioloog (Specialist)				
	Psycholoog				
	Maatschappelijk werker				
	Seksuoloog				
	Huisarts				
[Intensity7] o Ik heb geen ondersteuning nodig o Ik heb geen ondersteuning gekregen o Soms krijg ik genoeg ondersteuning	[Intro - Matrix] Krijg je voldoende ondersteuning van de volgende zorgverleners om voor je gezondheid en welzijn te zorgen?	30	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS] "How did you arrive at those answers?" [possible follow-up probe if interviewee cannot respond: "Did you think about specific instances or do you answer based on a general feeling?"	2b	Int1: Clear and quick responses mapped. Clear distinction with previous question and correct interpretation / comprehension of the item's intent. Int2: Good comprehension of the concept, thinks about communication and emotional support.
	Infectioloog / Specialist				
	Psycholoog				
	Maatschappelijk werker				
	Seksuoloog				
	Huisarts				

- o Vaak krijg ik genoeg ondersteuning - o Ik heb altijd genoeg ondersteuning - o Niet zeker	Tandarts				- Tandarts: raar om te zeggen dat zij ondersteunen.
	Gynaecoloog (<i>enkel voor vrouwen</i>)				
	Hiv verpleegkundige				
[Intensity8] - o Helemaal niet - o Bijna niet - o Gemiddeld - o Grotendeels - o Helemaal - o Niet zeker	Ben je zoveel als je wilt betrokken bij beslissingen over de hiv-zorg die je krijgt?	31	"Could you repeat in your own words what we are asking with this question?"	1a	Int1: Clear and quick mapping of response and correct interpretation / comprehension of the item's intents. Int2: Clear comprehension; refers to concept of shared-decision making.

5.5. COGNITIVE INTERVIEWING FOR THE FRENCH HIV-SPECIFIC SPADIS PRM MODULE - SCRIPTED PROBES AND TEMPLATE

The French HIV-specific SPADIS PRM module is divided into 5 sections, each introduced by a distinct instructing phrase. The tables below report the items and response categories which will be read to the interviewees, and the scripted probes that will be used to evaluate the cognitive processes involved in answering the questions. The tables are used as templates for recording and reporting the results of the cognitive interviews.

The cognitive interviews are introduced with general information on the exercise and a description of the objectives:

- Explain that this constitutes one of the final phases of the development of a questionnaire that assesses the well-being and functioning of people living with HIV in Belgium. This questionnaire or measurement instrument can be used in-hospital for monitoring purposes and to inform tailored health care and support. At a population level, these tools can also be used for monitoring purposes to inform policy-making in the field of health care and social security.
- Over the course of the interview, the questions will be read out loud and a print-out of the response options for the respective questions is presented to the interviewee.
- The interviewee is instructed to carefully think about the question and provide a response.
- After each question, a scripted probe (see tables below) or question is presented to the interviewee. These probes are designed to investigate the cognitive processes or thought processes that occur when the interviewee responds to each questions. By doing this, SPADIS researchers can evaluate whether or not the questions (and response options) are suitable for their respective purposes.

The following remarks are also given at the start of the interview:

- Interviewees can give any feedback they want about each question and their response options if something is not clear. Interviewee responses are not restricted to the scripted probes.
- Responses to the PRM questions are not recorded in any way, ensuring anonymity.

5.5.1. Section 1

Introduction : « The following questions refer to how often you have felt or experienced certain things in the last four weeks »

French translation of introduction : « Les questions suivantes concernent la fréquence à laquelle vous avez ressenti ou vécu certaines choses au cours des quatre dernières semaines »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Frequency1] ○ Jamais ○ Parfois ○ Souvent ○ Très souvent ○ Toujours	[Intro - Matrix] A quelle fréquence ressentez-vous les sentiments négatifs suivants :	-	-	-	
	La tristesse	1	"Was this difficult to answer?"	3a	Int1: Clear to answer Int2: Clear to answer; clear comprehension
	Le désespoir	2	"For you, what does [<i>le désespoir</i>] mean?"	1b	Int1: Good comprehension of concept Int2: Clear to answer and clear comprehension of the topics
	Des sentiments d'anxiété	3	"When answering this question, do you think back on specific moments where you felt anxious, or do you look back at last four weeks in a more general manner?"	2a – b	Int1: Good comprehension Int2: Thinks about specific instances, correct comprehension of intent
	Des sentiments dépressifs	4	"When we talk about [<i>sentiments dépressifs</i>], what kind of feelings does this refer to according to you? "	1b	Int1: Good comprehension of terms Int2: Easy to answer yet slight misapprehension of what depressive feelings refers to, although interviewee correct himself.
	La solitude	5	"When answering this question, do you think back on specific moments where you felt lonely, or do you approach this in a more general manner?"	2a – b	Int1: General feeling solitude/loneliness Int2: Recollection based on general feeling rather than specific instances.
	A quelle fréquence avez-vous eu une personne de confiance à qui parler de vos problèmes?	6	"Can you explain in your own words what we are asking with this question?"	1a	Int1: Good comprehension of concept – "vos problèmes" refers not just to HIV but to other problems. Int2: Easy to answer; doubts about timeframe.
	A quelle fréquence avez-vous des difficultés à prendre correctement votre traitement contre le VIH?	7	[General probe] "Was this hard or easy to answer?"	[General probe]	Int1: Easy to answer and clear. Int2: Not easy to answer; as interviewee feels this is very much dependent on external factors. However, intent of question is clear to interviewee.
	[Intro - Matrix] Vous êtes-vous déjà retenu de faire des choses importantes, en raison des réactions potentielles des autres face à votre statut VIH, dans l'un des contextes ou situations suivants : Débuter une relation intime	8	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS] "How did you arrive at those answers?"	2b	Int1: Easy to answer, with focus on general feeling rather than based on specific instances. Int2: Easy to answer. More of a general feeling. Timeframe should not be restricted to last four weeks.

	Postuler à un emploi		[possible follow-up probe if interviewee cannot respond: “Did you think about specific instances or do you answer based on a general feeling?”]		Adjustment: Present question in its current form but in section 5.
	Participer à des activités sociales				
[Frequency2] ○ Jamais ○ Parfois ○ Souvent ○ Très souvent ○ Toujours ○ Je ne divulgue pas mon statut VIH à cette/ces personne(s) ○ Non-applicable	[Intro - Matrix] Avez-vous déjà été confronté.e à des attitudes négatives ou avez-vous été traité différemment en raison de votre statut VIH par l'une des personnes suivantes ?	9	“How did you arrive at those answers?” [possible follow-up probe if interviewee cannot respond: “Did you think about specific instances or do you answer based on a general feeling?”]	2b	Int1: Easy to answer. No issues. Int2: In general, easy to answer.
	Amis	-	-		
	Famille	-	-		Int2: Difficult to answer; interviewee hasn't seen family in long time. Adjustment: add response option for “Not applicable”
	Partenaire	-	-		
	Collègues ou employeurs	-	-		
	Professionnels de santé du centre de référence VIH	10	“When we say [centre de référence VIH], can you explain the setting you think we refer to?”	1b	Int1: Immediately thinks of hospital setting in general, after repeating the question the distinction with HRC became clear immediately. Int2: HIV reference centre is not the clearest. Distinction was unclear. Adjustment: provide definition of HIV reference centre with an asterisk (*).
	Professionnels de santé hospitalier (en dehors du centre de référence VIH)	-	-		
	Professionnels de santé en dehors de l'hôpital	11	“Can you give us an example of the [professionnels de santé] that you think of when hearing this question?”	1a	
[Frequency3] ○ En permanence ○ Très souvent	A quelle fréquence, avez-vous oublié, par exemple, des choses qui sont arrivées dernièrement, des rendez-vous, où vous aviez mis vos affaires?	12	[General probe] “Was everything about this question clear?”	[general probe]	Int1: Easy to answer and good comprehension of the item's intent. Int2: Clear question; easily answered.
	A quelle fréquence, avez-vous eu des difficultés à avoir des activités demandant concentration et réflexion?	13	[General probe] “Was everything about this question clear?”	[general probe]	Int1: Needed to repeat question but interviewee assured that question is clear. Int2: Clear question, easily answered.

- o Souvent - o Quelquefois - o Rarement - o Jamais	A quelle fréquence, avez-vous eu des difficultés pour raisonner et résoudre des problèmes, par exemple, faire des projets, prendre des décisions, apprendre quelque chose de nouveau?	1 4	[General probe] "Was everything about this question clear?"	[general probe]	Int2: Clear question, easily answered.
	A quelle fréquence, le fait de vivre avec le VIH, a-t-il limité vos activités sociales (comme rendre visite à des amis ou participer à des activités de loisirs)?	1 5	"Could you explain in your own words how you interpret this question"?	1a	Int1: Clear comprehension of the subject of the question and easy to answer. Int2: A bit difficult; <i>loisirs</i> can mean a different thing, is a bit vague according to interviewee; yet this corresponds to the item's intent and is therefore valid.

5.5.2. Section 2

Introduction : « **The following questions ask you how good or satisfied you have felt about various aspects of your life over the past four weeks** »

French translation of introduction : « **Les questions suivantes vous demandent à quel point vous vous êtes senti bien ou satisfait de divers aspects de votre vie au cours des quatre dernières semaines** »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Satisfaction1] - o Pas du tout satisfait - o Pas satisfait - o Ni satisfait, ni insatisfait - o Satisfait - o Très satisfait	Dans quelle mesure êtes-vous satisfait.e de votre capacité à accomplir vos activités quotidiennes (par exemple, tâches ménagères, les courses, ou les soins personnels comme s'habiller, se laver, etc.)?	16	"In your own life, what kinds of tasks do you think about when you hear this question?"	1a	Int1: Easy to answer and clear comprehension of the item's intent and concepts. Int2: Easy to answer. Good comprehension and interpretation of item's intent and concepts..
	Dans quelle mesure êtes-vous satisfait de votre sommeil?	-	[General probe can be used]	[General probe]	Int2: Straightforward and quick answer
	Dans quelle mesure êtes-vous satisfait de votre santé?	-	[General probe can be used]	[general probe]	Int2: Straightforward and quick answer
	Dans quelle mesure êtes-vous satisfait de votre vie sexuelle?	-	[General probe can be used]	[general probe]	Int2: Straightforward and quick answer
	Dans quelle mesure êtes-vous satisfait.e de vos relations personnelles (partenaire, amis, famille)?	17	"Can you explain to us what you thought of before deciding to answer [<i>answer given by interviewee</i>]?"	4	Int1: Clear chain of thought in which objective assessment of relations is combined with own preference. Adequate mapping of response. Int2: Straightforward and quick answer. Clear comprehension of subject and easy mapping of response.
	Dans quelle mesure êtes-vous satisfait du soutien que vous recevez de votre partenaire, de vos amis et de votre famille?	18	"Can you tell us how you arrived at your answer?"	4	Int1: Adequate mapping of the responses and clear chain of thought to get there. Int2: Needed guidance to describe what kind of [<i>soutien</i>]. Upon asking the interviewee,

				support referred to multiple topics. Easy mapping of responses.
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5.5.3. Section 3

Introduction : « **The following questions ask about how completely you experienced or were able to do certain things in the last four weeks** »

French translation of introduction : « **Les questions suivantes portent sur la mesure dans laquelle vous avez pleinement vécu ou été en mesure de faire certaines choses au cours des quatre dernières semaines.** »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Intensity1] o Pas du tout o Un peu o Modérément o Beaucoup o Complètement	Avez-vous assez d'énergie dans la vie de tous les jours?	-	[General probe can be used]	[general probe]	Int1: Clear question and no issues in mapping responses.
	Votre état de santé vous empêche-t-il d'exercer une activité professionnelle ou de poursuivre une éducation?	19	"What do you think of when you hear us ask about [votre état de santé]"	1b	Int1: Correct comprehension of topic by thinking of functioning/health outcomes that go beyond HIV Int2: Correct comprehension of topic (health in general, not necessarily VIH).
		20	"When you hear [poursuivre une éducation], what types of education do you think about?"	1b	Int1: Correct interpretation of concept of education, i.e. not limited to secondary or tertiary school. Int2: Correct comprehension of topic; education not restricted to secondary or tertiary school.
	Avez-vous assez d'argent pour satisfaire vos besoins?	21	"Could you tell us in your own words what you think this question refers to?"	1a	Int1: Clear questions and correct interpretation / comprehension of item's intent. Int2: Seen as a requisite for being able to live independently. → Correct interpretation/ comprehension.
	Dans quelle mesure vos douleurs physiques vous ont-elles limité.e dans vos activités normales (y compris votre travail, vos tâches ménagères et vos activités sociales)?	22	"How sure are you of your answer?" [possible follow-up probe if interviewee cannot respond: "Could you tell us how you arrived at your answer? Did you recall specific moments or do you base your answer more on a general feeling?"	3a [2b]	Int1: Confident of answer; thinks more about a general feeling. Int2: Question is inappropriate, given that the interviewee has no physical pain. Adjustment: Move to section 4, following the question on intensity of pain.

5.5.4. Section 4

Introduction : « The following questions ask about how much you have experienced certain things in the last four weeks »

French translation of introduction : « Les questions suivantes portent sur l'intensité avec laquelle vous avez vécu certaines expériences au cours des quatre dernières semaines. »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Intensity2] ☐ Nulle ☐ Très faible ☐ Faible ☐ Moyenne ☐ Grande ☐ Très grande	Quelle a été l'intensité Dans quelle mesure avez-vous eu des de vos douleurs physiques?	-	-	-	Int1: Easy to answer. Int2: Unclear what we referred to, this question assumes that there is definitely pain. Adjustment: change item so it doesn't assume pain to be present in respondent.
[Intensity3] ☐ Très mauvaise ☐ Mauvaise ☐ Ni bonne, ni mauvaise ☐ Bonne ☐ Très bonne	Comment trouvez-vous votre capacité à vous déplacer par vous-même (marcher de courtes distances, monter quelques marches, vous pencher ou soulever des objets)?	23	"When hearing this question, is it clear what we mean with [<i> votre capacité à vous déplacer par vous-même</i>]?"	1a	Int1: Correct interpretation / comprehension of topic and item's intent. Int2: Clear question; easy to respond to.
	Comment trouvez-vous votre qualité de vie?	-	-	-	Int1: Easy and quick to answer. Int2: Clear question; easy to respond.
[Intensity4] ☐ Pas du tout ☐ Un peu ☐ Modérément ☐ Beaucoup ☐ Extrêmement [Put item on pain interference question here]	Dans quelle mesure êtes-vous gêné.e par des problèmes physiques liés à votre infection par le VIH?	24	"Can you repeat the question I just asked in your own words?"	1a	Int1: Easy to answer. Int2: Refers to medication and their side-effects almost immediately; not necessarily wrong.
	Dans quelle mesure votre vie a-t-elle un sens?	25	[General probe] "Was it easy or hard to answer this question?"	[General probe]	Int1: Easy and clear to answer. Int2: Finds it not at all easy to respond to this question. Distinguishes between different aspects of life (morally, financially, mentally, physically, and makes a general deduction); yet this is the correct interpretation of the item's intent.
	Dans quelle mesure craignez-vous l'avenir?	-	-	-	Int1: Easy and clear to answer.
	Dans quelle mesure la mort vous inquiète-t-elle?	-	-	-	Int1: Easy and clear to answer.
	Dans quelle mesure vous êtes-vous inquiété.e du fait de dire à quelqu'un que vous avez le VIH ?	26	"Can you explain to us how you arrived at your answer and who you think of when hearing [<i>à quelqu'un</i>]?"	1a – 4	Int1: Easy and clear to answer. Int2: Unique situation of interviewee makes it difficult to answer the scripted probe.
[Intensity5] ☐ Pas du tout ☐ Un peu ☐ Modérément	Dans quelle mesure êtes-vous gêné.e par les personnes qui vous reprochent votre séropositivité?	27	"Can you explain to us how you arrived at your answer and which persons you think of when trying to answer this question?"	1a – 4	Int1: Easy to answer and correct interpretation of concepts/item's intent. Int2: Quite easy to answer.

- o Beaucoup - o Extrêmement - o En général, je ne divulgue pas mon statut VIH				
[Intensity6] - o Non applicable - o Pas du tout - o Un peu - o Modérément - o Beaucoup - o Extrêmement	Avez-vous des difficultés à prendre soin de ceux qui dépendent de vous ? (comme vos enfants, parents, grands-parents, partenaires, animaux domestiques)	-	-	Int1: Easy to comprehend and answer to question Int2: Interviewee links this with all kind of other aspects of functioning. E.g. if interviewee is psychologically unwell or financially, than interviewee cannot provide care and support for those who dependent on the interviewee. This is a valid interpretation of the item's intent.

5.5.5. Section 5

Introduction : « **The following questions ask about how completely you experienced certain things, not restricted to the last four weeks** »

French translation of introduction : « **Les questions suivantes portent sur la mesure dans laquelle vous avez pleinement vécu certaines expériences, sans restriction aux quatre dernières semaines** »

Item and response options		Scripted probes			Notes
Response options	Item	#	Scripted probe	Cogn. process	
[Satisfaction1] o Pas du tout satisfait o Pas satisfait o Ni satisfait, ni insatisfait o Satisfait o Très satisfait o J'ignore si j'y ai accès o Pas besoin	Dans quelle mesure êtes-vous satisfait de l'endroit où vous vivez?	28	“When you hear [<i>/endroit où vous vivez</i>], what do you think about ? »	1a	Int1: Miscomprehension of the topic. Does not think about the home, neighbourhood or city/community, rather about life in general. Int2: Clear comprehension of intent.
	Dans quelle mesure êtes-vous satisfait.e de votre relation avec les professionnels de santé impliqué dans vos soins (par ex. attitude bienveillante et communication)	-	[General probe] “Was it easy or hard to answer this question”?	[General probe]	
	[Intro - Matrix] Dans quelle mesure êtes-vous satisfait.e de votre accès aux services ou aux professionnels de santé suivants ?	29	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS]	2b	Int1: Need another response option that says ‘never needed or wanted to have access’. Int2: Clear responses were provided, although unique given unique setting in which interviewee lives. Adjustment: Add response option: “Not applicable”.
	Infectiologue		“How did you arrive at those answers?” [bad probe]		
	Psychologue		[possible follow-up probe if interviewee cannot respond: “Did you think about specific instances or do you answer based on a general feeling?”]		
	Assistant.e sociale				
	Sexologue				
	Médecin généraliste				
Dentiste					

	Gynécologue (uniquement pour des femmes)			
	Infirmier.ère VIH			
[Intensity7] o Je n'ai pas besoin de soutien o Je n'ai eu aucun soutien o Parfois, j'ai suffisamment de soutien o Souvent, j'ai suffisamment de soutien o J'ai toujours suffisamment de soutien o Pas sûr	[Intro - Matrix] Recevez-vous suffisamment de soutien pour vous aider à gérer votre santé et votre bien-être de la part des professionnels de santé suivants ? Infectiologue Psychologue Assistant.e sociale Sexologue Médecin généraliste Dentiste Gynécologue (uniquement pour des femmes) Infirmier.ère VIH	30	[ASK AFTER HAVING GONE THROUGH ALL MATRIX ITEMS] "How did you arrive at those answers?" [possible follow-up probe if interviewee cannot respond: "Did you think about specific instances or do you answer based on a general feeling?"]	2b Int1: Quite clear, however distinction with previous question is a bit vague. Int2: HIV nurse question is unclear. In some case the interviewee thinks that nurses are <i>too aware</i> of what they are doing. Adjustment: Put key words of the item in bold.
[Intensity8] o Pas du tout o Un peu o Modérément o Beaucoup o Complètement o Pas sûr	Etes-vous impliqué.e autant que vous le souhaitez dans les décisions au sujet de vos soins pour le VIH?	31	"Could you repeat in your own words what we are asking with this question?"	1a Int1: Clear comprehension of concept. Int2: Kind of misinterpreted the question at first, but the probe questions were correctly answered; correct interpretation of intent of question.

IV. A final HIV-specific PRM module ready for psychometric testing

The SPADIS patient-reported measures (PRMs) consist of a generic PRM module and disease-specific PRM modules. The latter are developed through a step-wise, standardized process that involved patients (or for HIV: people living with HIV - PLWH) early on in the process. Their perspectives were collected through qualitative research methods (focus groups) and cross-matched with results (content analysis) of a literature review on existing disease-specific PRMs. For HIV, this comparative exercise informed the development of a first item set designed to measure functioning in a multidimensional and HIV-specific way that emphasizes **social participation** as one of the main dimensions and measurement outcomes. The item set has subsequently undergone expert validation and translation (Dutch and French), was presented to PLWH during cognitive interviews, and was finally translated into an electronic questionnaire (REDCap®). The methodological approach is detailed here (protocol paper).

The first dissemination and analyses of responses of PLWH will help to serve the following objectives:

- a) Perform (advanced) psychometric testing to validate the questions/items for the purpose of measuring functioning and well-being in a multidimensional, HIV-specific way with a focus on social participation. The results of these analyses will enable SPADIS researchers to fine-tune (shorten and/or tailor) the item sets to specific profiles of PLWH in Belgium.
- b) Perform a first measurement of social participation and associated factors of PLWH in Belgium.

In order to make meaningful and direct comparisons with other conditions for which the SPADIS project develops disease-specific modules (e.g. cancer), a generic PRM module was developed in parallel. This module, which was the results of collaborative effort between Sciensano researchers and (labour market) sociologists of the Vrije Universiteit Brussel (BRISPO), includes questions or items that enable the following investigations:

- a) Identifying inequalities by sociodemographic or -economic profiles.
- b) Comparing outcomes of generic HRQoL assessments across conditions and with the general public (EQ-5D)
- c) Comparing outcomes of generic social participation measures across SPADIS conditions.
- d) Performing detailed analysis of labour market positions and transitions, in line with Belgian administrative nomenclature ('socio-economic position') and data collected in work package 2 of the SPADIS project (data linkages). Specific items related to the work environment or employment conditions are also included.

1. The SPADIS PRM: HIV specific and generic PRM module

The SPADIS PRM for PLWH has **four main components**:

Language, informed consent & diagnosis	Respondents to select their language (English, Dutch or French), provide informed consent , and specify the diagnosis/-es of the condition(s) they live with.
Generic module: part I	The generic module of the SPADIS PRM is split up in two parts. In Part I , respondents are asked to provide sociodemographic and -economic background information.
HIV-specific module	The HIV-specific module presented only to PLWH (based on responses in component 1: diagnosis)
Generic module: part II	In part II of the generic module, respondents are asked to provide information related to their present or past labour market position and work environment , social participation , and health-related quality of life .

This document provides an overview of the items or questions that are included in the HIV-specific module and in the generic module (part I and II).

- (!) An electronic **test** version of the survey (REDCap) can be consulted here: https://redcap.link/SPADIS_HIV_Test. This version does not contain any compulsory items/questions.
- (!) Items that are **compulsory** are marked with a red asterix (*) in the overview tables below.

2. The generic module: part 1

(sub)-domains	topic	#	Item	Item response options
Personal factors (sociodemographic and -economic factors)	Age	1*	What is your age?	Dropdown list (18-99)
	Date of diagnosis	2*	In what year did you receive your final diagnosis?	Dropdown list (x – 2024)
		2a	In what month of this year did you receive your final diagnosis?	Dropdown list
	Gender and sex	3*	What sex were you assigned at birth, on your original birth certificate?	1. Female 2. Male 3. <i>Don't know</i> 4. <i>Prefer not to answer</i>
		4	To which gender do you most identify?	1. Female 2. Male 3. Transgender 4. Don't know 5. Prefer not to answer 6. Other: <i>[open text box]</i>
	Region	5*	In what region do you live?	1. Brussels Capital Region 2. Flemish region 3. Walloon region
	Household size and composition	6*	Which of the following best describes the composition of your current household?	1. Single without children 2. Single parent with child(ren) 3. Living with partner without children 4. Living with partner and child(ren) 5. Other: <i>[open text response]</i>
		6a	(conditional: If item 6 = 3 or 4) How would you describe your partner's current working status?	1. Employee (not civil servant) 2. Self-employed without employees 3. Self-employed with employees 4. Civil servant 5. Family worker 6. Career break 7. Unemployed/job seeking 8. On sick leave or invalidity 9. Full-time student 10. (early) retirement 11. Does the housekeeping, without benefits 12. Integration income (OCMW/CPAS) 13. Does not work and has a recognized handicap 14. Don't know 15. Other situation: describe <i>[Open text response]</i>
		7	(conditional: If item 6 = 2 or 4) How many children live in your household?	Dropdown list from 0-“10 or more”
		7a	(conditional on item 7) – How old are the children that live in your household?	Dropdown list from 0-35 for each child (number of children depends on answer for item 7)

	Educational level	8*	What is your highest attained education level?	1. Primary or no education 2. Lower secondary education (minimum 3 years) 3. Upper secondary education (minimum 6 years) 4. Higher education short type (2-3 years) (bachelor) 5. Higher education long type or university (master) 6. Doctorate with dissertation
	Income	9*	A household may have different sources of income and more than one household member may contribute to it. Thinking of your household's total monthly income, is your household able to make ends meet...?	1. With great difficulty 2. With difficulty 3. With some difficulty 4. Fairly easily 5. Easily 6. Very easily
	Ethnicity	10	Where were you born?	Dropdown list of all countries
		10a	Where was your mother born?	Dropdown list of all countries OR "Don't know"
		10b	Where was your father born?	Dropdown list of all countries OR "Don't know"
	Sexual orientation	11	Which of the following best represents how you think of yourself?	1. Lesbian or gay; 2. Straight, 3. Bisexual; 4. Don't know; 5. Prefer not to answer; 6. I use a different term

3. The HIV-specific module

Section 1: these questions are framed within a **four week** time period

#	Item	Response options
[Matrix - intro] How often do you have the following negative feelings?		
1*	Blue mood	1. Never 2. Seldom 3. Quite often 4. Very often 5. Always
2	Despair	
3	Feelings of anxiety	
4	Depressive feelings	
5	Loneliness	
6	How often have you had someone you trust to speak to about your problems?	1. None of the time 2. A little of the time 3. Some of the time 4. A good bit of the time 5. Most of the time 6. All of the time
7*	How often did you find it difficult to take your HIV medication properly?	
8	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments?	
9*	How much of the time did you have difficulty doing activities involving concentration and thinking?	
10	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?	
11*	How much of the time has living with HIV limited your social activities (like visiting with friends or participating in leisure activities)?	

Section 2: these questions are framed within a four week time period

#	Item	Response options
12	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?	1. Very dissatisfied 2. Dissatisfied 3. Neither satisfied nor dissatisfied 4. Satisfied 5. Very satisfied
13	How satisfied are you with your sleep?	
14*	How satisfied are you with your health?	
15	How satisfied are you with your sex life?	
16	How satisfied are you with your personal relationships (partner, friends and family)?	
17	How satisfied are you with the support you get from your partner, friends and family?	

Section 3: these questions are framed within a four week time period

#	Item	Response options
18	Do you have enough energy for everyday life?	1. Not at all 2. A little 3. Moderately 4. Mostly 5. Completely
19*	Does your health keep you from working at a job or going to school?	
20	Have you enough money to meet your needs?	

Section 4: these questions are framed within a four week time period

#	Item	Response options
21*	How much bodily pain have you generally had?	1. None 2. Very mild 3. Mild 4. Moderate 5. Severe 6. Very severe
22*	To what extent are you bothered by people blaming you for your HIV status?	1. Not at all 2. A little 3. A moderate amount 4. Very much 5. An extreme amount 6. I generally do not disclose my HIV status
23*	Do you find it hard to meet the needs of those you care for (such as children, parents, grandparents, partners, or pets)?	1. Not applicable 2. Not at all 3. A little 4. A moderate amount 5. Very much 6. Extremely
24	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	1. Very poor 2. Poor 3. Neither poor nor good 4. Good
25	How would you rate your quality of life?	

		5. Very good
26	(conditional: only if PLWH indicate having had bodily pain) How much did pain interfere with your normal activities (including work outside the home, housework and social activities)?	
27*	How much are you bothered by any physical problems related to your HIV infection?	1. Not at all
28	To what extent do you feel your life to be meaningful?	2. A little
29	How much do you fear the future?	3. A moderate amount
30	How much do you worry about death?	4. Very much
31	To what extent have you been worried about telling someone you have HIV?	5. An extreme amount

Section 5: These questions are **not framed within a specific time period**

#	Item	Response options
[Matrix - intro] Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following contexts or situations:		
32*	Starting an intimate relationship	1. Never
33*	Applying for a job	2. Seldom
34*	Participating in social activities such as visiting friends or other leisure activities	3. Quite often
		4. Very often
		5. Always
[Matrix - intro] Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by the following individuals?		
35	Friends	1. Never
36	Family	2. Seldom
37	Partner	3. Quite often
39	Colleagues or employer	4. Very often
39	Health professionals in the HIV reference centre*	5. Always
40	Health professionals in a hospital (outside of the HIV reference centre*)	6. I do not disclose my HIV-status to this/these individuals
41	Health professionals outside of the hospital	7. Not applicable
42	How satisfied are you with the conditions of your living place?	
43*	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?	1. Very dissatisfied
		2. Dissatisfied
		3. Neither satisfied nor dissatisfied
		4. Satisfied
		5. Very satisfied
[Matrix - intro] How satisfied are you with your access to the following healthcare services or professionals?		
45	Infectiologist	1. Very dissatisfied
46	Psychologist	2. Dissatisfied
47	Social support / worker	3. Neither satisfied nor dissatisfied
48	Sexologist	4. Satisfied
49	General practioner	5. Very satisfied
50	Dentist	6. I don't know if I have access
51	(conditional : for female sex only) Gynaecologist	

52	HIV nurse	7. No need
[Matrix - intro] Do you have enough support from the following health care professionals to help you manage your own health and wellbeing?		1. I do not need support
53	Infectiologist	2. I have had no support
54	Psychologist	3. I sometimes have enough support
55	Social support / worker	4. I often have enough support
56	Sexologist	5. I always have enough support
57	General practioner	6. Not sure
58	Dentist	
59	(conditional : for female sex only) Gynaecologist	
60	HIV nurse	
61	Are you involved as much as you want in decisions about your HIV care?	1. Not at all
		2. A little
		3. Moderately
		4. Mostly
		5. Completely
		6. Not sure

4. The generic module: part 2

Items are numbered consecutively, picking up where Part I of the module ends.

4.1. Past and present labour market positions

Domain / subdomains	#	Item	Item response options
Current labour market position	12*	Do you have at this moment a paid job (incl. self-employed job)?	1. Yes, paid job 2. Yes, I have a paid job but am currently on career break 3. Yes, a paid student job 4. No, no paid job
	13*	(conditional: if item 12 = 1 or 2) Which of the following best describes your paid job?	1. Employee (not civil servant) 2. Self-employed without employees 3. Self-employed with employees 4. Civil servant 5. I am a family worker
	13a	(conditional: if item 12 = 1 or 2) Which of the following best describes the type of work you do?	1. Mainly manual work 2. Mainly intellectual work 3. Manual and intellectual work

			4. Management
	13b	(conditional: if item 12 = 1 or 2) Can you briefly describe your function and the sector you work in?	[Open text response]
	14	(conditional: if item 12 = 4) You don't have a paid job nowadays. Which of the following profiles is applicable to you?°	1. Unemployed 2. On sick leave or invalidity 3. Full-time student 4. (early) Retirement 5. I do the housekeeping, without benefits 6. I receive OCMW / CPAS support 7. I don't work and receive an integration allowance due to handicap 8. Other situation: [Open text box]
	14a	(conditional: if item 14 = 1) Are you exempt from looking for work because of a health condition?	1. Yes 2. No
Past labour market position	15*	(conditional: if item 12 = 4) Did you ever have a paid job (incl. self-employed, and excluding student jobs)?	1. Yes, I once had a paid job 2. No, I have never had a paid job
	16	(conditional: if item 15 = 1) Which of the following best describes your last job?	1. Employee (not civil servant) 2. Self-employed without employees 3. Self-employed without employees 4. Civil servant 5. I was a family worker
	16a	(conditional: if item 15 = 1) Which of the following best describes the type of work you did in your last job?	1. Mainly manual work 2. Mainly intellectual work 3. Manual and intellectual work 4. Management
	16b	(conditional: if item 15 = 1) Can you briefly describe your function and the sector you worked in at your last job?	[Open text response]
	17a	(conditional: if item 15 = 1 or item 12 = 2) In what year did you stop working?	Dropdown list: before 1970 – 1970 – 1971 - ... - 2025
	17b	(conditional: if item 15 = 1 or item 12 = 2) In which month of that year did you stop working?	Dropdown list: January - December
	18	(conditional: if item 12 = 1) Have you (temporarily) stopped your paid job since your diagnosis?	[Multiple options are possible] 1. No, I never stopped my paid job 2. Yes, I stopped my paid job and became unemployed 3. Yes, I stopped my paid job and became inactive due to sick leave or invalidity 4. Yes, I stopped my paid job and became a housemaker, without benefits 5. Yes, I stopped my paid job and received OCMW / CPAS support

			6. Yes, I stopped my paid job and was recognized as having a handicap
	19a-e	(conditional: if item 18 = 2 - 6) How long did you work interruption last?	Dropdown list: amount of months (e.g. ... - 10 months – 11 months – 12 months (1 year) – 13 months - ...)
Work place adaptation to facilitate RTW	20	(conditional: if item 18 = 2 - 6) At the moment when you returned to work, in which capacity did you do so?	1. Full-time work 2. Part-time work, but at least 4/5ths 3. Part-time work, but at least 50% or half-time 4. Part-time work, less than 50% or half-time
	20a	(conditional: if item 18 = 2 - 6) At the moment when you returned to work, did you receive any other work adjustments to take into account any limitations or difficulties linked to your health condition at the time (e.g. flexible hours, reduced workload)?	1. Yes, I received work adjustments 2. No, I did not receive work adjustments
	20b	(conditional: if item 20a = 1) Can you briefly describe the main work adjustment(s)?	[Open text box]
Volunteering work	21	Have you ever participated in volunteering work?	1. Yes, before my diagnosis but not anymore 2. Yes, before my diagnosis and I still do 3. Yes, after my diagnosis but not anymore 4. Yes, after my diagnosis and I still do 5. No, never

4.2. Job or work environment characteristics

These items will refer to an individual's current job or past job for people who currently do not have a paid job.

Factor	#	Item	Item response options
Work content	22	Does your job involve learning new things?	1. Never 2. Rarely 3. Sometimes 4. Often 5. Always
	23	In your main paid job are you able to change or choose the order of your working tasks?	
Working conditions	a-b	In your main paid job are you able to change or choose your methods of work?	
	24	Does your main job require you to work at a very high speed?	
	a-c	Does your main job require you to work very hard?	
		Does your main job require a lot of physical exertion?	
Working times	25	How often do you work between 5pm & 10pm, at night, on a Saturday, Sunday, or public holiday?	
Employment conditions > Employment stability	26	[Not for self-employed individuals] What kind of employment contract do you have in your main paid job?	1. Unlimited duration 2. Limited duration > 6 months 3. Limited duration < 6 months 4. Employment agency

Employment conditions > Working times	27	Does your work schedule change regularly beyond your control? If yes, how long before the change are you usually notified?	5. Other 1. My schedule does not change regularly outside my control. 2. Change communicated several weeks in advance. 3. Change communicated several days in advance. 4. Change communicated one day in advance. 5. Change communicated the same day.
	28	How many actual hours per week do you usually work in your main paid job?	Slide bar (0 hours – 80 hours or more)
Employment relations	29 a-c	[Matrix question] Do you agree with the following statements? (a) My immediate supervisor helps to get the job done. (b) My colleagues help to get the work done. (c) In general, I feel supported in the relationships I have with the people I work with.	1. Strongly agree 2. Tend to agree 3. Neither agree, nor disagree 4. Tend to disagree 5. Strongly disagree 6. I don't have an immediate supervisor or colleagues

4.3. Work ability and self-efficacy

Factor	#	Item	Item response options
Work-ability	30	Let's assume that your work ability would receive a score of 10 at its best. What score would you give your current work ability? (0 means that you are currently unable to do any work).	0: completely unable to work – 10: Work ability at its best
Self-efficacy		[Matrix question] How well do these descriptions fit you as a person?	1. Fits perfectly 2. Fits quite well 3. Fits a little bit 4. Does not fit
	31	"I am always able to solve difficult problems, if I try hard enough"	
	32	"If people work against me, I find a way of achieving what I want"	
	33	"It is easy for me to stick to my plans and reach my objectives"	
	34	"I feel confident that I can handle unexpected events"	
	35	"When I have a problem, I can usually find several ways of solving it"	
	36	"Regardless of what happens, I usually manage"	

4.4. Social participation

Factor	#	Item	Item response options
Social participation outside of the labour market	37	[matrix] How often in the past four weeks did you participate in the following activities: - Sport activities with other individuals - Hobbies with other individuals - In-person interactions with friends	1. 4 times or more per week 2. 2 to 3 times per week 3. Once a week 4. Less than once a week 5. I did not participate in this kind of activity

		- In-person interactions with family members - Activities organized by patient organizations or associations - Socializing with members of your community, in your neighbourhood - Volunteering activities	
	38	Have you felt that you wanted to participate in these activities more often?	1. Very much so, 2. Quite a bit 3. Somewhat 4. A little 5. Not at all
	39a	(conditional: if item 38 = 1 - 4) Why do you feel you were unable to participate in these activities as often as you would like?	[Open text box]
	39b	(conditional: if item 37 = 3 – 5 and item 38 = 5) Is there a reason why you do not feel like you would like to participate in these activities more often?	[Open text box]

4.5. Health-related quality of life and comorbidities

Factor	#	Item [origin of item]	Item response options
HRQoL	Main question: Under each heading, please tick the ONE box that best describes your health TODAY [EQ-5D, HIS]		
	46	MOBILITY	☐ "I have no problems in walking about" ☐ "I have slight problems in walking about" ☐ "I have moderate problems in walking about" ☐ "I have severe problems in walking about" ☐ "I am unable to walk about"
	47	SELF-CARE	☐ "I have no problems washing or dressing myself" ☐ "I have slight problems washing or dressing myself" ☐ "I have moderate problems washing or dressing myself" ☐ "I have severe problems washing or dressing myself" ☐ "I am unable to wash or dress myself"
	47	USUAL ACTIVITIES (e.g. work, study, housework, family or leisure activities)	☐ "I have no problems doing my usual activities" ☐ "I have slight problems doing my usual activities" ☐ "I have moderate problems doing my usual activities" ☐ "I have severe problems doing my usual activities" ☐ "I am unable to do my usual activities"
	49	PAIN / DISCOMFORT	☐ "I have no pain or discomfort" ☐ "I have slight pain or discomfort" ☐ "I have moderate pain or discomfort" ☐ "I have severe pain or discomfort" ☐ "I have extreme pain or discomfort"

	50	ANXIETY / DEPRESSION	○ "I am not anxious or depressed" ○ "I am slightly anxious or depressed" ○ "I am moderately anxious or depressed" ○ "I am severely anxious or depressed" ○ "I am extremely anxious or depressed"
Subjective health	51	How is your health in general?	1. Very good 2. Good 3. Fair 4. Poor 5. Very poor
Comorbidities	52	Do you have any other conditions or diseases that were diagnosed by a physician?	[Multiple responses can be selected] 1. No 2. Injury resulting from an accident (e.g. back or limb injury, burns) 3. Musculoskeletal disease affecting the back, limbs or other body part (e.g. recurring joint or muscle pain, sciatica, rheumatism, arthritis) 4. Cardiovascular disease (e.g. hypertension, coronary heart disease) 5. Respiratory disease (e.g. recurring respiratory tract infections, asthma, emphysema) 6. Mental disorder (e.g. depression, burnout, anxiety, insomnia) 7. Neurological or sensory disease (e.g. ear or eye disease, migraine, epilepsy) 8. Digestive disease or disorder (e.g. gastritis, gallstones, liver or pancreatic disease, frequent constipation) 9. Urogenital disease (e.g. urinary tract infection, gynaecological disease, prostate condition) 10. Skin disease (e.g. allergic or other rash, varicose veins) 11. Tumour or cancer 12. Endocrine or metabolic disease (e.g. diabetes, severe obesity, gout) 13. Blood disease (e.g. anaemia, other blood disorders or deficiencies) 14. Congenital disorder 15. Other condition or disease

V. Annexes

Annex 1. Overview of instruments identified in the literature review

Instrument	Zhang et al. ¹	Cooper et al. ²	Wen et al. et al. ³	Additional search >2021
ACTG-SF-21		x		
CST-HIV				x
EQ-5D (generic)	x		x	
EUROHIS QoL-8			x	
FAHI			x	
HAQ-DI			x	
HAT-QOL	x		x	
HIV QoL Scale-1			x	
HIV QoL Scale-2			x	
HIV QoL Scale-3			x	
HIV QoL Scale-4			x	
HIV/AIDSQoL			x	
HIV-QL31		x	x	
HIV-SQUAD		x		
HOPES			x	
HUI3			x	
ISSQoL			x	
MOS (generic)			x	
MOS-HIV	x	x	x	
MOS-SF-20 (Generic)			x	
MQOL-HIV		x	x	
MVQoLI			x	
PozQoL			x	
PROMIS-29 (generic)			x	
PROQOL-HIV		x	x	
QWB scale (generic)			x	
SF-12 (generic)	x		x	
SF-36 (generic)	x		x	
SF-HDQ				x
WHOQOL HIV			x	
WHOQOL-BREF (generic)	x			
WHOQOL-HIV BREF	x	x	x	

¹ Zhang Y, He C, Peasgood T, Hulse ESG, Fairley CK, Brown G, Ofori-Asenso R, Ong JJ. Use of quality-of-life instruments for people living with HIV: a global systematic review and meta-analysis. *J Int AIDS Soc.* 2022;25(3):e25902. doi:10.1002/jia2.25902

² Cooper V, Clatworthy J, Harding R, Whetham J; Emerge Consortium. Measuring quality of life among people living with HIV: a systematic review of reviews. *Health Qual Life Outcomes.* 2017;15:220. doi:10.1186/s12955-017-0778-6

³ Wen H, Yang Z, Zhu Z, Han S, Zhang L, Hu Y. Psychometric properties of self-reported measures of health-related quality of life in people living with HIV: a systematic review. *Health Qual Life Outcomes.* 2022;20:5. doi:10.1186/s12955-021-01910-w

Annex 2. Adapted COSMIN checklist



Adapted_COSMIN_
Checklist.pdf

Annex 3. Results of the COSMIN rating exercise



HIV_COSMIN_rating
_results.xlsx

Annex 4. Overview of subtopics and ICF codes for the topic of social relationships

Topics and subtopics	FG	HRC	#PROMs	WHO-HIV	MOS-HIV	PROQoL-HIV	CST-HIV	SF-HDQ
Social relationships	x	x	4	x		x	x	x
Acceptance by others								
b152 - Emotional functions			1	x				
e410 - Individual attitudes of family members			1	x				
e415 - Individual attitudes of extended family members			1	x				
e420 - Individual attitudes of friends			1	x				
e425 - Individual attitudes of acquaintances			1	x				
Asking for help			1					x
No ICF code			1					x
Being open about disease	x		0				x	
No ICF code	x		0				x	
d660 - Assisting others	x		0					
d740 - Formal relationships		x	0					
d750 - Informal social relationships	x	x	0					
d760 - Family relationships	x	x	0					
d770 - Intimate relationships		x						
d9205 - Socializing	x		0					
e310 - Immediate family	x		0					
e315 - Extended family	x		0					
e320 - Friends	x		0					
e420 - Individual attitudes of friends	x		0					
e425 - Individual attitudes of acquaintances, peers, colleagues, neighbours and community members	x		0					
Friends			1	x				
e320 - Friends			1	x				
Intimate relationships	x	x	4	x		x	x	x
d770 - Intimate relationships		x	4	x		x	x	x
d7700 - Romantic relationships			1			x		
d7702 - Sexual relationships			2				x	x
Loneliness	x		1					x
b152 - Emotional functions			1					x
d750 - Informal social relationships	x		0					
d760 - Family relationships	x		0					
d910 - Community life	x		0					
d9205 - Socializing	x		1					x
e310 - Immediate family	x		0					
e325 - Acquaintances, peers, colleagues	x		0					
Personal relationships	x		2	x		x		
d660 - Assisting others	x		0					
d750 - Informal social relationships	x		2	x		x		
d760 - Family relationships	x		2	x		x		
d7600 - Parent-child relationships	x		0					
d770 - Intimate relationships	x		1	x				
d7702 - Sexual relationships	x		0					
d910 - Community life			1			x		
d9205 - Socializing	x		2	x		x		
e310 - Immediate family	x		1			x		
e315 - Extended family	x		1			x		
e320 - Friends	x		0					
e420 - Individual attitudes of friends	x		0					
e425 - Individual attitudes of acquaintances	x		0					
Support	x		1					x
d750 - Informal social relationships			1				x	
d760 - Family relationships			1				x	
d770 - Intimate relationships			1				x	
e310 - Immediate family		x	1				x	
e315 - Extended family		x	1				x	
e320 - Friends		x	1				x	

e325 - Acquaintances, peers, colleagues		1	x
e340 - Personal care providers and personal assistants		1	x
Supporting others			
d660 - Assisting others	x	1	x
d850 - Non-remunerative employment	x	0	

Annex 5. Expert validation survey



Sciensano surveys -
SPADIS_ Expert valid

Annex 6. Outcomes of the translation process: reconciliation exercise and the backward translations

Reconciliation exercise: from English to Dutch

Item	ref	Original item (English)	Preferred translation of item		Reconciliation	
			A or B	Decision criteria	Decision option	Translated item
1	21	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?	A	1.6	6	Ben je tevreden met je vermogen om alledaagse activiteiten te verrichten (bijvoorbeeld uitvoeren van huishoudelijke taken, boodschappen doen, of zelfzorg zoals aankleden en wassen, enz.)?
2	40	Do you find it hard to care for those who depend on you (such as children, parents, grandparents, partners, or pets)?	A = B	/	/	Vind je het moeilijk om te zorgen voor degenen die van jou afhankelijk zijn (zoals kinderen, ouders, grootouders, partners of huisdieren)?
3	17	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments?	Required no translation			
4	18	How much of the time did you have difficulty doing activities involving concentration and thinking?	Required no translation			
5	19	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?	Required no translation			
6	1 - 4	How often do you have the following negative feelings: (a) blue mood, (b) despair, (c) feelings of anxiety, (d) depressive feelings, (e) loneliness?	A = B	/	/	Hoe vaak heb je de volgende negatieve gevoelens: een sombere stemming, wanhoop, angstgevoelens, depressieve gevoelens, eenzaamheid?
7	27	Do you have enough energy for everyday life?	Required no translation			
8	22	How satisfied are you with your sleep?	Required no translation			
9	31	How much bodily pain have you generally had?	Required no translation but adapted nonetheless			
10	30	how much did pain interfere with your normal activities (including work outside the home and housework and social activities)?	A	4.3, 4.1	1	Hoeveel lichamelijke pijn heb je? In welke mate ben je door pijn gehinderd in je normale activiteiten (zowel werk buitenshuis als huishoudelijk werk en sociale activiteiten zoals vrienden bezoeken of vrijetijdsactiviteiten)?
11	32	How well are you able to get around (walking short distances, climbing a few flights of stairs, bending or lifting)?	B	1.6, 3.4, 4.2	2	Hoe goed kan je jezelf verplaatsen (korte afstanden wandelen, enkele trappen oplopen, bukken of tillen)?
12	23	How satisfied are you with your health?	Required no translation			
13	33	How would you rate your quality of life?	Required no translation			
14	24	How satisfied are you with your sex life?	Required no translation – In welke mate ben je tevreden met je seksleven? (Decision 9; decision criteria: 4.2, 1.6)			
15	34	How much are you bothered by any physical problems related to your HIV infection?	A = B	/	8	Hoeveel last heb je van lichamelijke problemen die verband houden met je hiv-infectie?
16	35	To what extent do you feel your life to be meaningful?	Required no translation			
17	36	How much do you fear the future?	A	1.1, 1.6, 4.2	6	In welke mate heb je angst voor de toekomst?
18	37	How much do you worry about death?	A	1.1, 1.6, 4.2	6	In welke mate maak je je zorgen over de dood?
19	38	To what extent are you bothered by people blaming you for your HIV status?	B	1.5	4	In welke mate heb je last van mensen die je de schuld geven van je hiv-status?

20	8 - 10	Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following contexts or situations: (a) starting an intimate relationship, (b) applying for a job, (c) participating in social activities	B	1.6, 1.5, 4.3	4	Heb je jezelf ooit weerhouden van belangrijke zaken vanwege mogelijke reacties van anderen op uw hiv-status, in een van de volgende situaties: (a) het beginnen van een intieme relatie, (b) solliciteren voor werk, (c) deelnemen aan sociale activiteiten zoals vrienden bezoeken en vrijetijdsactiviteiten?
21	11 - 16	Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following individuals? (a) Friends, (b) Family, (c) partner, (d) colleagues or employers (e) health professionals in the HIV reference centre, (f) health professionals in a hospital (not in HIV reference centre), (g) health professionals outside the hospital	A = B	/	/	Heb je ooit negatieve houdingen ervaren of bent je anders behandeld vanwege je Hiv-status door een van de volgende personen? (a) vrienden, (b) familie, (c) partner, (d) collega's of werkgevers, (e) zorgverleners in het hiv referentie centrum, (f) zorgverleners in een ziekenhuis (niet in een hiv referentie centrum), (g) zorgverleners buiten het ziekenhuis
22	20	How much of the time has living with HIV limited your social activities (like visiting with friends or participating in leisure activities)?	A = B	/	/	Hoe vaak heeft je gezondheid je gehinderd bij je sociale activiteiten (zoals vriendenbezoeken of vrijetijdsactiviteiten)?
23	28	Does your health keep you from working at a job or going to school? (required no Dutch translation)	A = B	/	/	Verhindert je gezondheid dat je kan gaan werken of naar school kan gaan?
24	25	How satisfied are you with your personal relationships (partner, friends and family)?	A = B	/	/	Hoe tevreden ben je met je persoonlijke relaties (partner, vrienden en familie)?
25	26	How satisfied are you with the support you get from your partner, friends and family?	A = B	/	/	Hoe tevreden ben je met de steun die je krijgt van je partner, vrienden en familie?
26	6	How often have you had someone you trust to speak to about your problems?	A	1.6	1	Hoe vaak was er iemand die je vertrouwt om te praten over je problemen?
27	39	To what extent have you been worried about telling someone you have HIV?	A	4.3	1	In welke mate maak je je zorgen om iemand te vertellen dat je hiv hebt?
28	7	How often do you find it difficult to take your HIV medication properly?	A = B	/	/	Hoe vaak vind je het moeilijk om je hiv-medicatie op de juiste manier in te nemen?
29	29	Have you enough money to meet your needs?	Required no translation			
30	41	How satisfied are you with the conditions of your living place? (Required no French translation)	A	2.1, 2.2	1	Hoe tevreden ben je over de omstandigheden van de plaats waar je leeft?
31	43- 49	How satisfied are you with your access to the following healthcare services or professionals? (a) Infectiologist, (b) Psychologist, (c) Social support/worker, (d) Sexologist, (e) General Practitioner, (f) Dentist, (g) gynaecologist (women only) (h) HIV nurse	A = B	/	/	Hoe tevreden ben je over je toegang tot de volgende zorgverleners? (a) Infectioloog, (b) Psycholoog, (c) Maatschappelijk werker, (d) Seksuoloog, (e) Huisarts, (f) Tandarts, (g) Gynaecoloog (alleen voor vrouwen), (h) hiv-verpleegkundige
32	50- 56	Do you have enough support from the following health care professionals to help you manage your own health and wellbeing? (a) Infectiologist, (b) Psychologist, (c) Social support/worker, (d) Sexologist, (e) General Practitioner, (f) Dentist, (g) Gynaecologist (for women), (h) HIV nurse	B	1.6	2	Krijg je voldoende ondersteuning van de volgende zorgverleners om voor je gezondheid en welzijn te zorgen? (a) Infectioloog, (b) Psycholoog, (c) Maatschappelijk werker, (d) Seksuoloog, (e) Huisarts, (f) Tandarts, (g) Gynaecoloog (alleen voor vrouwen), (h) Hiv-verpleegkundige
33	42	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?	B	1.6, 1.1, 1.3, 1.4	4	Hoe tevreden ben je over uw relaties met de zorgverleners die betrokken zijn bij je zorg (bijvoorbeeld zorgzame houding, communicatie met zorgverleners)?
34	57	Are you involved as much as you want in decisions about your HIV care?	B	1.1, 1.4, 4.2	6	Ben je zoveel als je wilt betrokken bij beslissingen over de hiv-zorg die je krijgt?

Reconciliation exercise: from English to French

Item	ref	Original item (English)	Preferred translation of item		Reconciliation	
			A or B	Decision criteria	Decision option	Translated item
1	21	How satisfied are you with your ability to perform your daily living activities (for example, doing housework, doing errands such as grocery shopping, or self-care such as dressing, bathing etc.)?	A	1.5, 1.6	1	Dans quelle mesure êtes-vous satisfait.e de votre capacité à accomplir vos activités quotidiennes (par exemple, faire le ménage, les courses, ou les soins personnels comme s'habiller, se laver, etc.)?
2	40	Do you find it hard to care for those who depend on you (such as children, parents, grandparents, partners, or pets)?	B	3.1	6	Avez-vous des difficultés à prendre soin de ceux qui dépendent de vous ? (comme vos enfants, parents, grands-parents, partenaires, animaux domestiques)
3	17	How much of the time did you forget things that happened recently, for example, where you put things and when you had appointments?	Avez-vous oublié des choses qui sont arrivées dernièrement, par exemple des rendez-vous ou là où vous aviez mis vos affaires?			
4	18	How much of the time did you have difficulty doing activities involving concentration and thinking?	Required no translation			
5	19	How much of the time did you have difficulty reasoning and solving problems, for example, making plans, making decisions, learning new things?	Required no translation			
6	1 - 4	How often do you have the following negative feelings: (a) blue mood, (b) despair, (c) feelings of anxiety, (d) depressive feelings, (e) loneliness?	C			A quelle fréquence ressentez-vous les sentiments négatifs suivants : (a) la tristesse, (b) le désespoir, (c) des sentiments d'anxiété, (d) des sentiments dépressifs, (e) la solitude
7	27	Do you have enough energy for everyday life?	Required no translation			
8	22	How satisfied are you with your sleep?	Required no translation			
9	31	How much bodily pain have you generally? How much bodily pain did you have?	Required no translation			
10	30	how much did pain interfere with your normal activities (including work outside the home and housework and social activities)? Replace interfere by limit	A	1.6 1.1	5	Dans quelle mesure vos douleurs physiques vous ont-elles limité.e dans vos activités normales (y compris votre travail, vos tâches ménagères et vos activités sociales)?
11	32	How well are you able to get around by yourself (walking short distances, climbing a few flights of stairs, bending or lifting)?	A et B	3.1 1.6	6	Comment trouvez-vous votre capacité à vous déplacer par vous-même (marcher de courtes distances, monter quelques marches, vous pencher ou soulever des objets)?
12	23	How satisfied are you with your health?	Required no translation			
13	33	How would you rate your quality of life?	Required no translation			
14	24	How satisfied are you with your sex life?	Required no translation			
15	34	How much are you bothered by any physical problems related to your HIV infection?	A	4.3	1	Dans quelle mesure êtes-vous gêné.e par des problèmes physiques liés à votre infection par le VIH ?
16	35	To what extent do you feel your life to be meaningful?	Required no translation Dans quelle mesure pensez-vous que votre vie à un sens?			
17	36	How much do you fear the future?	A et B	1.6 1.1	5	Dans quelle mesure craignez-vous l'avenir?
18	37	How much do you worry about death?	B	3.1	2	Dans quelle mesure la mort vous inquiète-t-elle ?
19	38	To what extent are you bothered by people blaming you for your HIV status?	A		1	Dans quelle mesure êtes-vous gêné.e par les personnes qui vous reprochent votre séropositivité ?

20	8 - 10	Have you ever withheld yourself from important matters, because of potential reactions of others to your HIV-status, in one of the following contexts or situations: (a) starting an intimate relationship, (b) applying for a job, (c) participating in social activities	B	1.1	4	Vous êtes-vous déjà retenu de faire des choses importantes, en raison des réactions potentielles des autres face à votre statut VIH, dans l'un des contextes ou situations suivants : (a) débiter une relation intime, (b) postuler à un emploi, (c) participer à des activités sociales ?
21	11 - 16	Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following individuals? (a) Friends, (b) Family, (c) partner, (d) colleagues or employers (e) health professionals in the HIV reference centre, (f) health professionals in a hospital that is not a HIV reference centre, (g) health professionals outside the hospital	B	1.1	6	Avez-vous déjà été confronté.e à des attitudes négatives ou avez-vous été traité différemment en raison de votre statut VIH par l'une des personnes suivantes ? (a) Amis, (b) Famille, (c) Partenaire, (d) Collègues ou employeurs, (e) Professionnels de santé du centre de référence VIH, (f) Professionnels de santé hospitalier (en dehors du centre de référence VIH), (g) Professionnels de santé en dehors de l'hôpital.
22	20	How much of the time has living with HIV limited your social activities (like visiting with friends or participating in leisure activities)?	A		3	A quelle fréquence, le fait de vivre avec le VIH, a-t-il limité votre vie sociale (comme rendre visite à des amis ou participer à des activités de loisirs)?
23	28	Does your health keep you from working at a job or going to school? Follow vocational training?	B			Votre état de santé vous empêche-t-il d'exercer une activité professionnelle ou de suivre une formation?
24	25	How satisfied are you with your personal relationships (partner, friends and family)?	A		1	Dans quelle mesure êtes-vous satisfait.e de vos relations personnelles (partenaire, amis, famille)?
25	26	How satisfied are you with the support you get from your partner, friends and family?	A		1	Dans quelle mesure êtes-vous satisfait du soutien que vous recevez de votre partenaire, de vos amis et de votre famille?
26	6	How often have you had someone you trust to speak to about your problems?	A	1.1	1	A quelle fréquence avez-vous une personne de confiance à qui parler de vos problèmes?
27	39	To what extent have you been worried about telling someone you have HIV?	A et B	1.1	6	Dans quelle mesure vous êtes-vous inquiété.e du fait de dire à quelqu'un que vous avez le VIH ?
28	7	How often do you find it difficult to take your HIV medication properly?	A		1	A quelle fréquence avez-vous des difficultés à prendre correctement votre traitement contre le VIH?
29	29	Have you enough money to meet your needs?	Required no translation			
30	41	How satisfied are you with the conditions of your living place?				
31	43- 49	How satisfied are you with your access to the following healthcare services or professionals? (a) Infectiologist, (b) Psychologist, (c) Social support/worker, (d) Sexologist, (e) General Practitioner, (f) Dentist, (g) gynaecologist (women only) (h) HIV nurse	A		3	Dans quelle mesure êtes-vous satisfait.e de votre accès aux services ou aux professionnels de santé suivants ? : (a) infectiologue, (b) psychologue, (c) assistant.e sociale, (d) sexologue, (e) médecin généraliste, (f) dentiste, (g) gynécologue (femme uniquement), (h) infirmier.ère VIH
32	50- 56	Do you have enough support from the following health care professionals to help you manage your own health and wellbeing? (a) Infectiologist, (b) Psychologist, (c) Social support/worker, (d) Sexologist, (e) General Practitioner, (f) Dentist, (g) Gynaecologist (for women), (h) HIV nurse	B	1.1 1.6	4	Recevez-vous suffisamment de soutien pour vous aider à gérer votre santé et votre bien-être de la part des professionnels de santé suivants : (a) infectiologue, (b) psychologue, (c) assistant.e sociale, (d) sexologue, (e) médecin généraliste, (f) dentiste, (g) gynécologue (women only) (h) infirmier.ère VIH
33	42	How satisfied are you with your relationships with the health professionals involved in your care (e.g. caring attitudes, communication with health professionals)?	B		2	Dans quelle mesure êtes-vous satisfait.e de votre relation avec les professionnels de santé impliqué dans vos soins (par ex. attitude bienveillante et communication)
34	57	Are you involved as much as you want in decisions about your HIV care?	B		4	Etes-vous impliqué.e autant que vous le souhaitez dans les décisions au sujet de vos soins pour le VIH ?

Backward translation: from Dutch to English

Item	ref	NL item	Backtranslated item (ENG)
1	21	Ben je tevreden met je vermogen om alledaagse activiteiten te verrichten (bijvoorbeeld uitvoeren van huishoudelijke taken, boodschappen doen, of zelfzorg zoals aankleden en wassen, enz.)?	Are you satisfied with your ability to perform activities of daily living (for example doing housework, going shopping, or self-care such as getting dressed and washing, etc.)?
2	40	Vind je het moeilijk om te zorgen voor degenen die van jou afhankelijk zijn (zoals kinderen, ouders, grootouders, partners of huisdieren)?	Do you find it difficult to look after those who are dependent on you (such as children, parents, grandparents, partners or pets)?
6	1 - 4	Hoe vaak heb je de volgende negatieve gevoelens: een sombere stemming, wanhoop, angstgevoelens, depressieve gevoelens, eenzaamheid?	How often do you have the following negative feelings: a low mood, despair, anxiety, feelings of depression, loneliness?
9	31	Hoeveel lichamelijke pijn heb je?	How much physical pain do you have?
10	30	In welke mate ben je door pijn gehinderd in je normale activiteiten (zowel werk buitenshuis als huishoudelijk werk en sociale activiteiten zoals vrienden bezoeken of vrijetijdsactiviteiten)?	To what extent are you hindered by pain in your normal activities (both work outdoors as well as housework and social activities such as visiting friends or leisure activities)?
11	32	Hoe goed kan je jezelf verplaatsen (korte afstanden wandelen, enkele trappen oplopen, bukken of tillen)?	How well can you move around (walk short distances, walk up a few steps, bend or lift)?
14	24	In welke mate ben je tevreden met je seksleven?	To what extent are you satisfied with your sex life?
15	34	Hoeveel last heb je van lichamelijke problemen die verband houden met je hiv-infectie?	How much do you suffer from physical problems related to your HIV infection?
17	36	In welke mate heb je angst voor de toekomst?	To what extent are you fearful of the future?
18	37	In welke mate heb je angst voor de dood?	To what extent are you afraid of death?
19	38	In welke mate heb je last van mensen die je de schuld geven van je hiv-status?	To what extent are you bothered by people blaming you for your HIV status?
20	8 - 10	Heb je jezelf ooit weerhouden van belangrijke zaken vanwege mogelijke reacties van anderen op uw hiv-status, in een van de volgende situaties: (a) het beginnen van een intieme relatie, (b) solliciteren voor werk, (c) deelnemen aan sociale activiteiten zoals vrienden bezoeken en vrijetijdsactiviteiten?	Have you ever held back from doing important things due to other people's potential reaction to your HIV status, in one of the following situations: (a) beginning an intimate relationship, (b) applying for work, (c) taking part in social activities such as visiting friends and leisure activities?
21	11 - 16	Heb je ooit negatieve houdingen ervaren of bent je anders behandeld vanwege je Hiv-status door een van de volgende personen? (a) vrienden, (b) familie, (c) partner, (d) collega's of werkgevers, (e) zorgverleners in het hiv referentie centrum, (f) zorgverleners in een ziekenhuis (niet in een hiv referentie centrum), (g) zorgverleners buiten het ziekenhuis	Have you ever experienced negative attitudes or have you been treated differently because of your HIV status by one of the following people? (a) friends, (b) family, (c) partner, (d) colleagues or employers, (e) caregivers in the HIV reference centre, (f) caregivers in a hospital (not in an HIV reference centre) (g) caregivers outside the hospital
22	20	Hoe vaak heeft je gezondheid je gehinderd bij je sociale activiteiten (zoals vriendenbezoeken of vrijetijdsactiviteiten)?	How often has your health hindered your social activities (such as visiting friends or leisure activities)?
23	28	Verhindert je gezondheid dat je kan gaan werken of naar school kan gaan?	Does your health prevent you from going to work or school?
24	25	Hoe tevreden ben je met je persoonlijke relaties (partner, vrienden en familie)?	How satisfied are you with your personal relationships (partner, friends and family)?
25	26	Hoe tevreden ben je met de steun die je krijgt van je partner, vrienden en familie?	How satisfied are you with the support you get from your partner, friends and family?
26	6	Hoe vaak was er iemand die je vertrouwt om te praten over je problemen?	How often have you had someone you trust to talk to about your problems?

27	39	In welke mate maak je je zorgen om iemand te vertellen dat je hiv hebt?	To what extent do you worry about telling someone that you have HIV?
28	7	Hoe vaak vind je het moeilijk om je hiv-medicatie op de juiste manier in te nemen?	How often do you find it difficult to take your HIV medication in the right way?
30	41	Hoe tevreden ben je over de omstandigheden van de plaats waar je leeft?	How satisfied are you with the condition of the place where you live?
31	43-49	Hoe tevreden ben je over je toegang tot de volgende zorgverleners? (a) Infectioloog, (b) Psycholoog, (c) Maatschappelijk werker, (d) Seksuoloog, (e) Huisarts, (f) Tandarts, (g) Gynaecoloog (alleen voor vrouwen), (h) hiv-verpleegkundige	How satisfied are you with your access to the following caregivers? (a) Infectious diseases specialist, (b) Psychologist, (c) Social worker, (d) Sex therapist, (e) General Practitioner, (f) Dentist, (g) Gynaecologist (for women only), (h) HIV nurse
32	50-56	Krijg je voldoende ondersteuning van de volgende zorgverleners om voor je gezondheid en welzijn te zorgen? (a) Infectioloog, (b) Psycholoog, (c) Maatschappelijk werker, (d) Seksuoloog, (e) Huisarts, (f) Tandarts, (g) Gynaecoloog (alleen voor vrouwen), (h) Hiv-verpleegkundige	Do you get enough support from the following caregivers to care for your health and well-being? (a) Infectious diseases specialist, (b) Psychologist, (c) Social worker, (d) Sex therapist, (e) General Practitioner, (f) Dentist, (g) Gynaecologist (for women only), (h) HIV nurse
33	42	Hoe tevreden ben je over uw relaties met de zorgverleners die betrokken zijn bij je zorg (bijvoorbeeld zorgzame houding, communicatie met zorgverleners)?	How satisfied are you with your relationships with the carers who are involved in your care (for example a considerate approach, communication with caregivers)?
34	57	Ben je zoveel als je wilt betrokken bij beslissingen over de hiv-zorg die je krijgt?	Are you as involved as you want to be in decisions about the HIV care that you receive?

Reconciliation exercise: from French to English

Item	ref	FR Item	Backtranslated item (ENG)
1	21	Dans quelle mesure êtes-vous satisfait.e de votre capacité à accomplir vos activités quotidiennes (par exemple, tâches ménagères, les courses, ou les soins personnels comme s'habiller, se laver, etc.)?	How satisfied are you with your ability to accomplish your daily tasks (for example, cleaning, doing groceries, personal care activities like dressing yourself or washing yourself, etc.)?
2	40	Avez-vous des difficultés à prendre soin de ceux qui dépendent de vous ? (comme vos enfants, parents, grands-parents, partenaires, animaux domestiques)	Do you have any difficulties to take care of your dependents? (dependents include, but are not limited to, children, parents, grand-parents, romantic partners, pets, etc.)
6	1 - 4	A quelle fréquence ressentez-vous les sentiments négatifs suivants : (a) la tristesse, (b) le désespoir, (c) des sentiments d'anxiété, (d) des sentiments dépressifs, (e) la solitude	With what frequency do you experience the following negative feelings: (a) sadness, (b) hopelessness, (c) anxiety, (d) depressed, (e) loneliness
10	30	Dans quelle mesure vos douleurs physiques vous ont-elles limité.e dans vos activités normales (y compris votre travail, vos tâches ménagères et vos activités sociales)?	To what extent has the physical pain you've experienced limited you in your regular activities (such as during work, during chores, during social activities)?
11	32	Comment trouvez-vous votre capacité à vous déplacer par vous-même (marcher de courtes distances, monter quelques marches, vous pencher ou soulever des objets)?	How would you evaluate your mobility (for example, walking short distances, climbing a set of stairs, bending over or picking up objects)?
15	34	Dans quelle mesure êtes-vous gêné.e par des problèmes physiques liés à votre infection par le VIH ?	To what extent are you bothered by physical problems related to your HIV infection?
17	36	Dans quelle mesure craignez-vous l'avenir?	To what extent do you fear the future?
18	37	Dans quelle mesure la mort vous inquiète-t-elle ?	To what extent does death worry you?
19	38	Dans quelle mesure êtes-vous gêné.e par les personnes qui vous reprochent votre séropositivité ?	To what extent are you bothered by people who complain about your HIV status?
20	8 - 10	Vous êtes-vous déjà retenu de faire des choses importantes, en raison des réactions potentielles des autres face à votre statut VIH, dans l'un des contextes ou situations	Have you ever decided to not participate in important activities due to your HIV status in the following contexts or situations: (a) starting a sexual relationship, (b) applying for a job, (c) participating in social activities?

		suivants : (a) débiter une relation intime, (b) postuler à un emploi, (c) participer à des activités sociales ?	
21	11 - 16	Avez-vous déjà été confronté.e à des attitudes négatives ou avez-vous été traité différemment en raison de votre statut VIH par l'une des personnes suivantes ? (a) Amis, (b) Famille, (c) Partenaire, (d) Collègues ou employeurs, (e) Professionnels de santé du centre de référence VIH, (f) Professionnels de santé hospitalier (en dehors du centre de référence VIH), (g) Professionnels de santé en dehors de l'hôpital.	Have you ever encountered negative attitudes or been treated different by the following people due to your HIV status? (a) friends, (b) famille, (c) romantic partner, (d) colleagues or your employer, (e) healthcare professionals at the HIV reference center, (f) healthcare professionals in a hospital setting (outside of HIV reference centers), (g) healthcare professionals outside of a hospital setting.
22	20	A quelle fréquence, le fait de vivre avec le VIH, a-t-il limité vos activités sociales (comme rendre visite à des amis ou participer à des activités de loisirs)?	With what frequency does living with HIV impact your social life? (such as going to visit friends or participating in hobbies)?
23	28	Votre état de santé vous empêche-t-il d'exercer une activité professionnelle ou de poursuivre une éducation?	Does your health status impede you from engaging in your profession or following any type of training?
24	25	Dans quelle mesure êtes-vous satisfait.e de vos relations personnelles (partenaire, amis, famille)?	To what extent are you satisfied with your personal relations (for example, partners, friends, family)?
25	26	Dans quelle mesure êtes-vous satisfait du soutien que vous recevez de votre partenaire, de vos amis et de votre famille?	To what extent are you satisfied by the support you receive from your romantic partner, your friends and your family?
26	6	A quelle fréquence avez-vous eu une personne de confiance à qui parler de vos problèmes?	With what frequency do you have a person you trust you can speak to about your problems?
27	39	Dans quelle mesure vous êtes-vous inquiété.e du fait de dire à quelqu'un que vous avez le VIH ?	To what extent are you nervous to tell someone you are HIV positive?
28	7	A quelle fréquence avez-vous des difficultés à prendre correctement votre traitement contre le VIH?	To what extent do you have issues taking your HIV treatment as prescribed?
31	43-49	Dans quelle mesure êtes-vous satisfait.e de votre accès aux services ou aux professionnels de santé suivants ? : (a) infectiologue, (b) psychologue, (c) assistant.e sociale, (d) sexologue, (e) médecin généraliste, (f) dentiste, (g) gynécologue (femme uniquement), (h) infirmier.ère VIH	To what extent are you satisfied with your access to the following healthcare professionals? : (a) infectiologist (b) psychologist, (c) social worker, (d) sex therapist, (e) general practitioner, (f) dentist, (g) gynecologist (women only)
32	50-56	Recevez-vous suffisamment de soutien pour vous aider à gérer votre santé et votre bien-être de la part des professionnels de santé suivants : (a) infectiologue, (b) psychologue, (c) assistant.e sociale, (d) sexologue, (e) médecin généraliste, (f) dentiste, (g) gynécologue (women only) (h) infirmier.ère VIH	Do you receive sufficient support from the following healthcare professionals to facilitate you managing your health and wellbeing? : (a) infectiologist (b) psychologist, (c) social worker, (d) sex therapist, (e) general practitioner, (f) dentist, (g) gynecologist (women only)
33	42	Dans quelle mesure êtes-vous satisfait.e de votre relation avec les professionnels de santé impliqué dans vos soins (par ex. attitude bienveillante et communication)	To what extent are you satisfied with your relationship with the healthcare professionals involved in your care? (for example, their kindness and communication).
34	57	Etes-vous impliqué.e autant que vous le souhaitez dans les décisions au sujet de vos soins pour le VIH ?	Are you as involved as you would want to be in the decision made about your HIV care?

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ABOUT SCIENSANO

Sciensano is a public and independent scientific institute dedicated to human and animal health and their environment.

We conduct scientific research and surveillance, and support the government and various agencies and organisations with health expertise, advice and services.

Sciensano attaches great importance to the One Health principle, which emphasises that the health of humans, animals and their environment are interconnected.

Therefore, our research is interdisciplinary and integrates different perspectives.

With more than a thousand employees and 120 years of scientific expertise, Sciensano is a renowned science institute, part of several international scientific networks.

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